

COMPUTATIONAL ILLUMINATION

Computational Imaging – Spring 2024

Julio Marco, Ana Serrano, Albert Redo-Sanchez

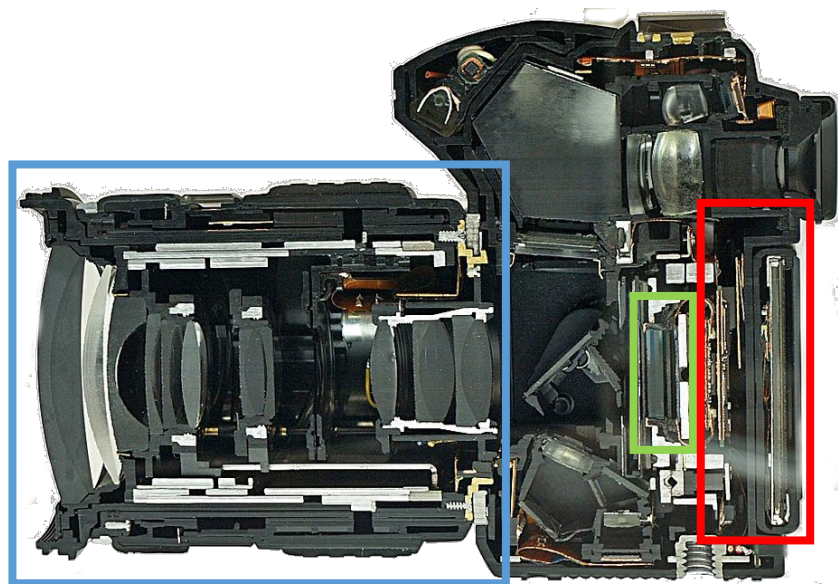
Master in Robotics, Graphics and Computer Vision – Universidad de Zaragoza

Previously on **COMPUTATIONAL IMAGING**

The modern photography pipeline



post-capture processing



optics and
optical controls



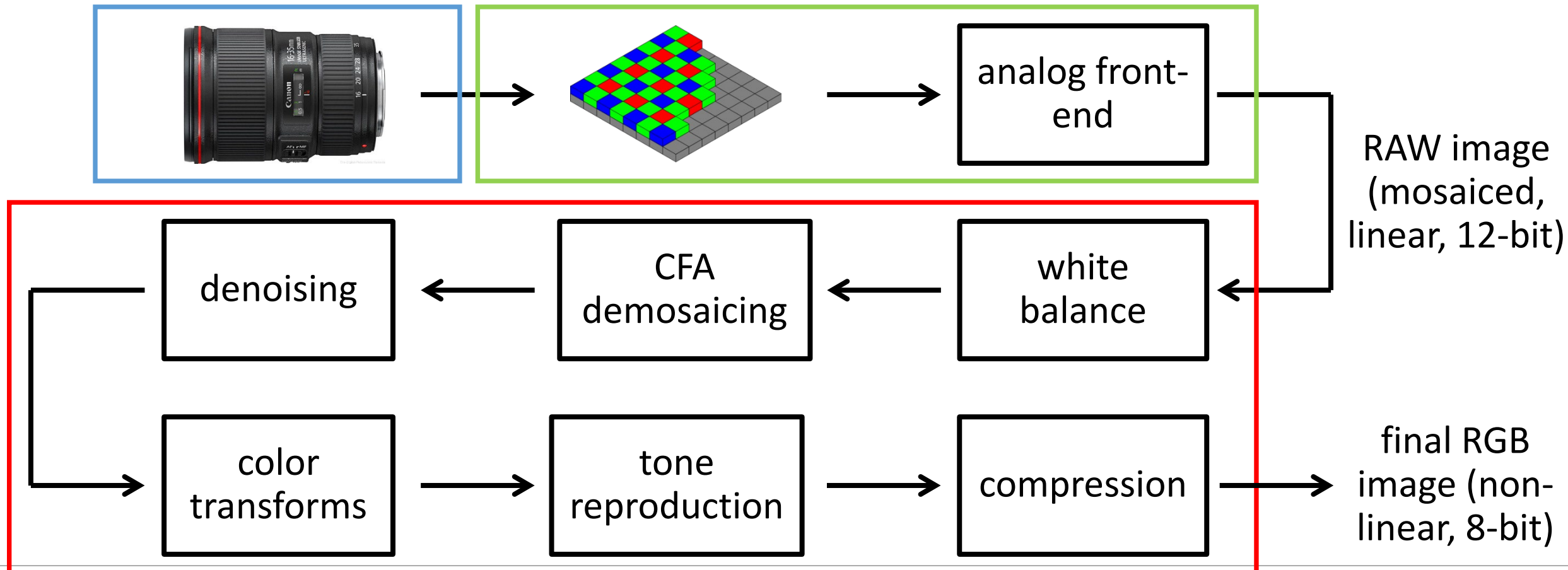
sensor, analog
front-end, and
color filter array



in-camera image
processing
pipeline

The (in-camera) image processing pipeline

The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



Plenoptic imaging

- ✓ Basis:
- ✓ Digital photography (Lecture #1)
- ✓ Mathematical foundations (Lecture #2)

Transient imaging (Lecture #8)

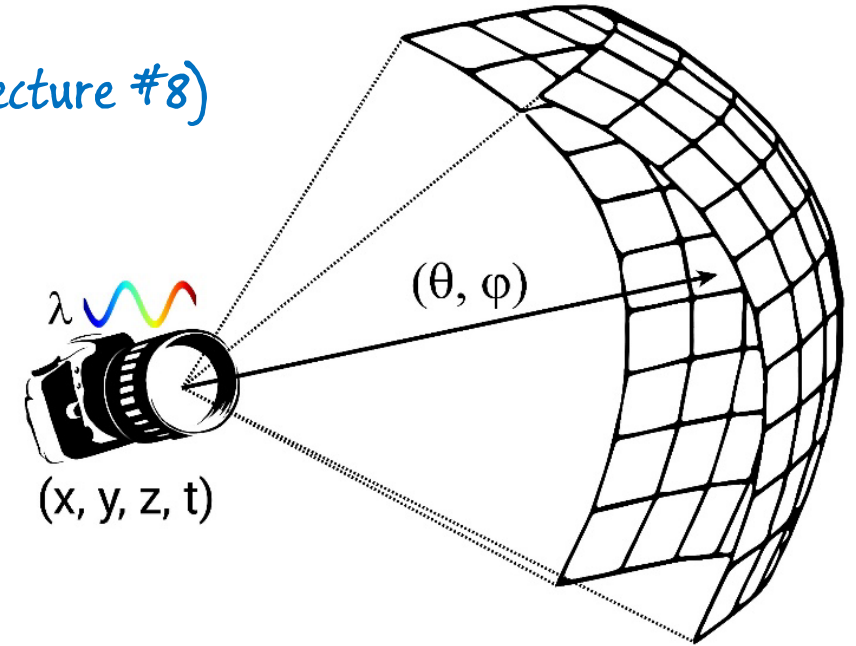
Hyperspectral imaging (Lecture #7)

$$P = P(\theta, \phi, \lambda, t, \underline{x, y, z})$$

✓ Light field imaging (Lecture #4)

✓ High dynamic range imaging (Lecture #5)

✓ Computational displays (Lecture #6)



- Plus common tools and techniques:
- ✓ Coded photography (Lecture #3)
- ✓ Compressive sensing (Lecture #3)
- Computational illumination (Lecture #7)

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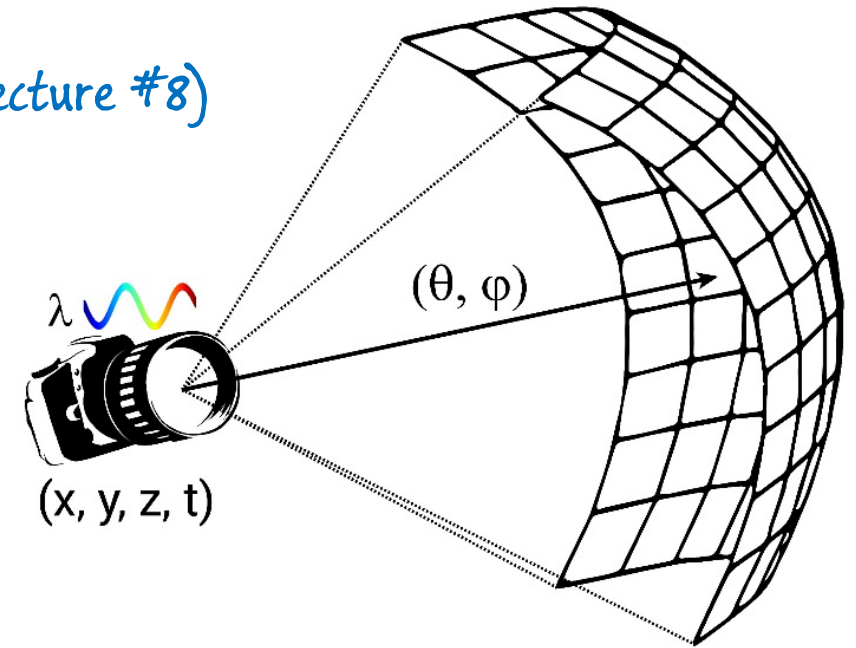
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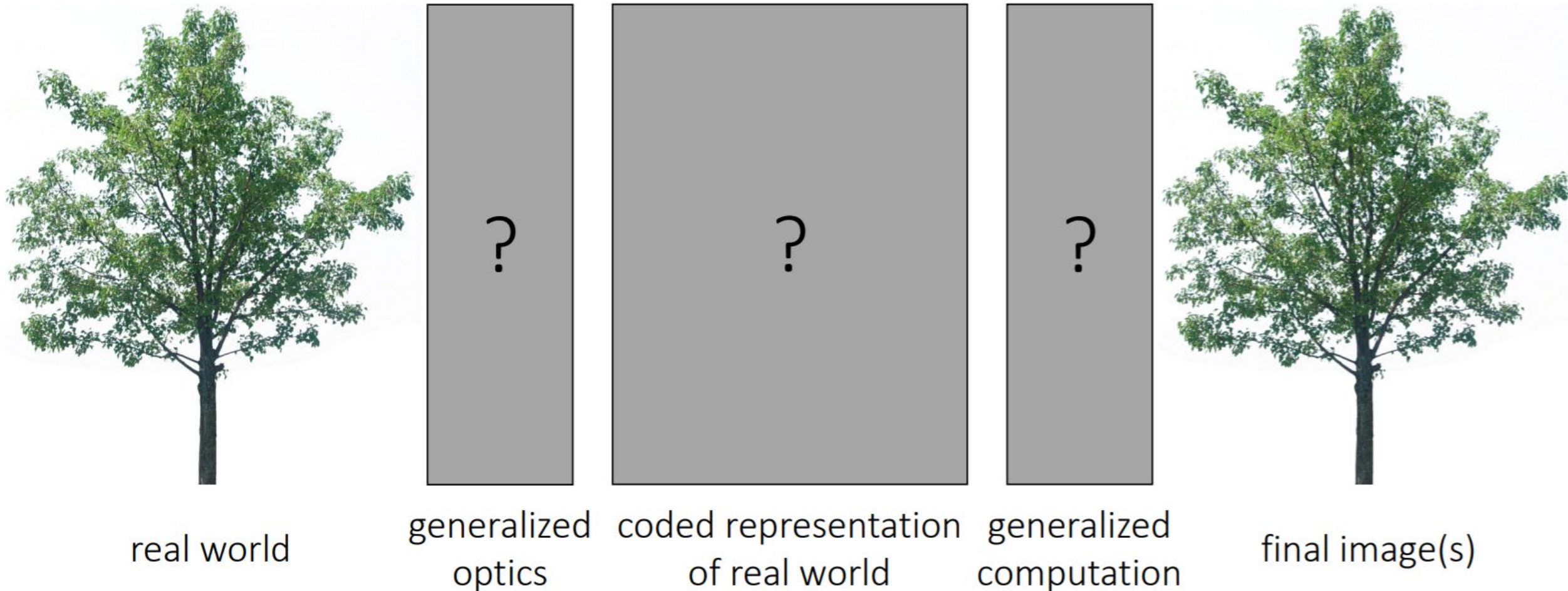
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Computational photography



- Generalized optics encode world into intermediate representation.
- Generalized computation decodes representation into multiple images.

WAIT!
I THOUGHT THIS COURSE WAS ABOUT
COMPUTATIONAL IMAGING

THIS IS ONLY **COMPUTATIONAL PHOTOGRAPHY!**

Comp. Imaging vs Comp. Photography

- Computational Photography is **passive**
 - You don't alter the scene
 - Just capture the reflected field of the scene
- But can you do more?
- Why not developing **active** systems?

The pipeline – *Conventional (Digital) Photography*



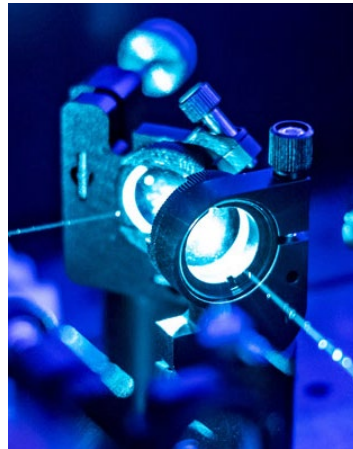
Camera



The pipeline – *Computational Photography*



Generalized
Optics



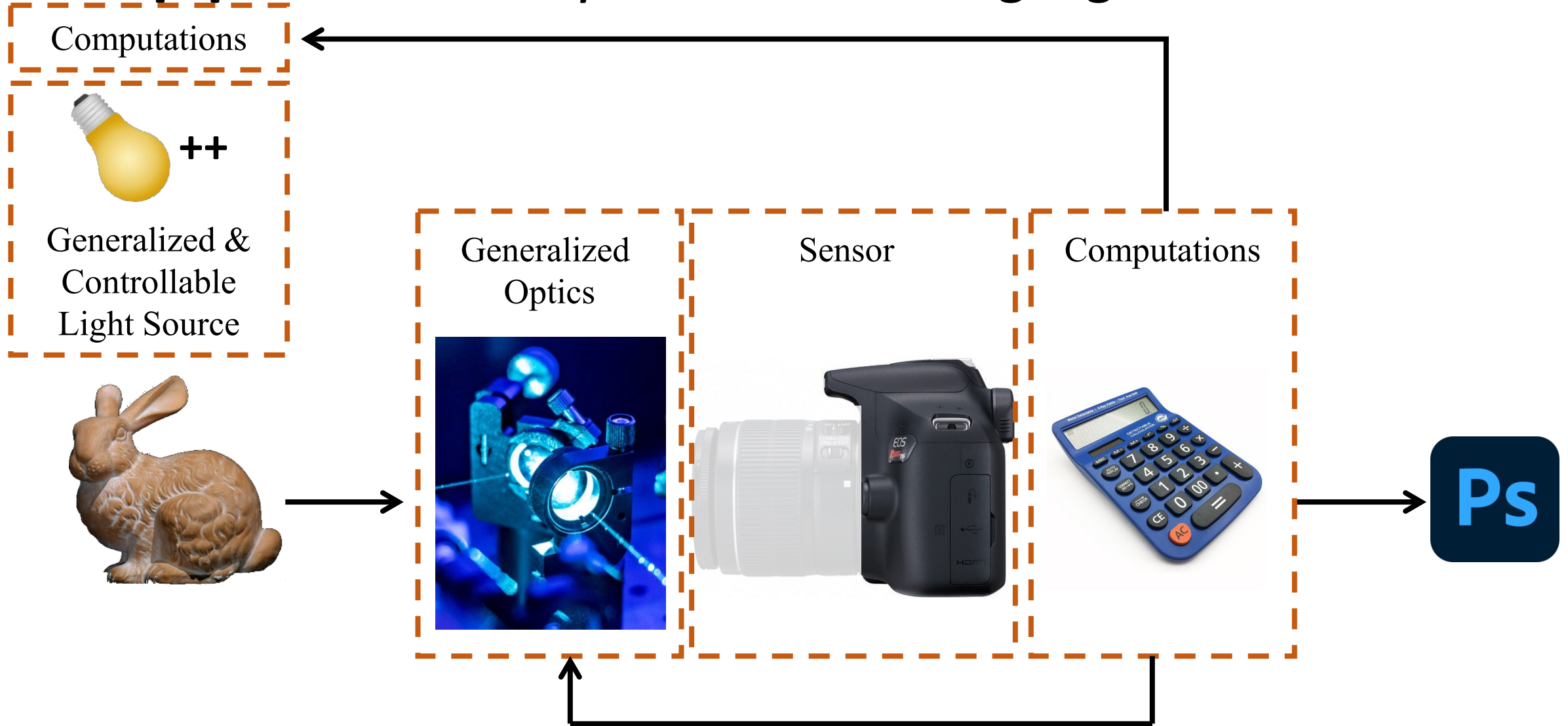
Sensor



Computations



The pipeline – *Computational Imaging*



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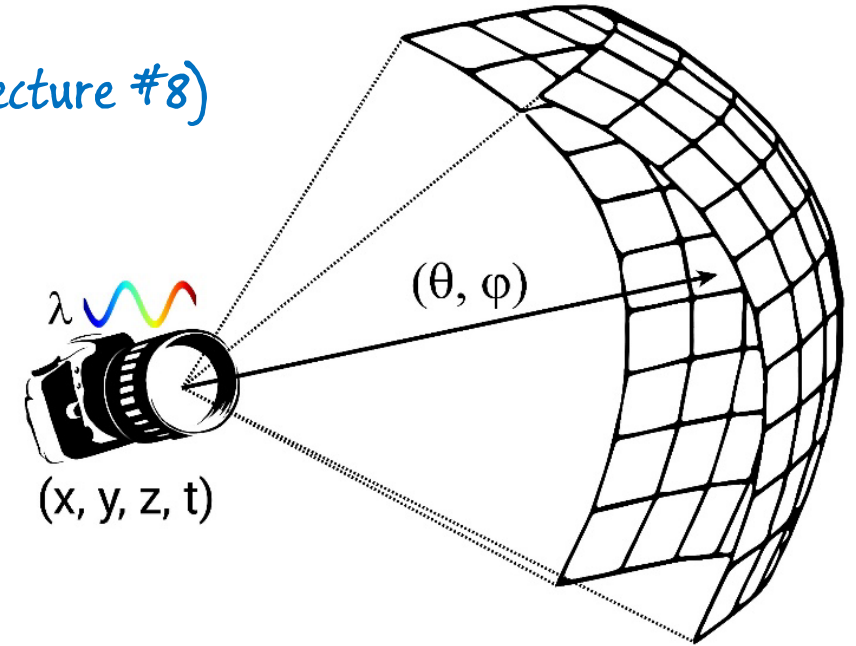
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Questions?

Passive vs Active (Illumination) Systems

Active systems are ubiquitous even in conventional photography.

Brainstorming: Any example?

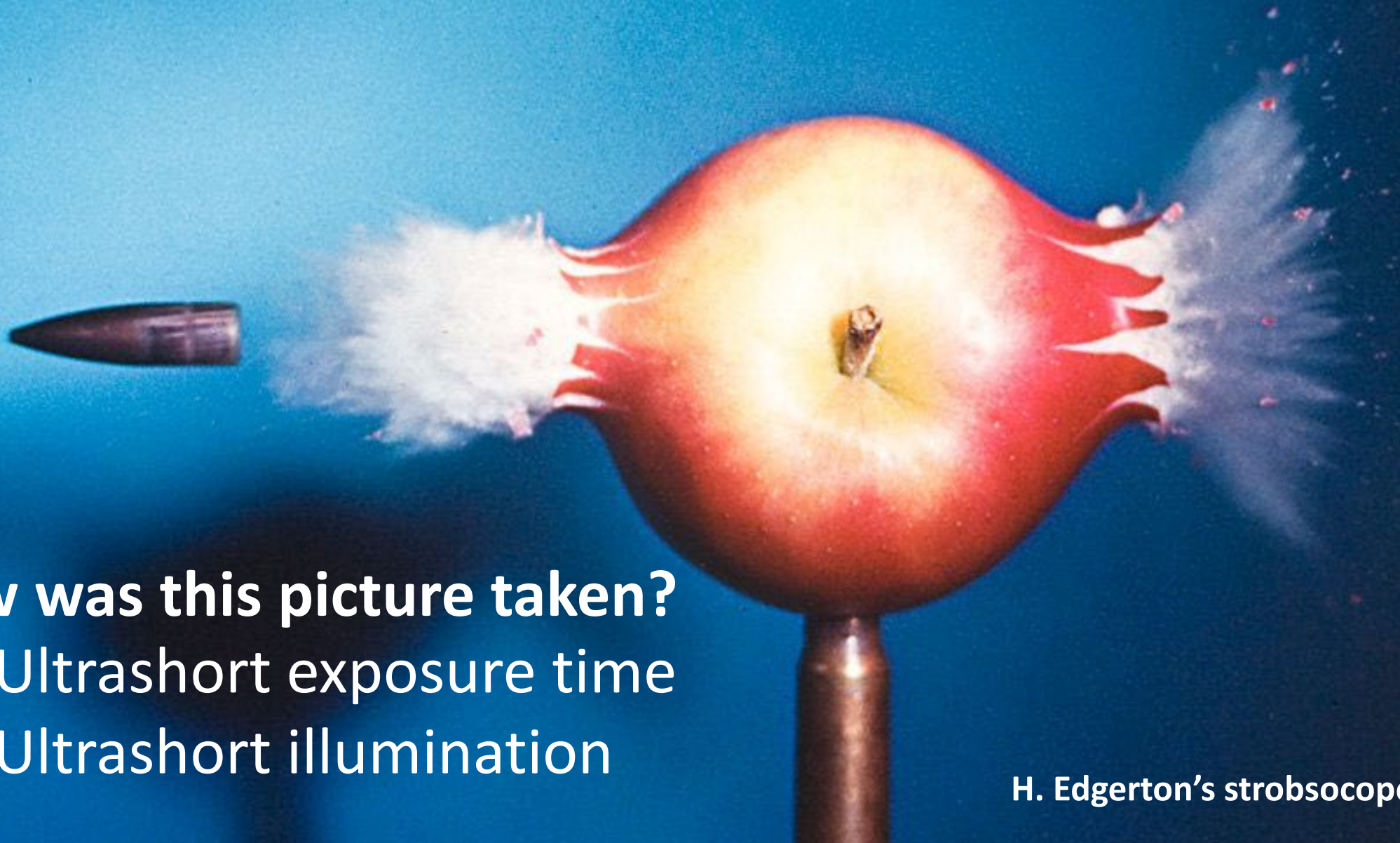


Passive vs Active (Illumination) Systems

Active systems are ubiquitous even in conventional photography.



Passive vs Active (Illumination) Systems



Q: How was this picture taken?

- a) Ultrashort exposure time
- b) Ultrashort illumination

H. Edgerton's stroboscope (1930)

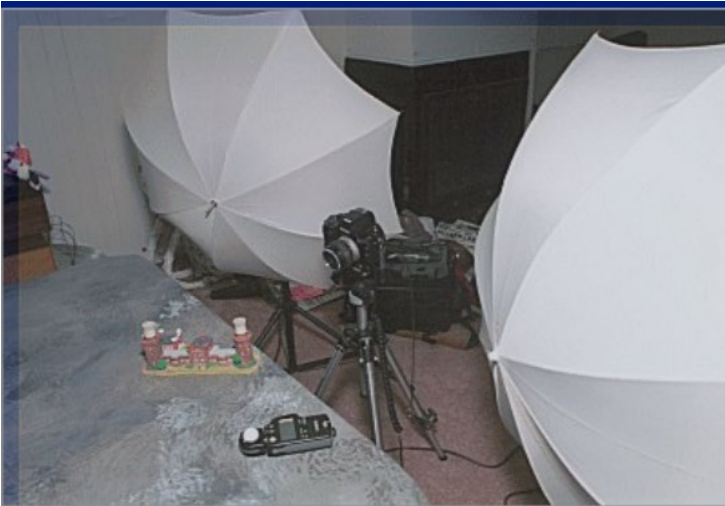
Passive vs Active (Illumination) Systems



H. Edgerton's stroboscope (1930)

Active Illumination Systems

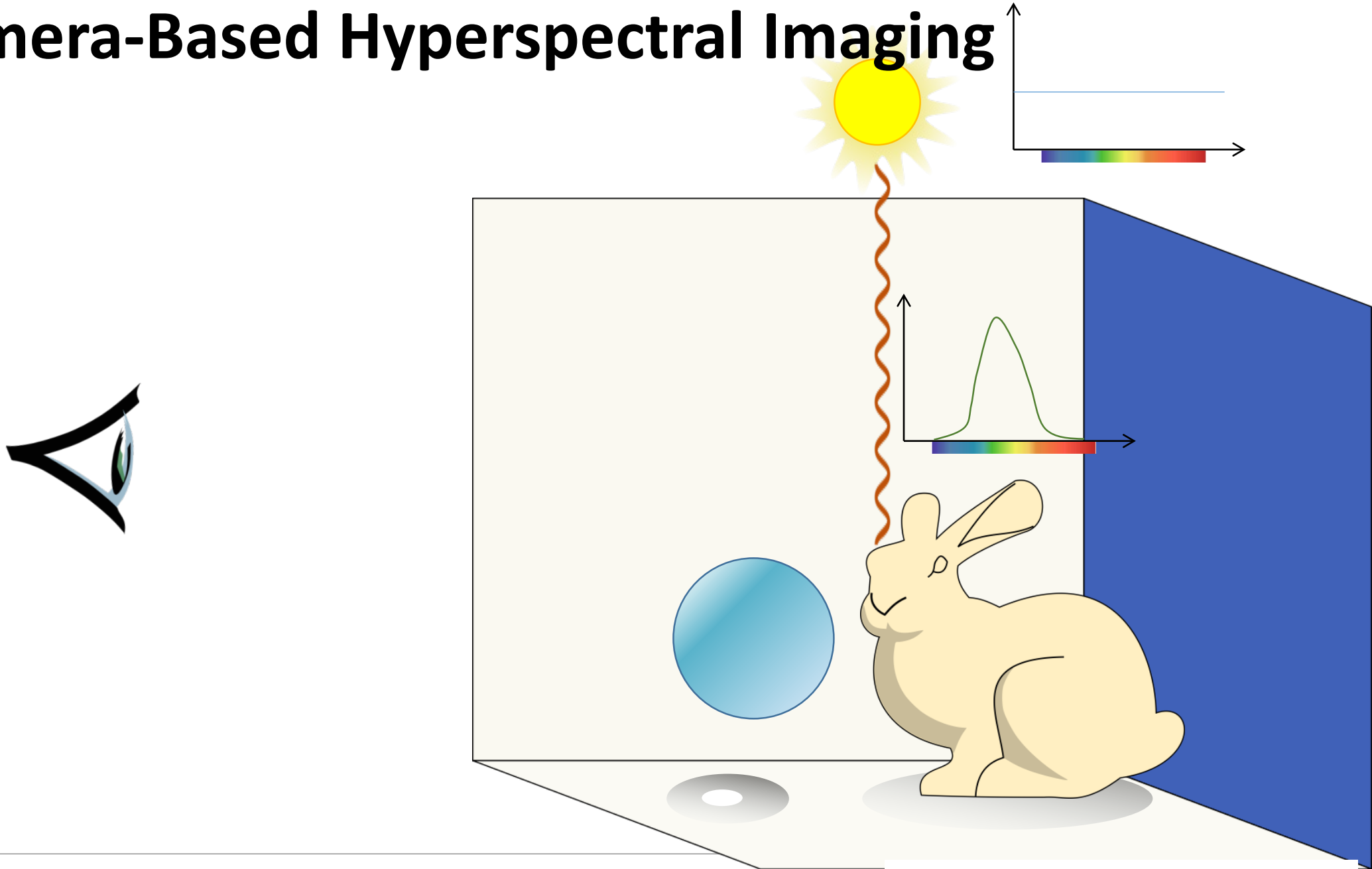
- What can we control?
 - Position
 - Time
 - Emission
 - ... anything else?



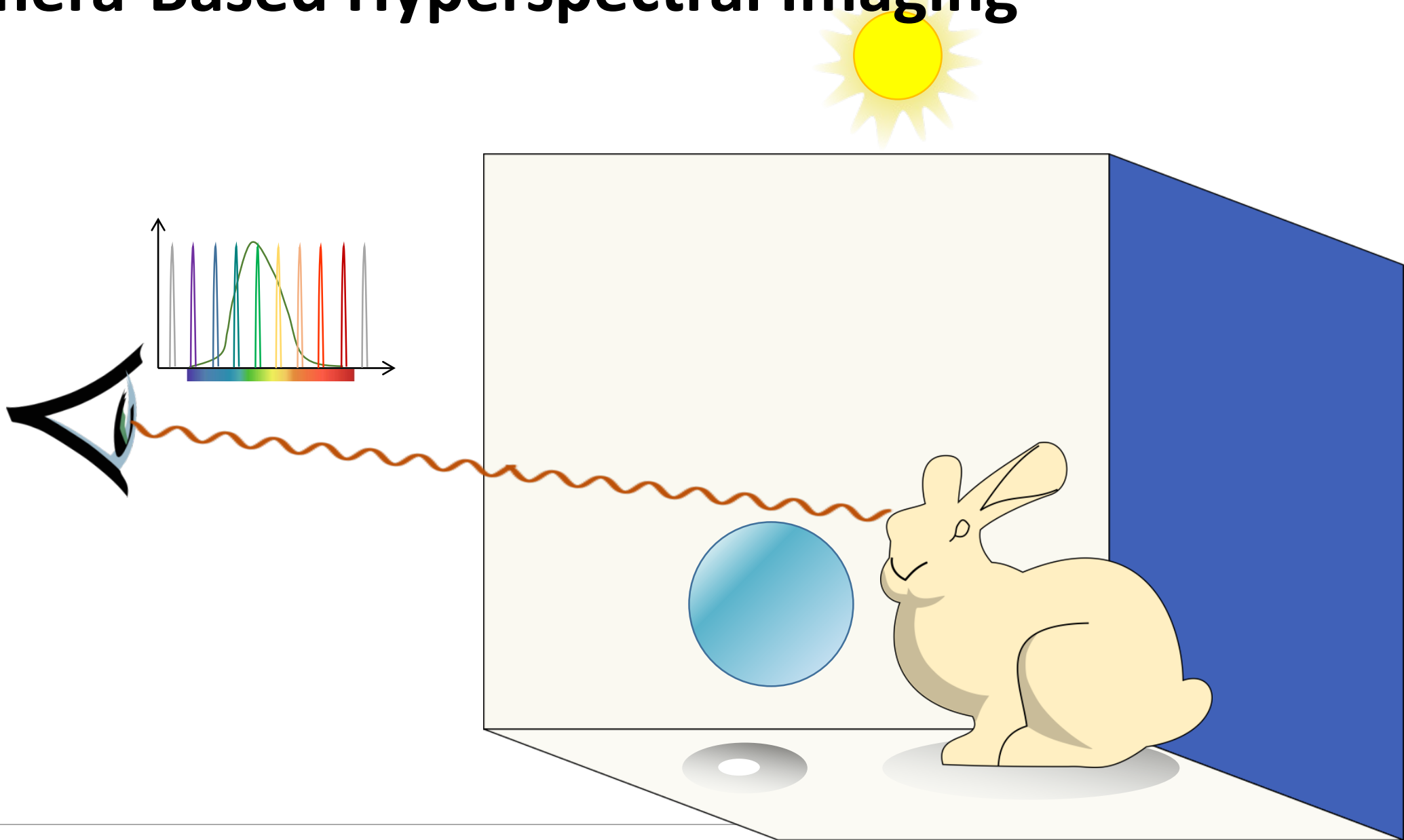
Active Illumination Systems

... and what can we do with active illumination?

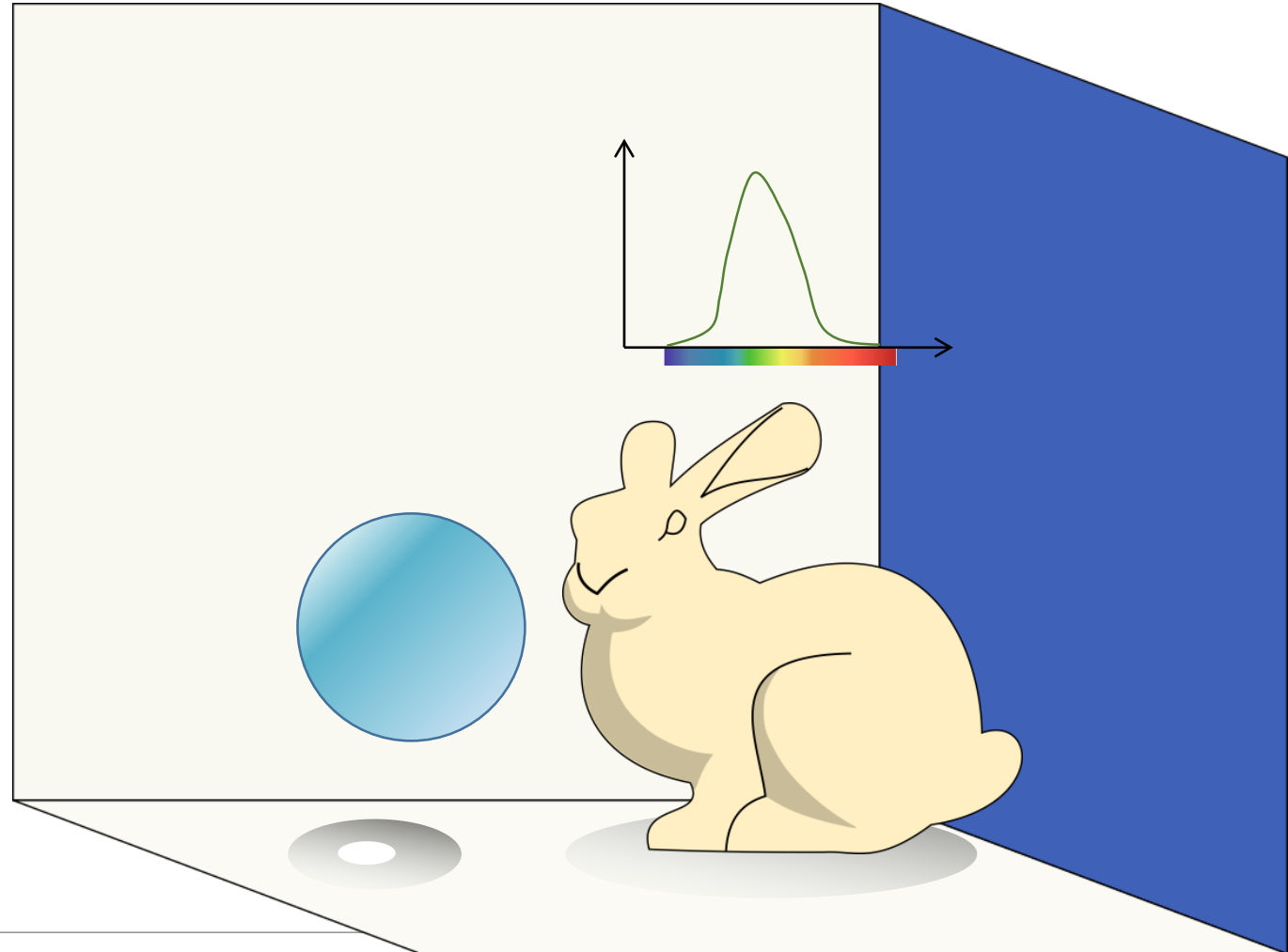
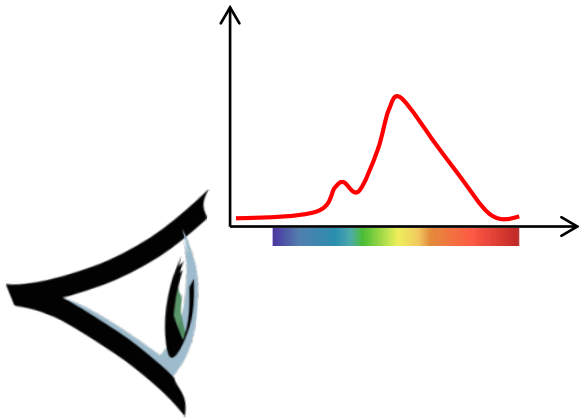
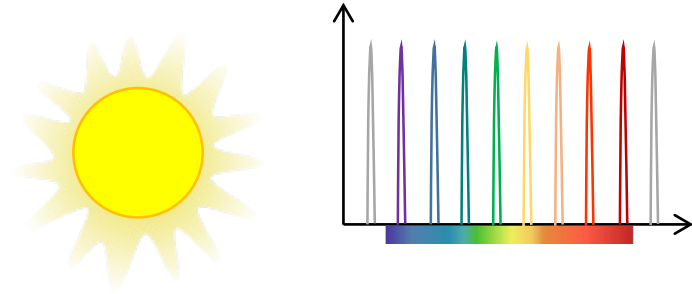
Camera-Based Hyperspectral Imaging



Camera-Based Hyperspectral Imaging



Active Hyperspectral Imaging



Active Illumination Systems

... and what can we do with active illumination?

- Photos under low-light conditions
- Stroboscopic Imaging (*freezing time*)
- Active Hyperspectral Imaging
- ...







- **Q:** How would you integrate real live-action footage (e.g. an actor) into a virtual environment?

A1: Scan the actor + MoCap -> Render all as 3D

A2: Relight the actor inside the virtual environment

The Light Stage

<https://youtu.be/dkEL1IfFLZE>

Active Illumination Systems

... and what can we do with active illumination?

- Photos under low-light conditions
- Stroboscopic Imaging (*freezing time*)
- Active Hyperspectral Imaging
- Relighting
- ...

Questions?

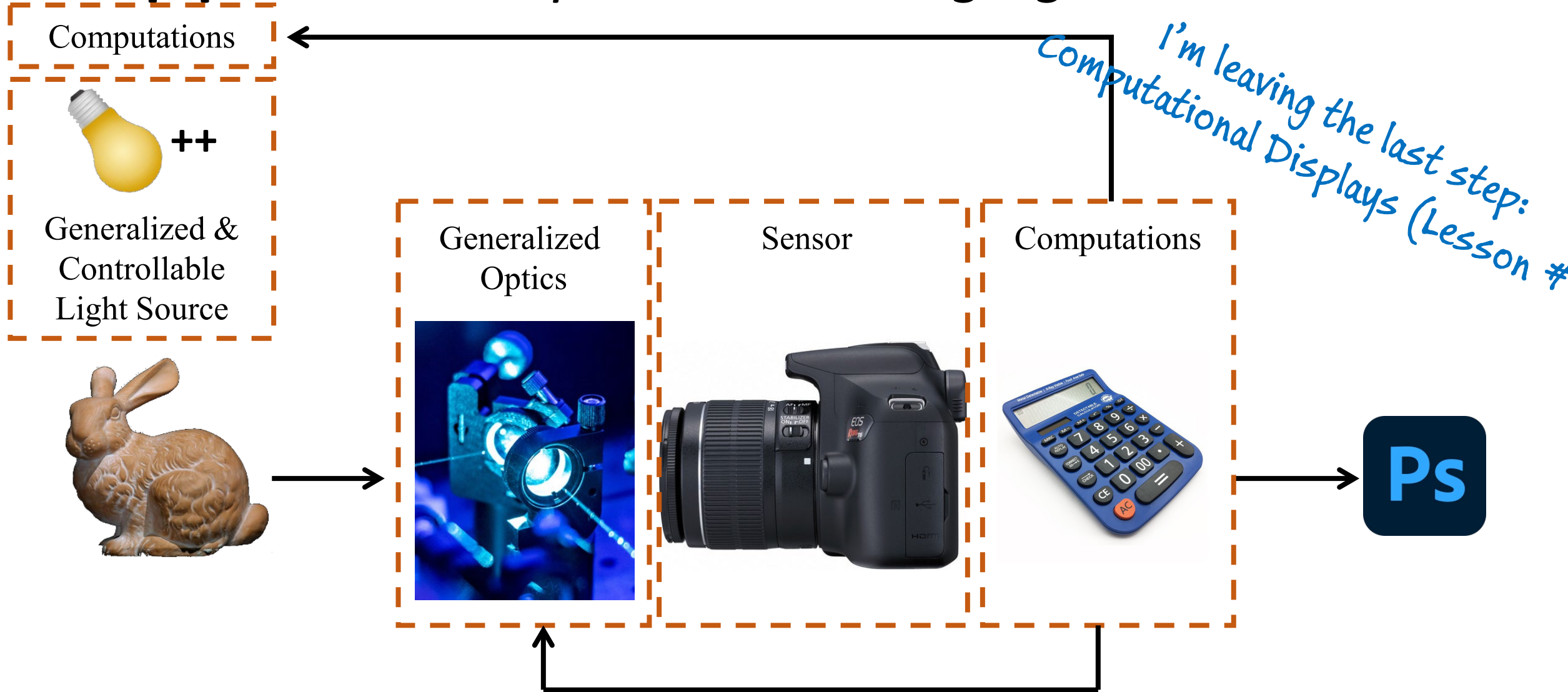
Recap

- **Active Illumination** is very useful for a good number of applications.
- In fact, we have been using it almost since the invention of photography!
- But still, no **computational illumination** here (or at least just some hints)

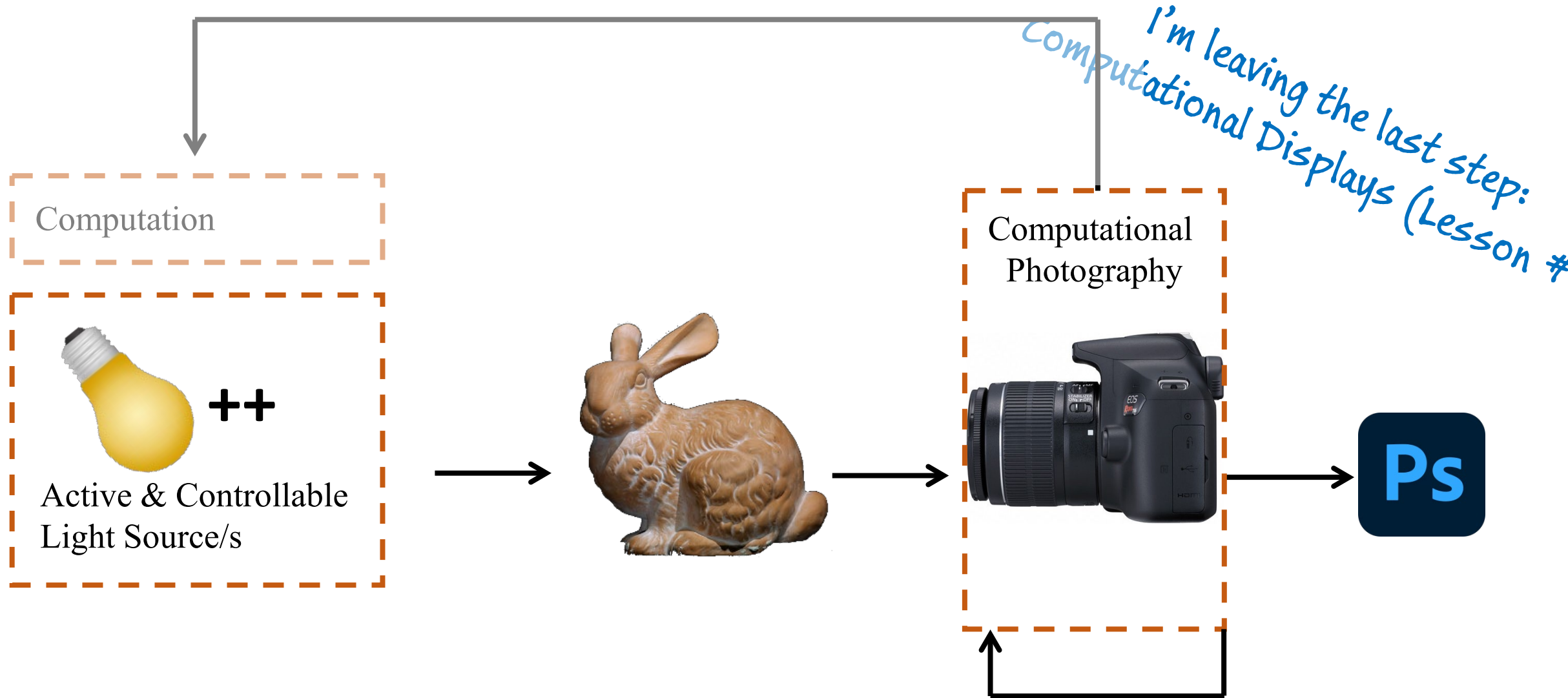
Today...

1. We will see what we can **control in illumination**
 - And analyze what kind of light sources
2. See some **computational imaging applications** that leverage active sources.
3. And capture and exploit the **light transport matrix** (computational light transport)

The pipeline – Computational Imaging



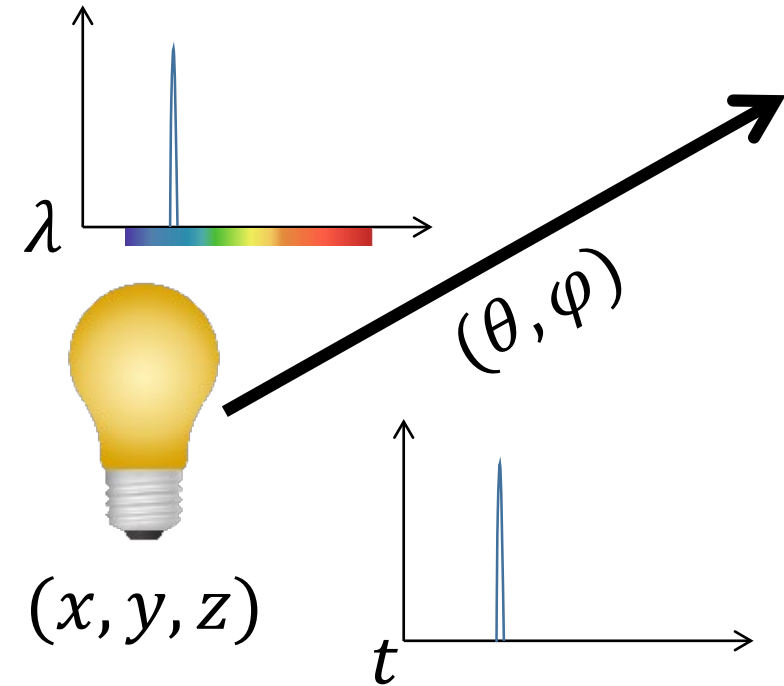
The pipeline – *Computational Imaging*



Active & Controllable Illumination

What can we control?

1. Power
2. Position
3. Directional Emission
4. Color (a.k.a. wavelength)
5. Temporal Emission Profile



All together...

$$P(\theta, \phi, \lambda, t, x, y, z)$$

Does this ring a bell?

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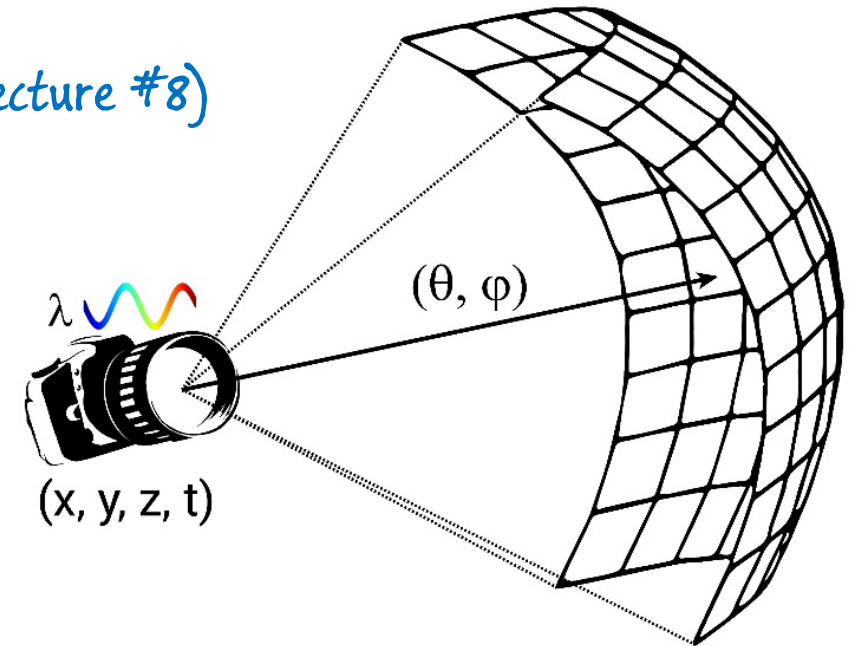
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Active & Controllable Illumination

How can we control $P(\theta, \phi, \lambda, t, x, y, z)$?

1. Power
 2. Position
 3. Directional Emission
 4. Color (a.k.a. wavelength)
 5. Temporal Emission Profile
- Pretty Straight Forward...*
- a) Laser, b) Projector, c) Mask/Coded Illumination, ...*
- a) Calibrated Light Source, b) Narrow-band LED/Laser, ...*
- a) Pulsed/Stroboscopic Light, b) Temporally Modulated Light, ...*

Active & Controllable Illumination

Quick Question?

1. What is the most versatile light source for this?

My take: Narrow-Band Ultrashort Pulsed Laser

But... expensive, difficult to handle, long operating times, eye safety?

Cheaper choices:

a) A well-calibrated projector

b) A decent mask (e.g. an LCOS display) in front of the light source

Questions?

The pipeline – *Computational Imaging*

What if we take a few pictures with different illumination and use them to compute a new/better image?



Flash/No-Flash Images

Low-light conditions - *The perils of active illumination...*



Red Eye

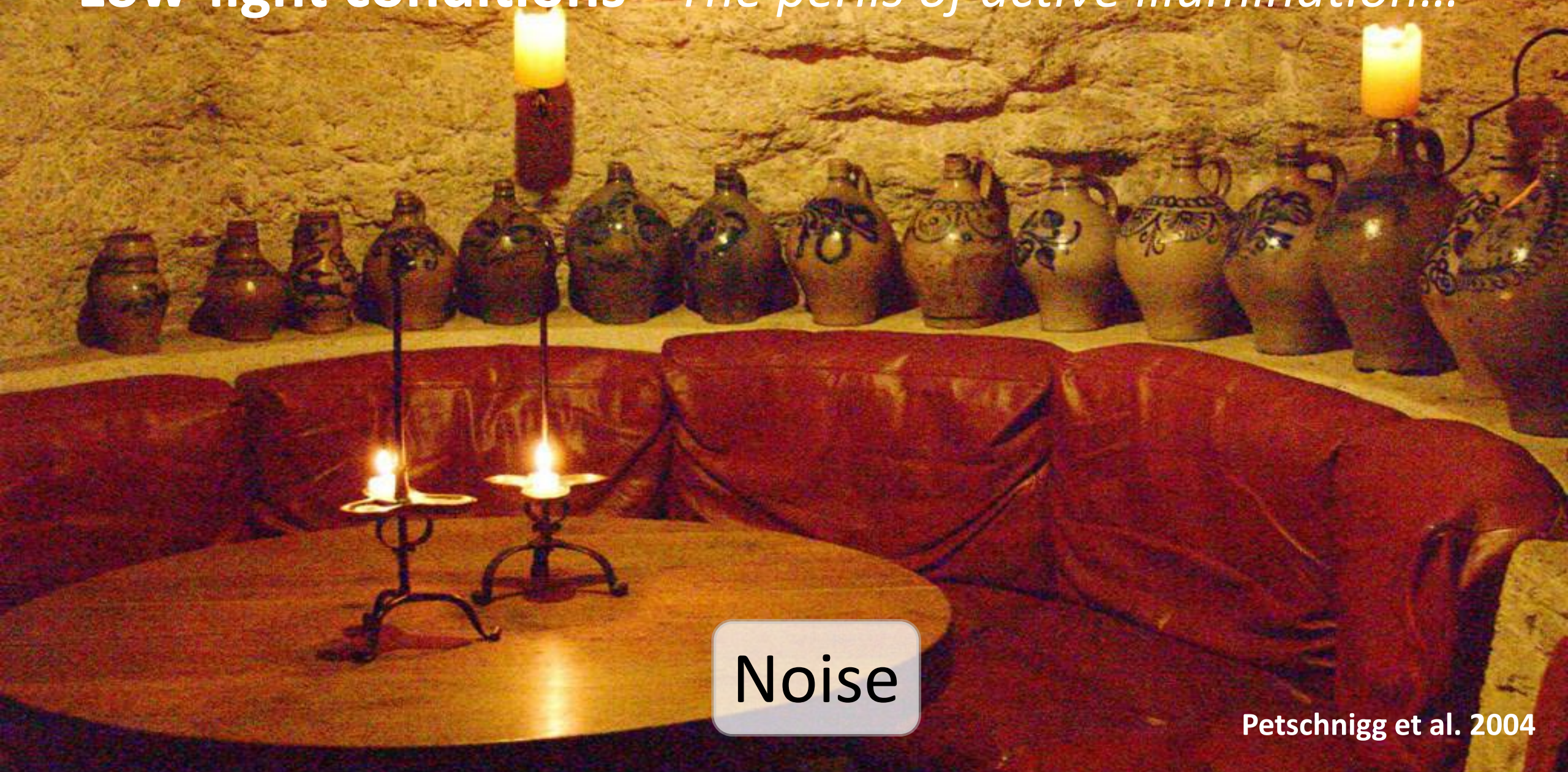
Petschnigg et al. 2004

Low-light conditions - *The perils of active illumination...*

Unflattering Lighting

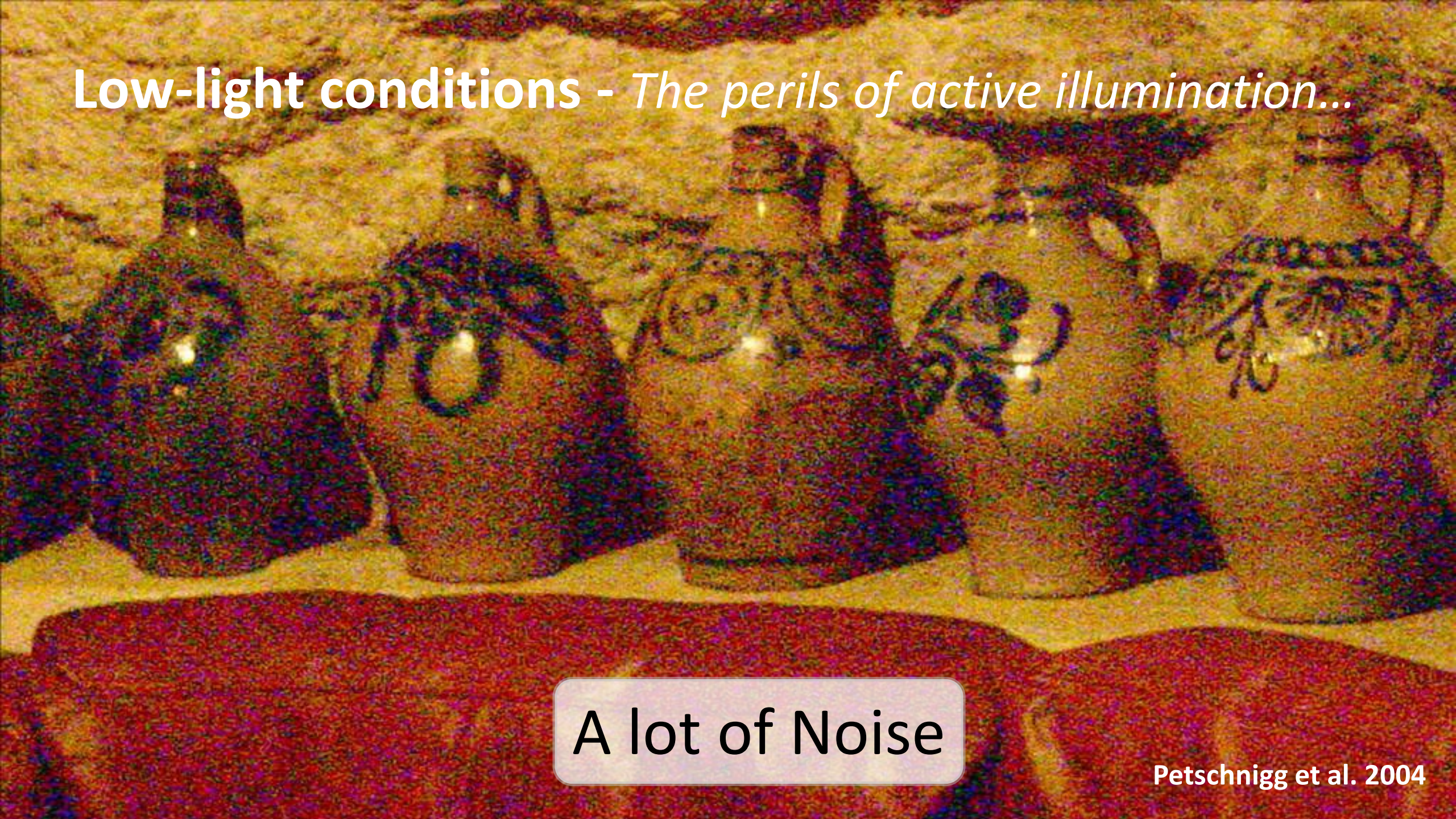
Petschnigg et al. 2004

Low-light conditions - *The perils of active illumination...*



Noise

Low-light conditions - *The perils of active illumination...*



A lot of Noise

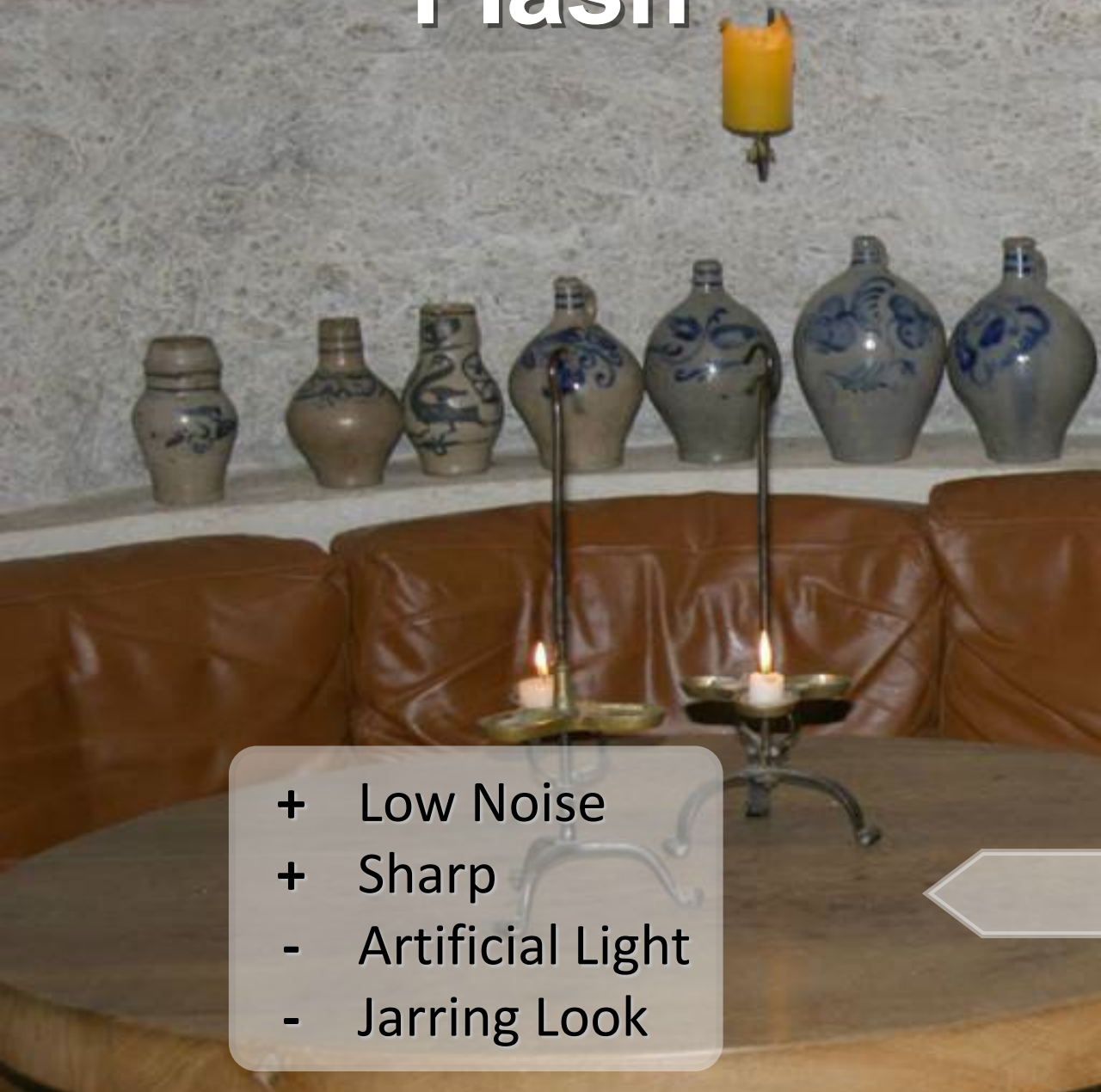
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Ruined Ambiance

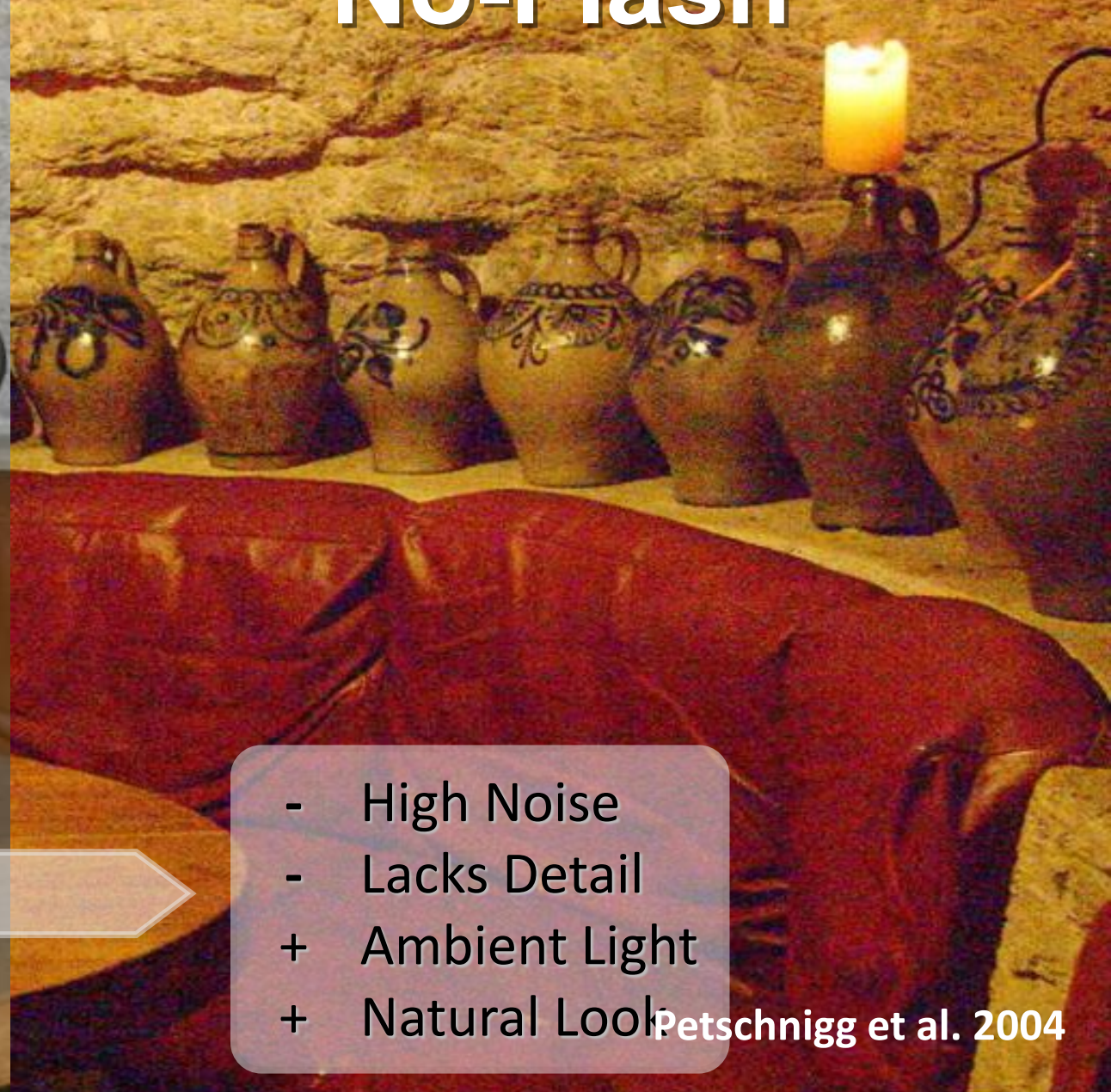
Petschnigg et al. 2004

Flash



- + Low Noise
- + Sharp
- Artificial Light
- Jarring Look

No-Flash



- High Noise
- Lacks Detail
- + Ambient Light
- + Natural Look

Flash-No Flash guided denoising



Denoising Result



No-Flash

Petschnigg et al. 2004

Flash-No Flash guided denoising



Denoising Result

Active Illumination + Computation - *Recap*

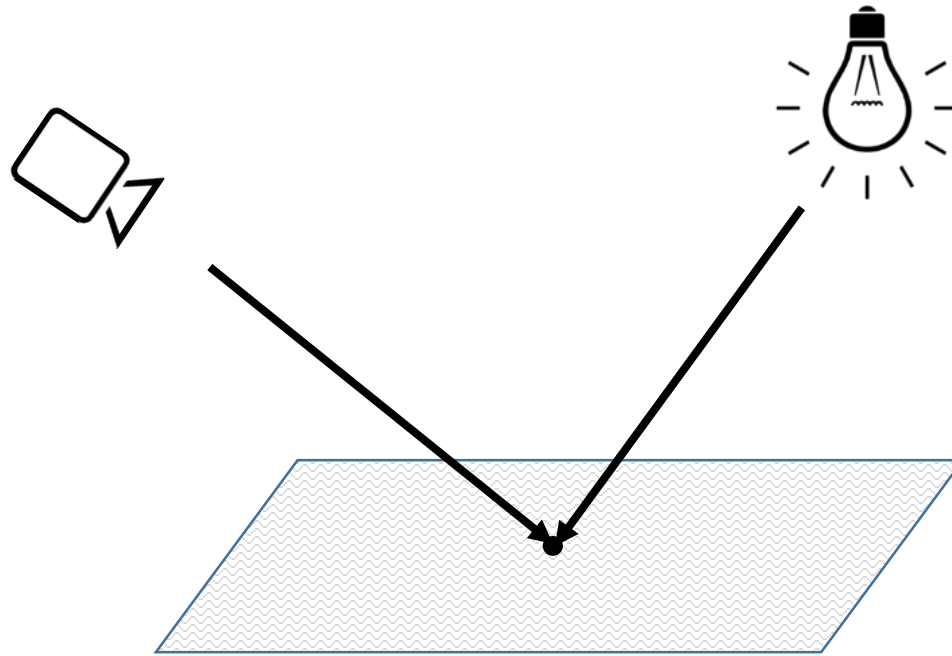
- **Example I: Flash-No-Flash Photography**

By adding a little bit of computation after capture, and *coding* illumination, we can significantly improve photography in low-light conditions

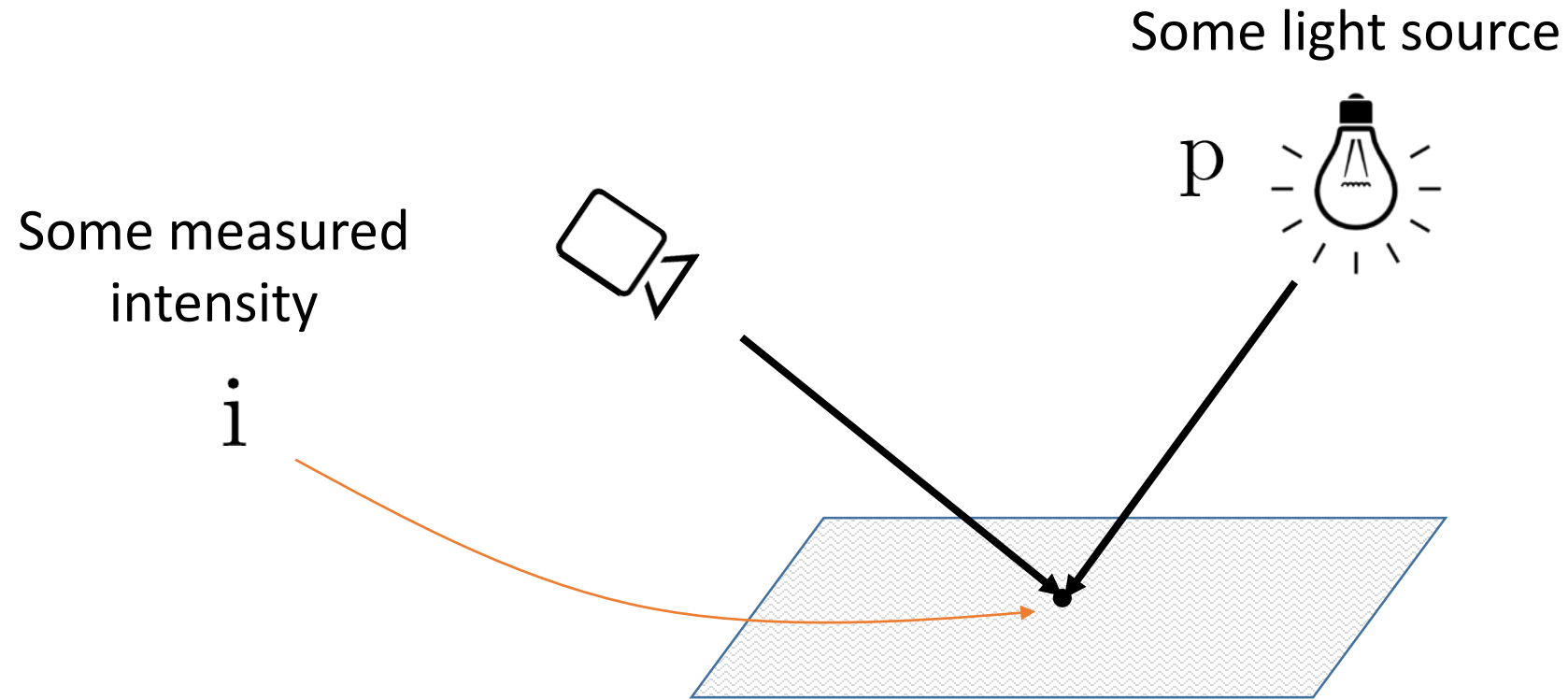
- Can we do more?

The Light Transport Matrix Formulation

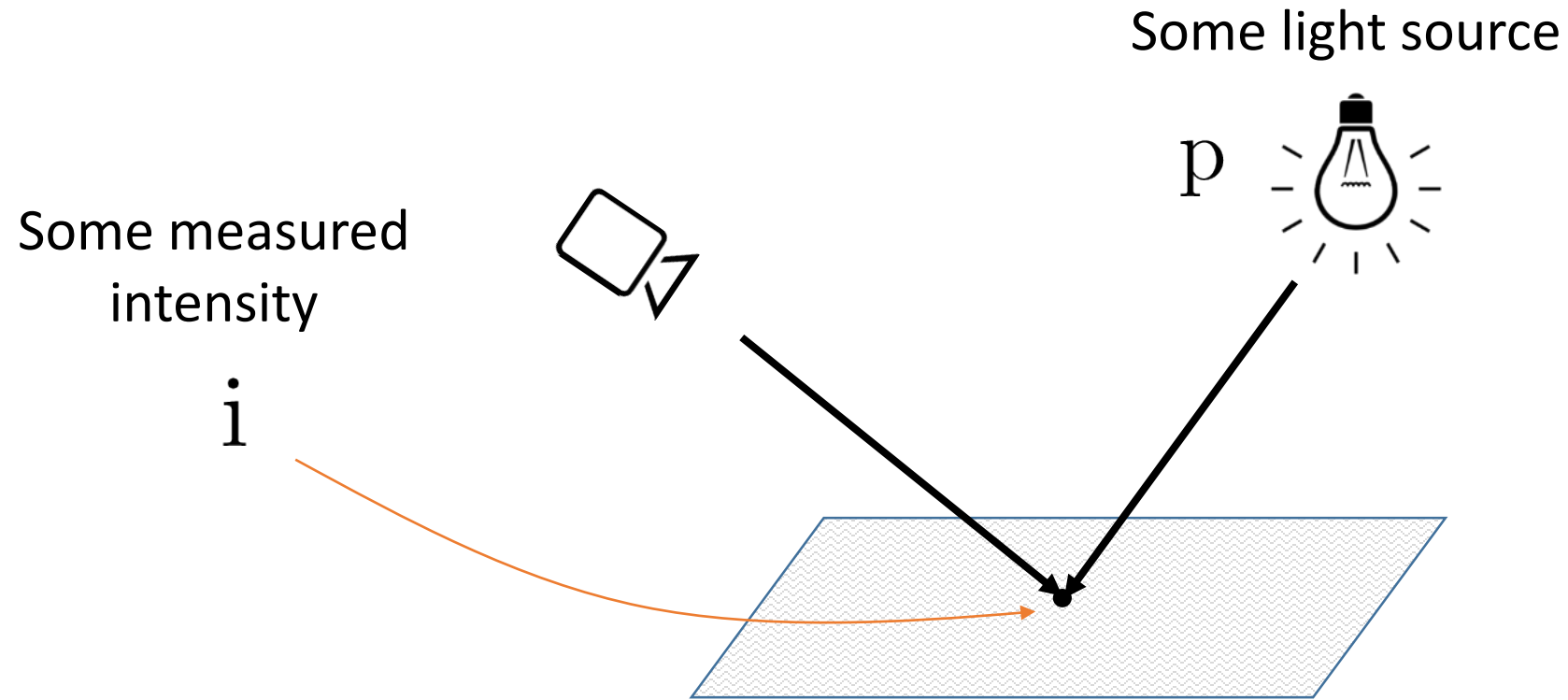
Light transport basics



Light transport basics



Light transport basics



Some coefficient t that **defines light transport linearly**

$$i = tp$$

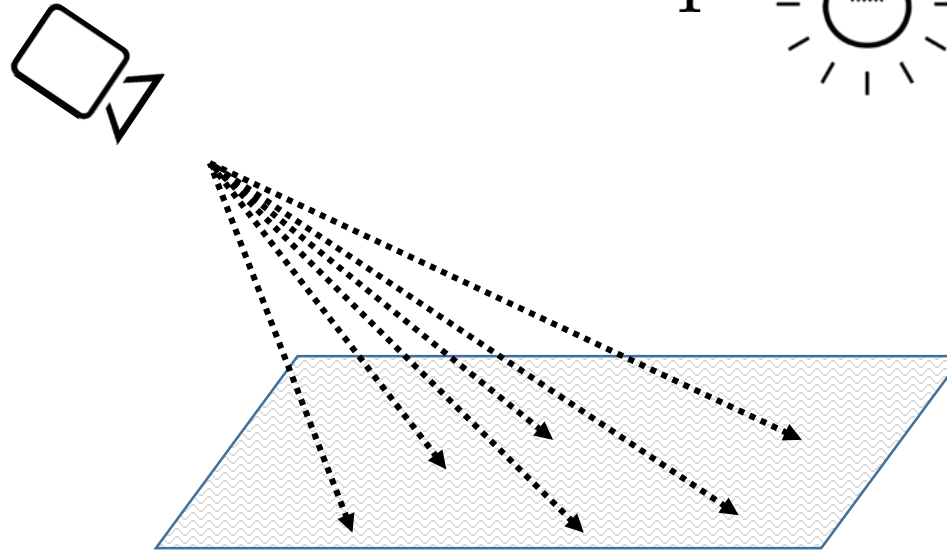
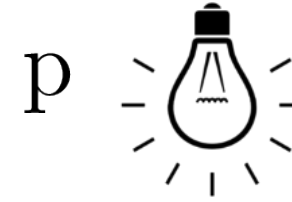
Light transport basics

Many measured
intensities

$\mathbf{i}$

(e.g. camera pixels)

Some light source



Some **coefficients** $\mathbf{t}$ (column vector) that define light transport linearly

$$\mathbf{i} = \mathbf{t}p$$

Light transport basics

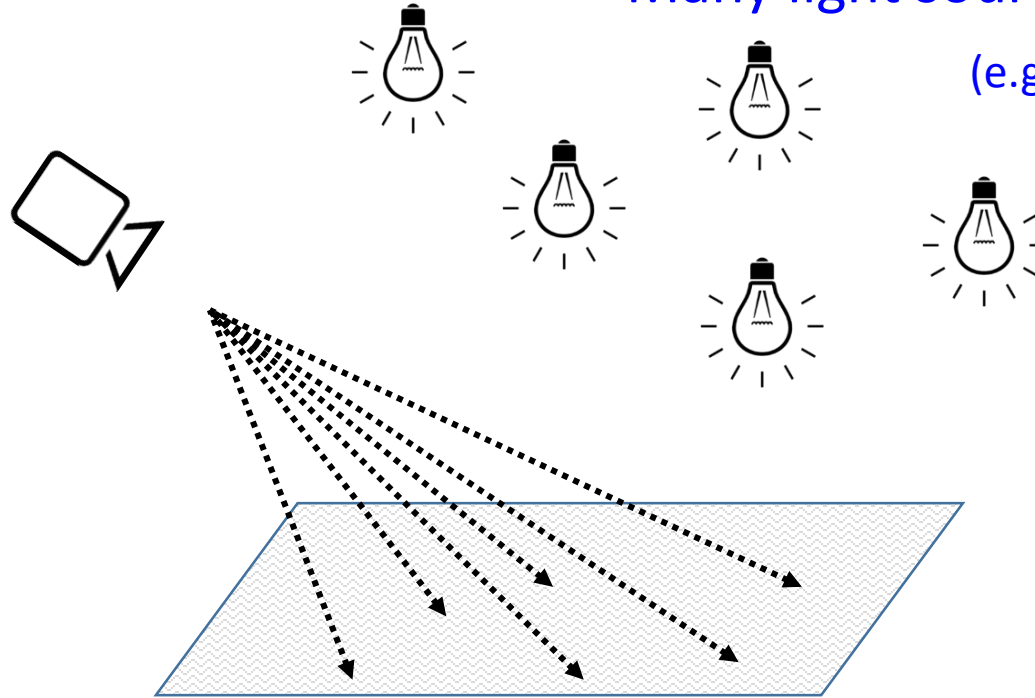
Many measured
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$\mathbf{i}$

(e.g. camera pixels)

Many light sources **$\mathbf{p}$**

(e.g. lamps in a room)



Some **coefficients** **$\mathbf{T}$** (matrix) that define light transport linearly

$$\mathbf{i} = \mathbf{T}\mathbf{p}$$

The Light Transport Equation

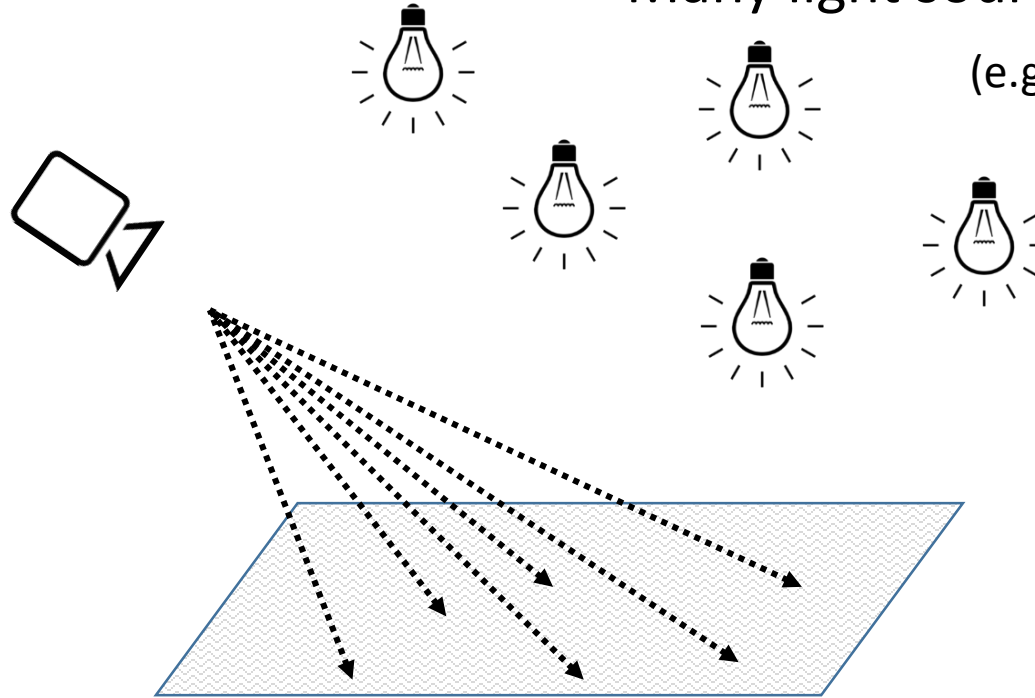
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$$\mathbf{i} = \mathbf{T}\mathbf{p} \quad \text{Light Transport Equation}$$

[Ng et al. 2003]

The Light Transport Equation

The Light Transport Equation establishes a linear transport relationship between a set of light sources $\mathbf{p}$ and a set of observations $\mathbf{i}$.

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$\mathbf{i}$ - *Observations* ($N \times 1$): Made by a sensor (e.g. image pixels, a LIDAR measurement)

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$\mathbf{i}$ - *Observations* ($N \times 1$): Made by a sensor (e.g. image pixels, a LIDAR measurement)

$\mathbf{p}$ - *Light sources* ($M \times 1$): Illuminate the scene (e.g. light bulbs, lasers, projector pixels)

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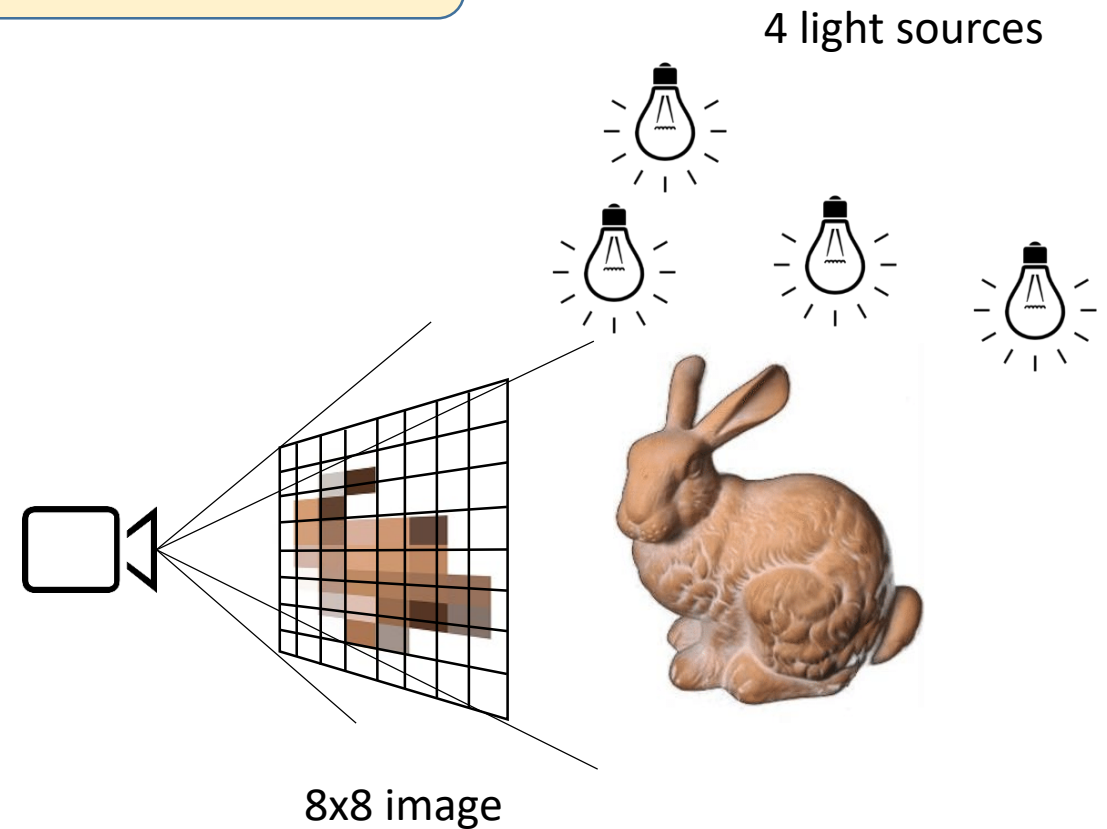
$\mathbf{T}$ - *Light transport matrix* ($N \times M$): Encodes transport through the scene between $\mathbf{p}$ and $\mathbf{i}$

The Light Transport Equation

$$\mathbf{i} = \mathbf{T}\mathbf{p} \quad \text{Light Transport Equation}$$

[Ng et al. 2003]

- An example of a typical use case:

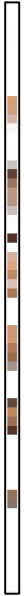


The Light Transport Equation

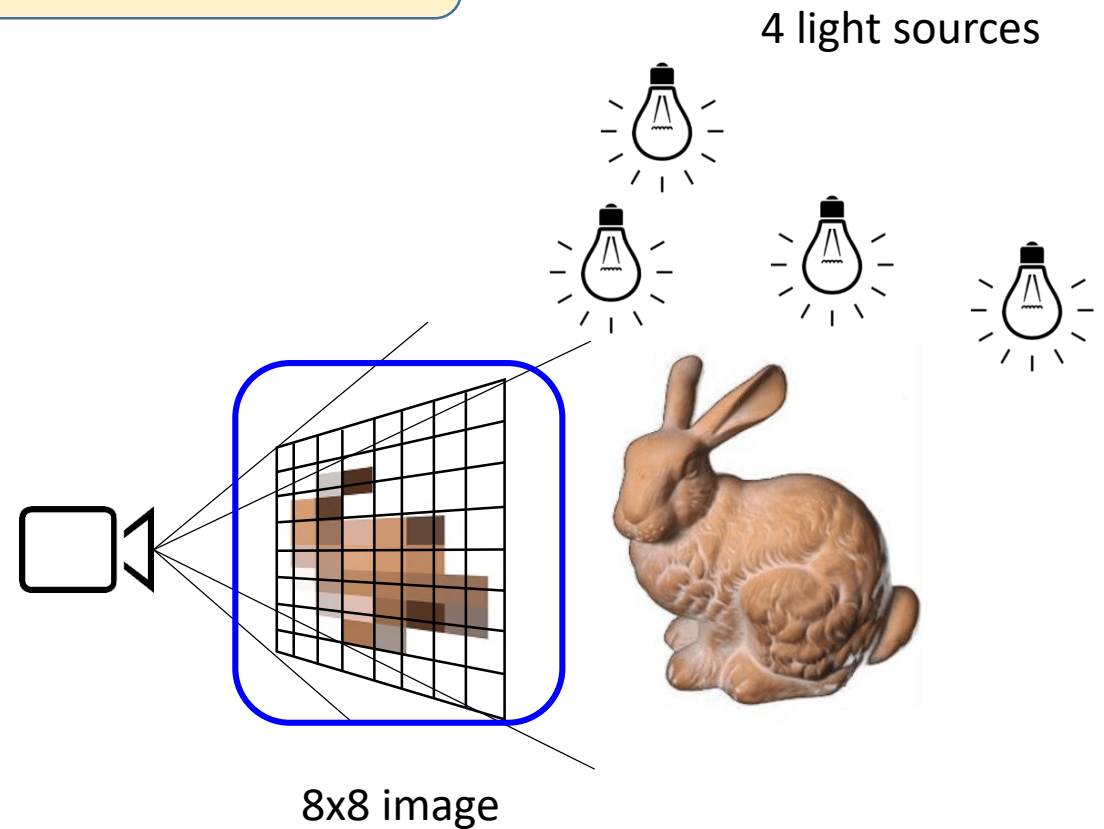
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$\mathbf{i}$
(64x1)



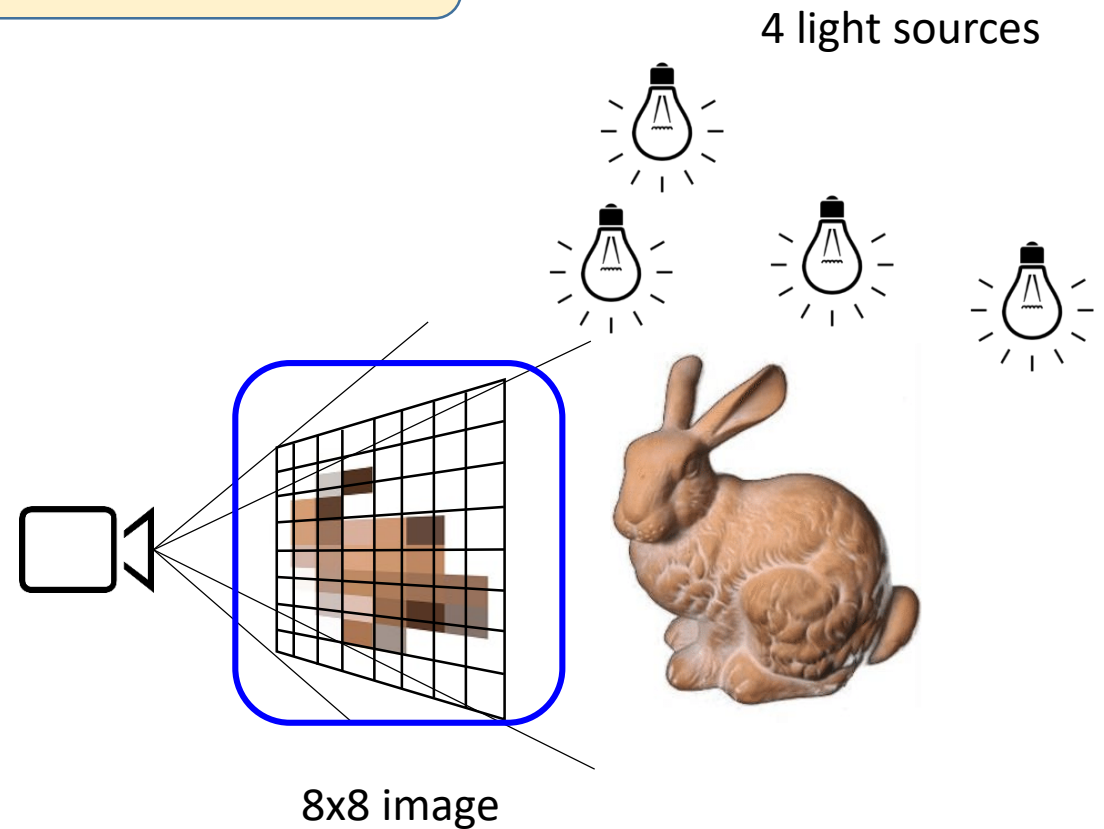
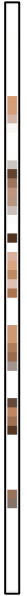
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- An example of a typical use case:

Known $\mathbf{i}$
(64x1)

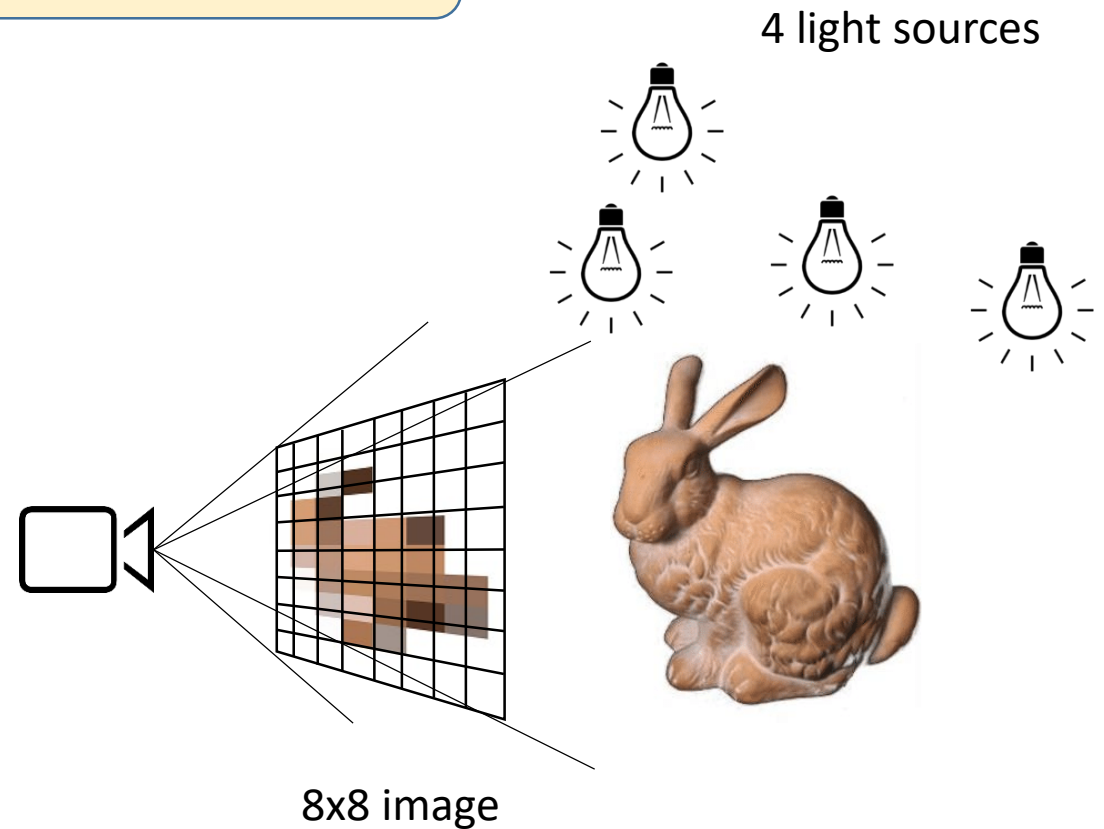
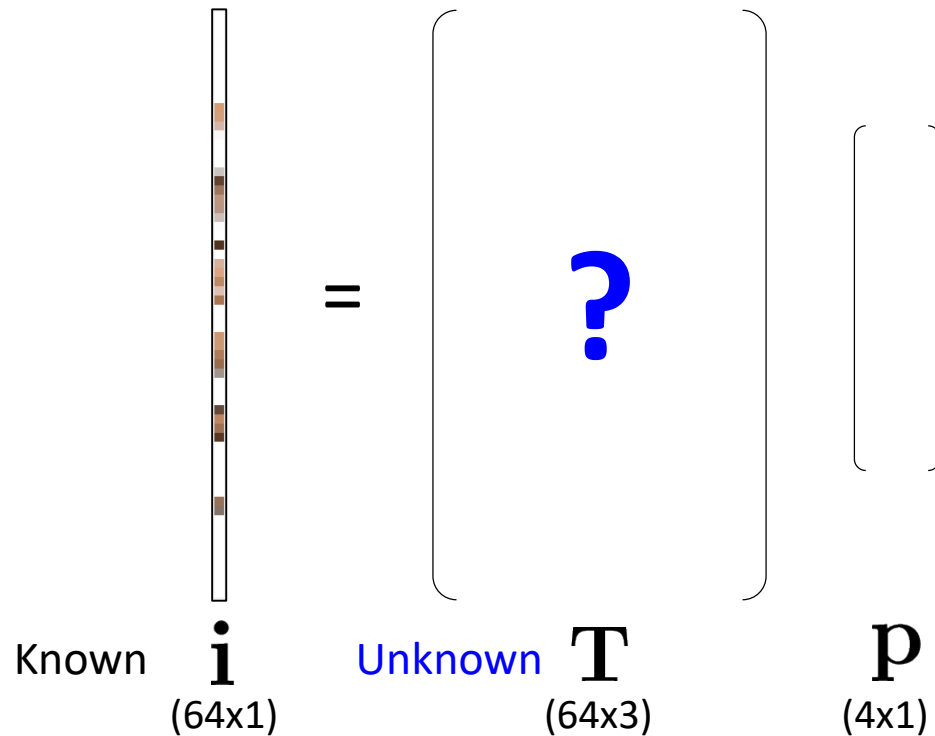


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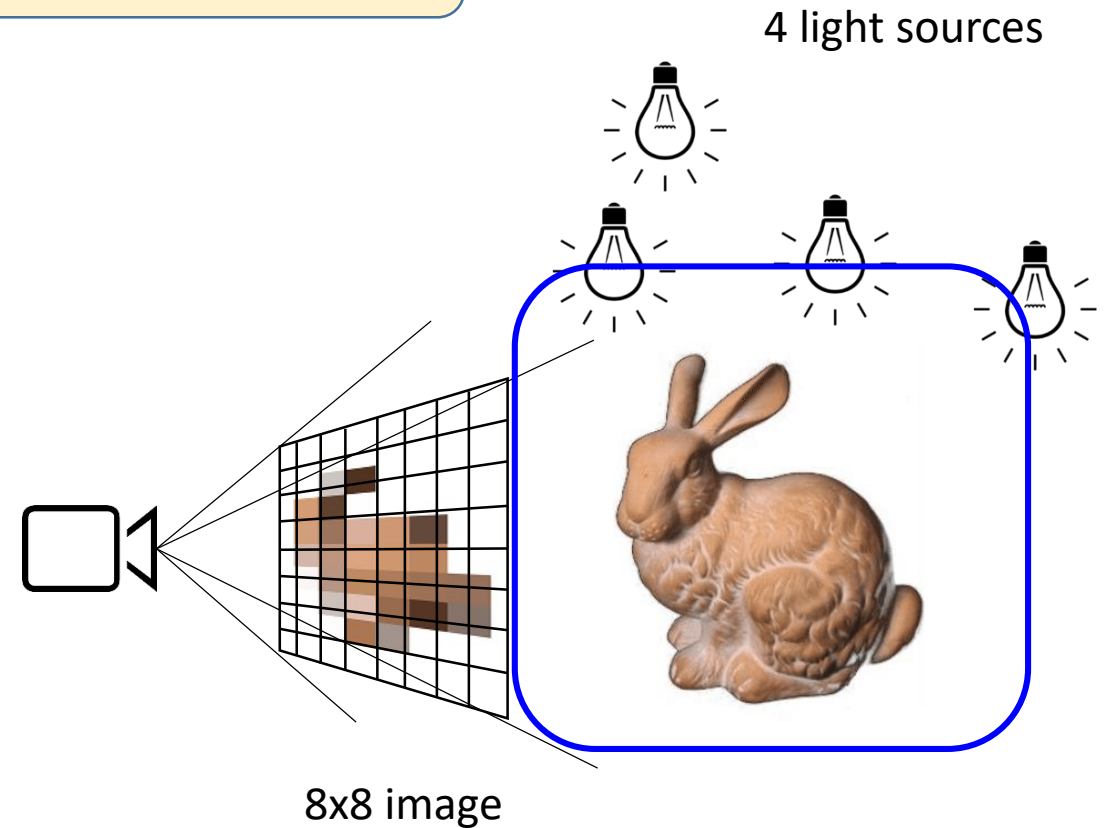
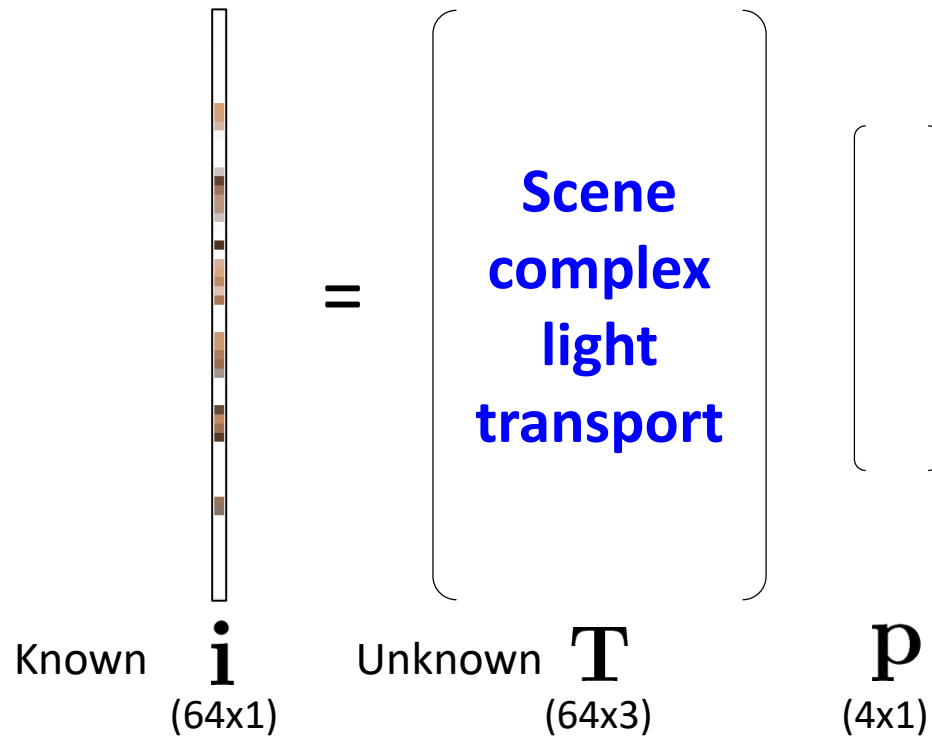


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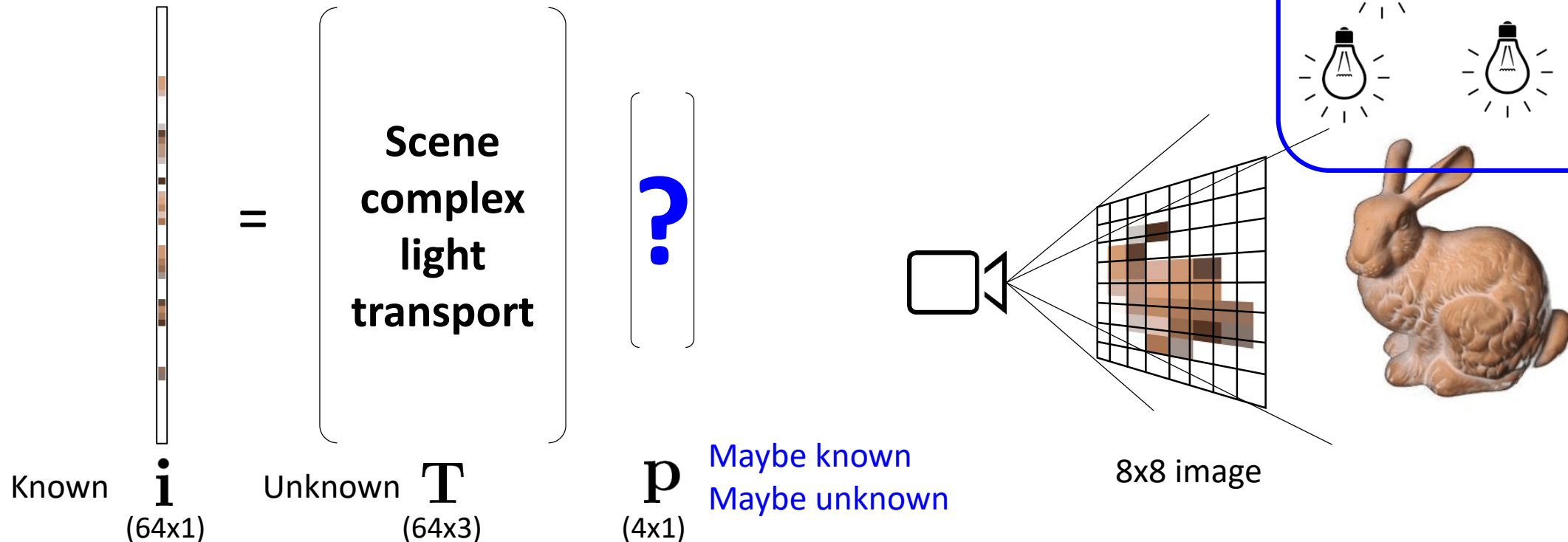


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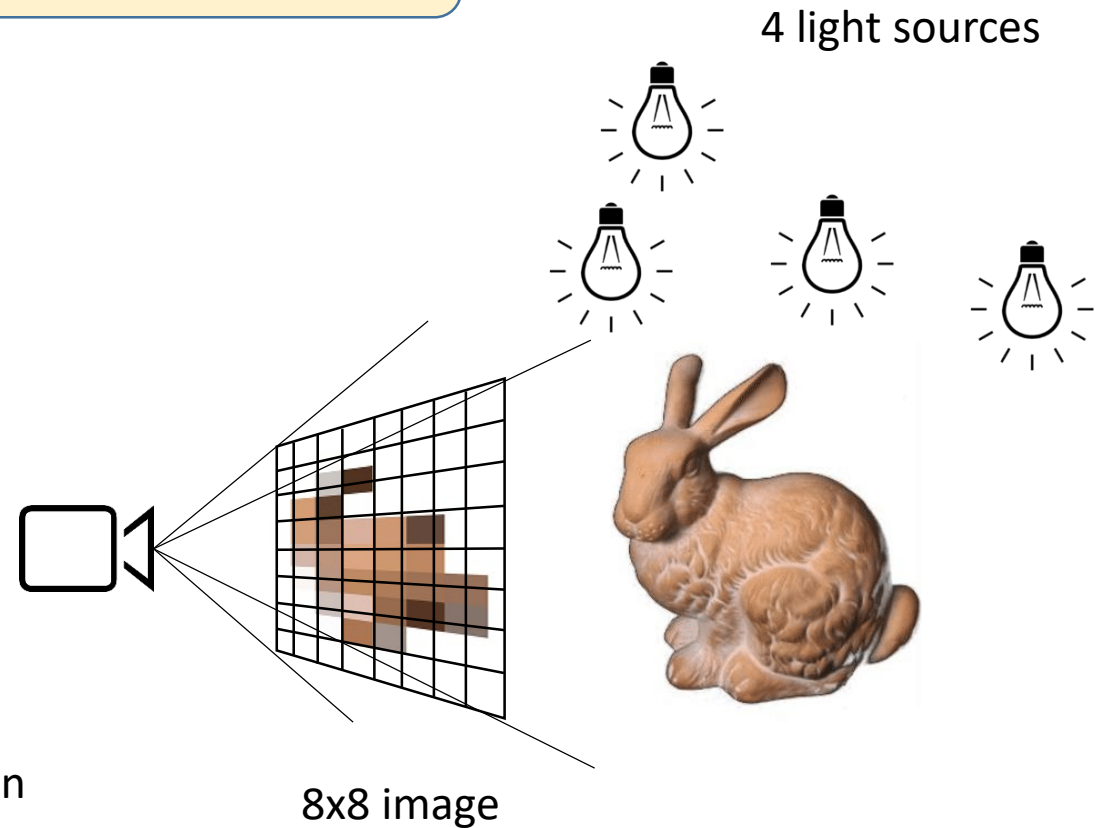
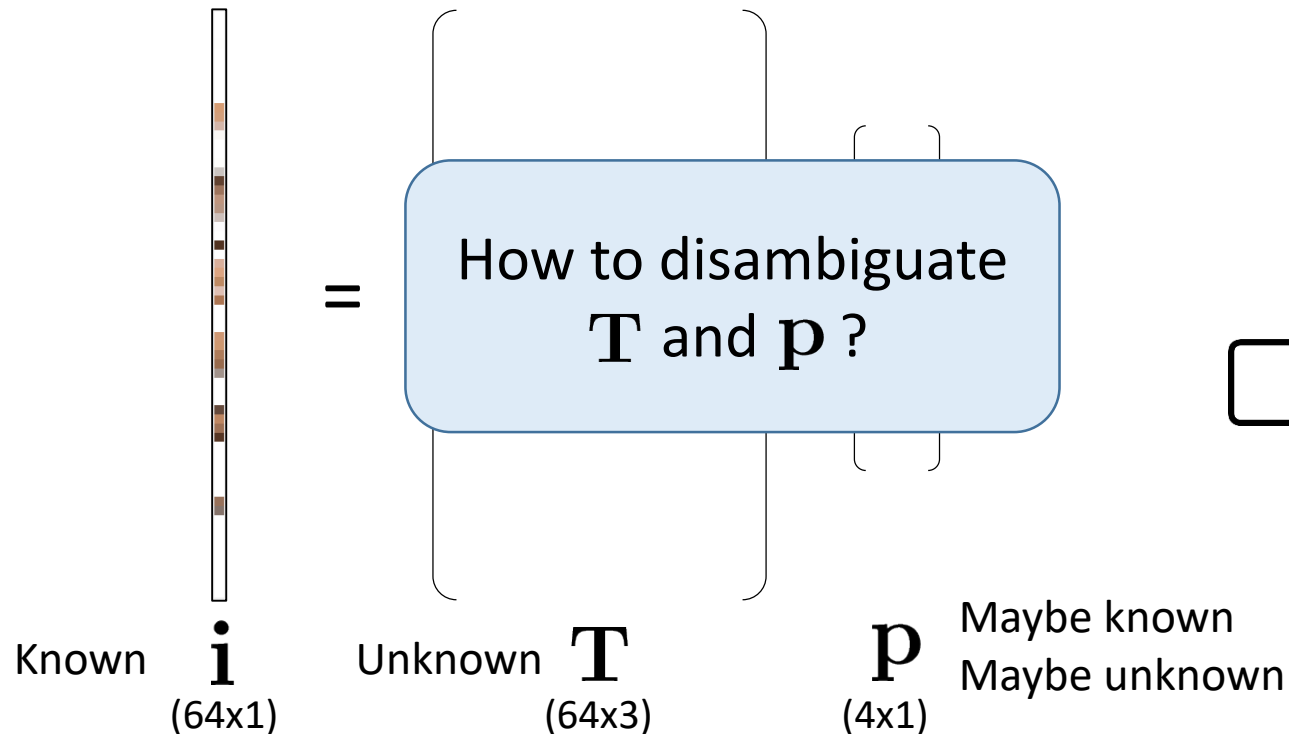


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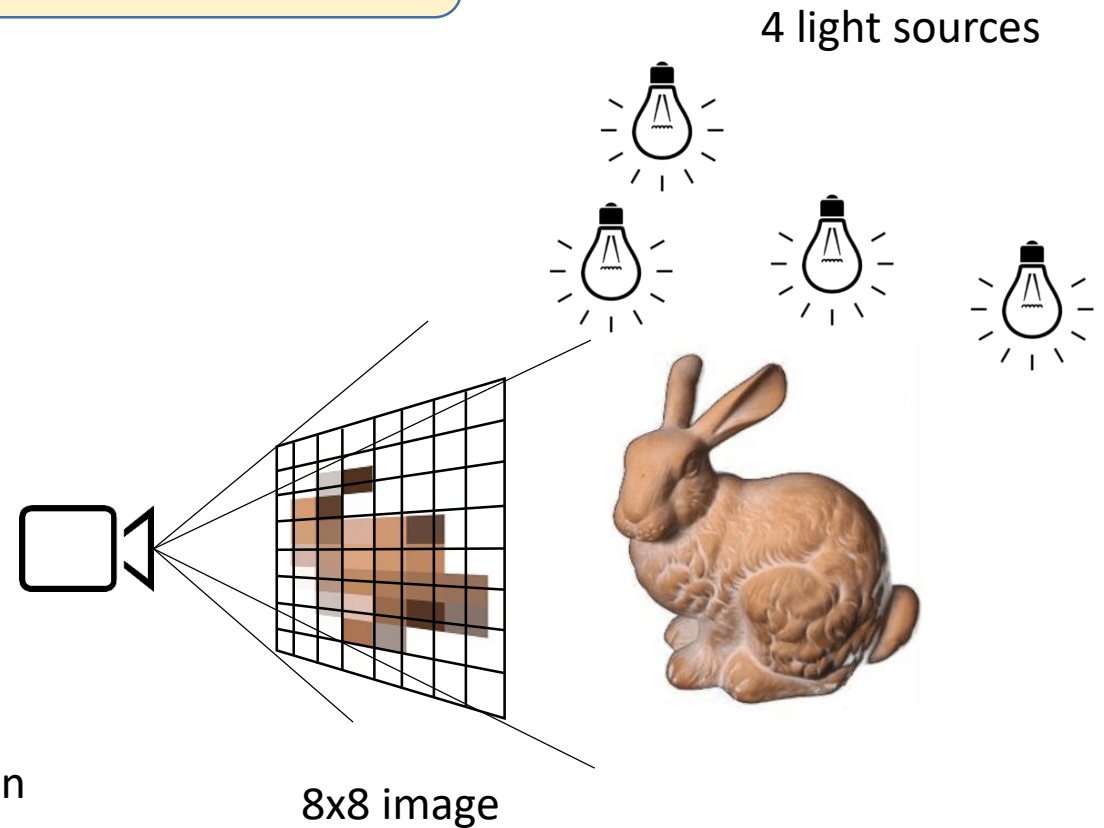
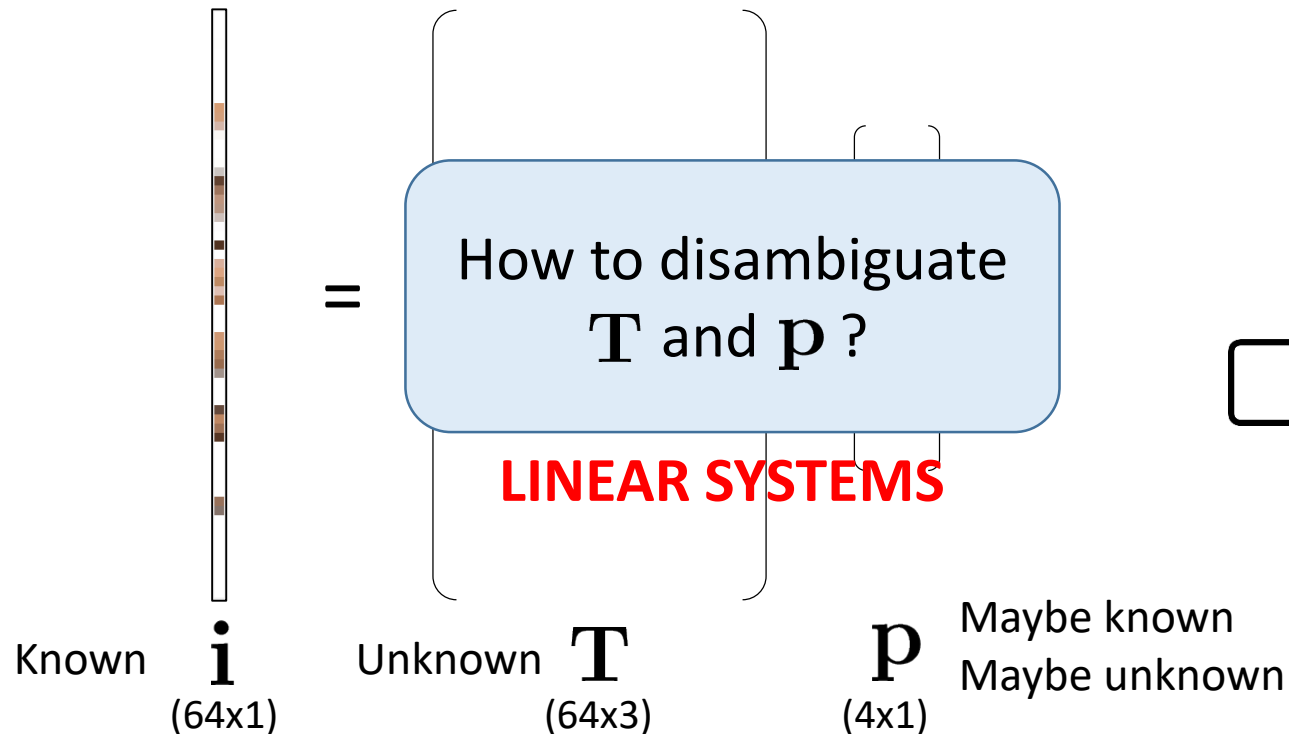


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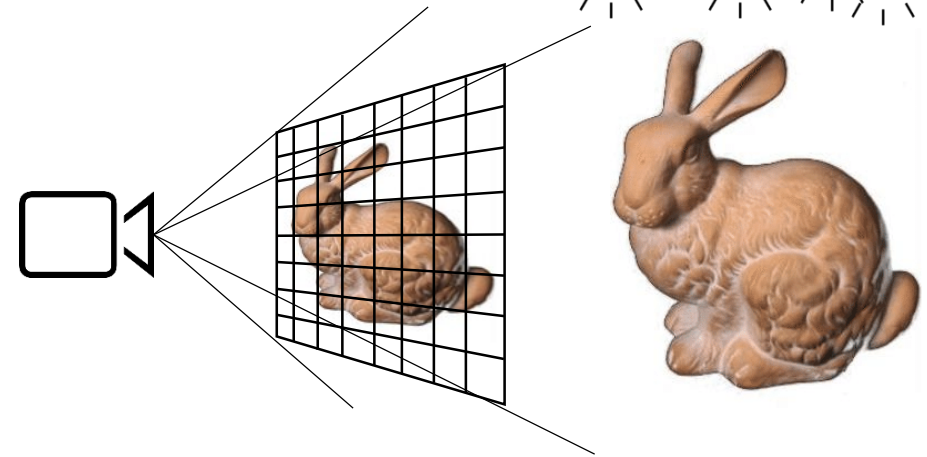
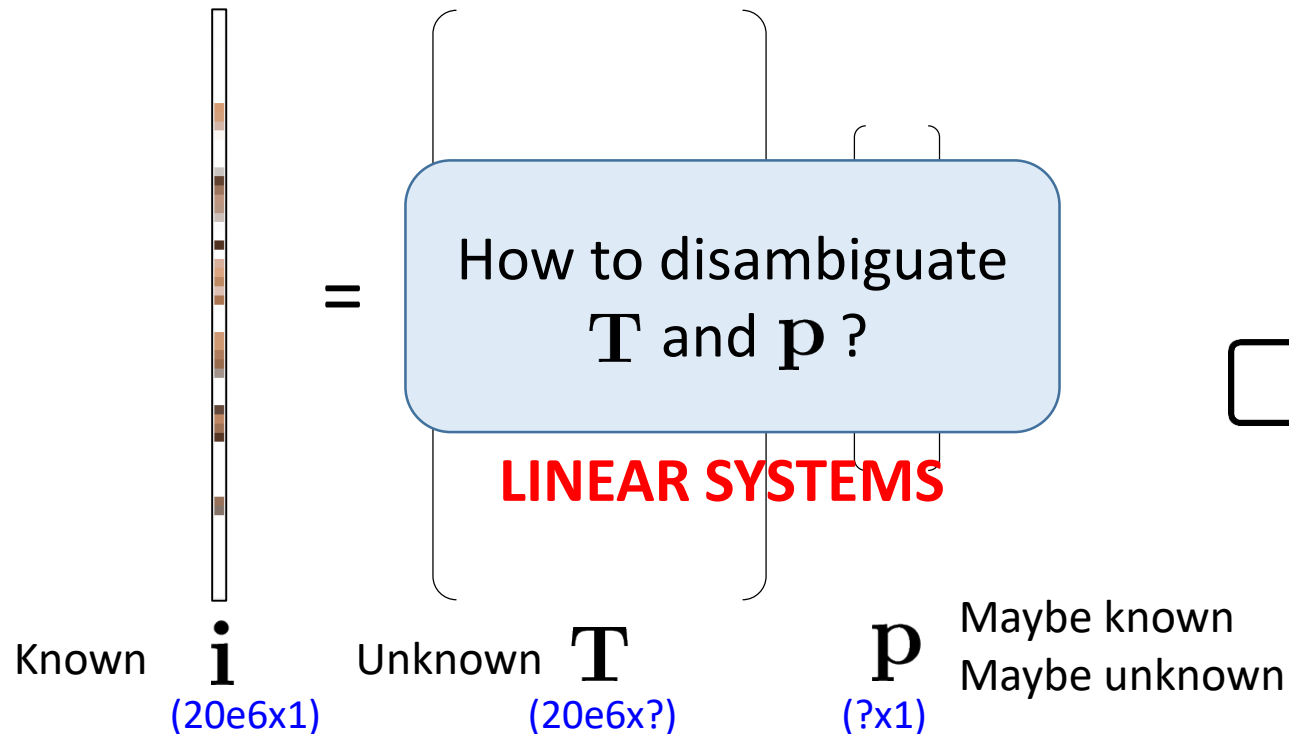


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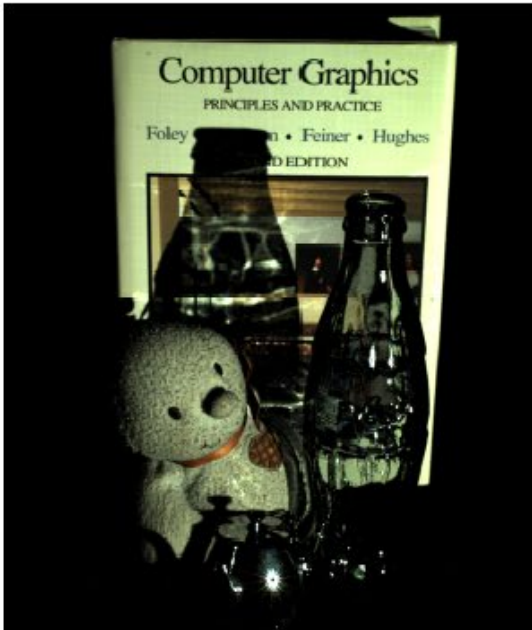
20MP image

LTE: Applications

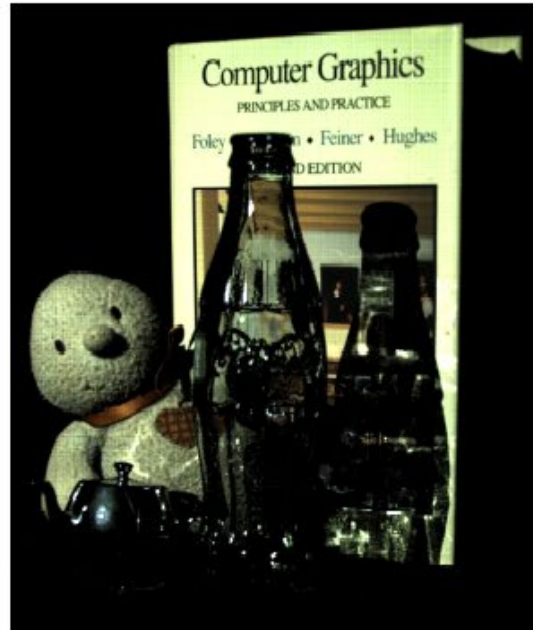
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[Ng et al. 2003]

Dual Photography [Sen et al. 2005]



(a)



(b)



(c)



(d)

LTE: Applications

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[Ng et al. 2003]

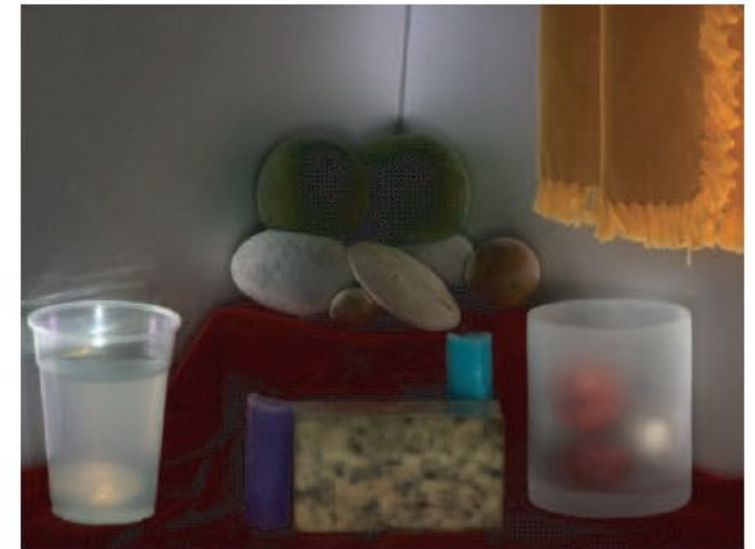
Global illumination separation [Nayar et al. 2006]



(a) Scene

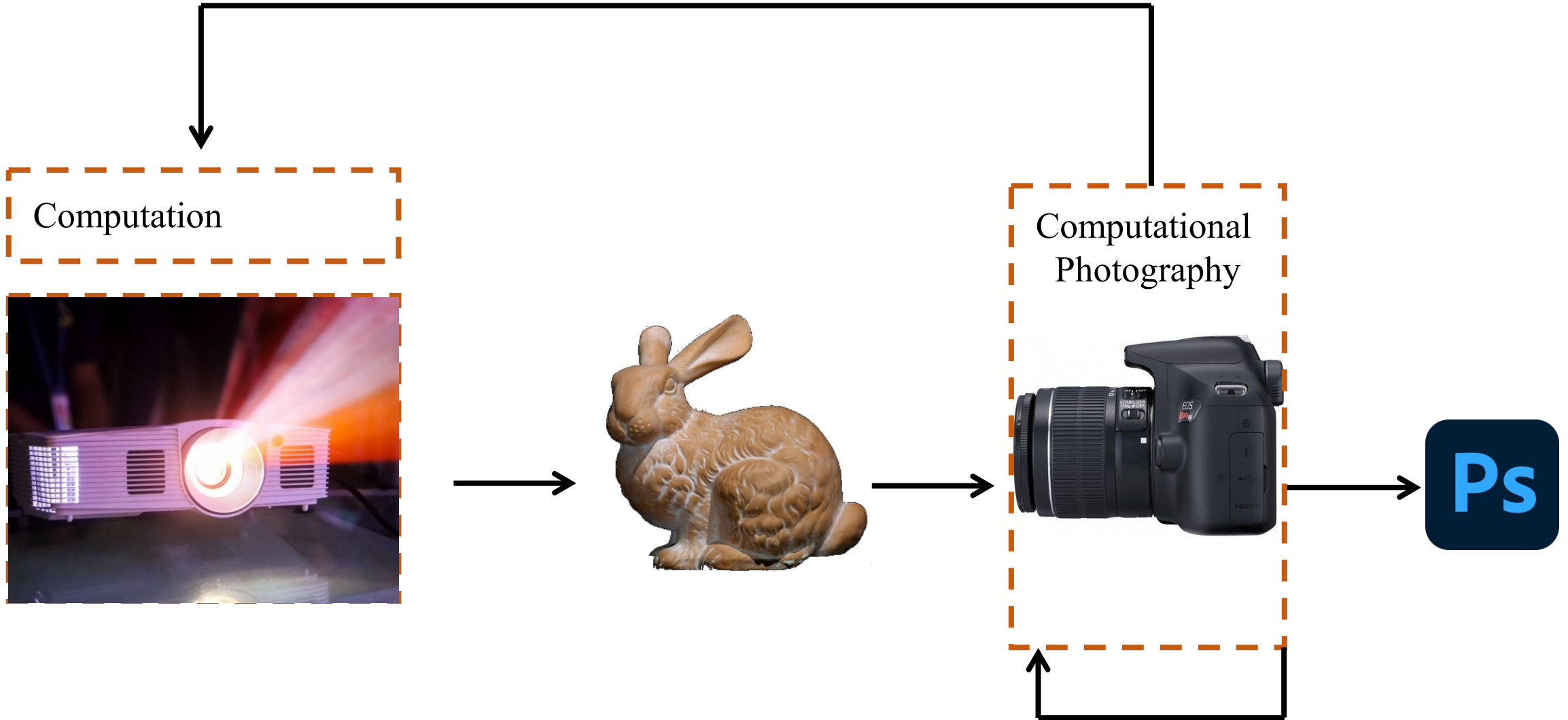


(b) Direct Component



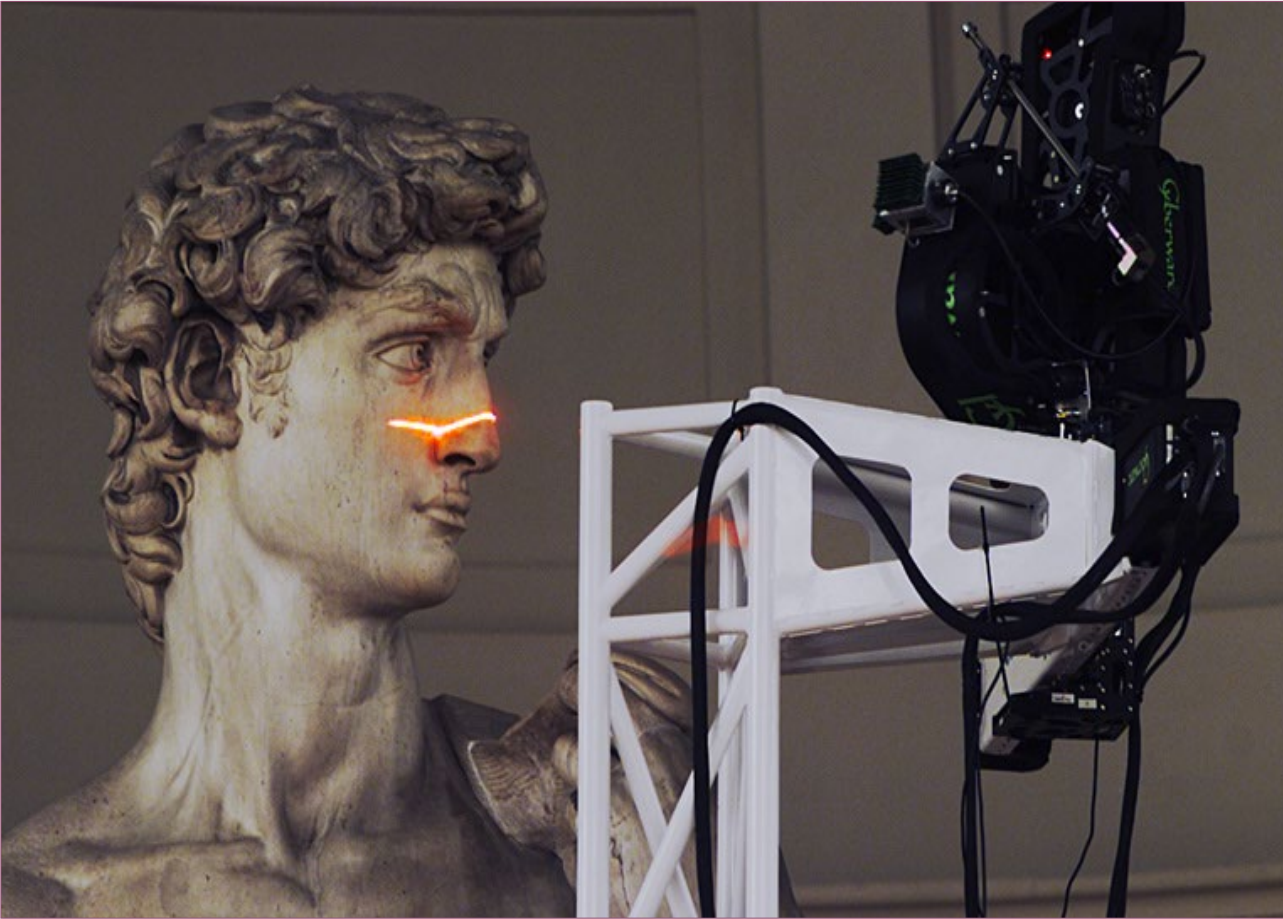
(c) Global Component

The pipeline – *Computational Imaging*

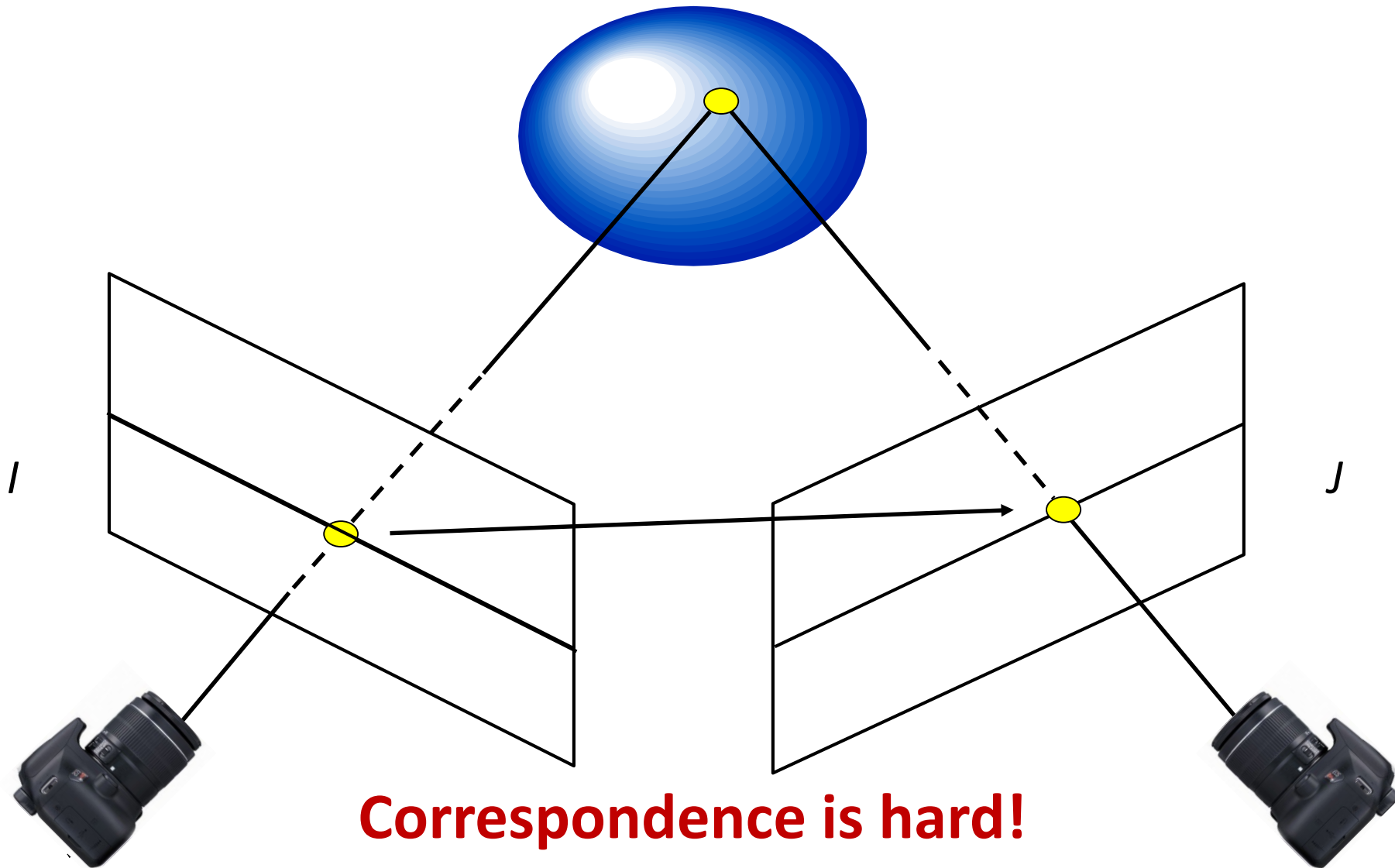


Structured Light

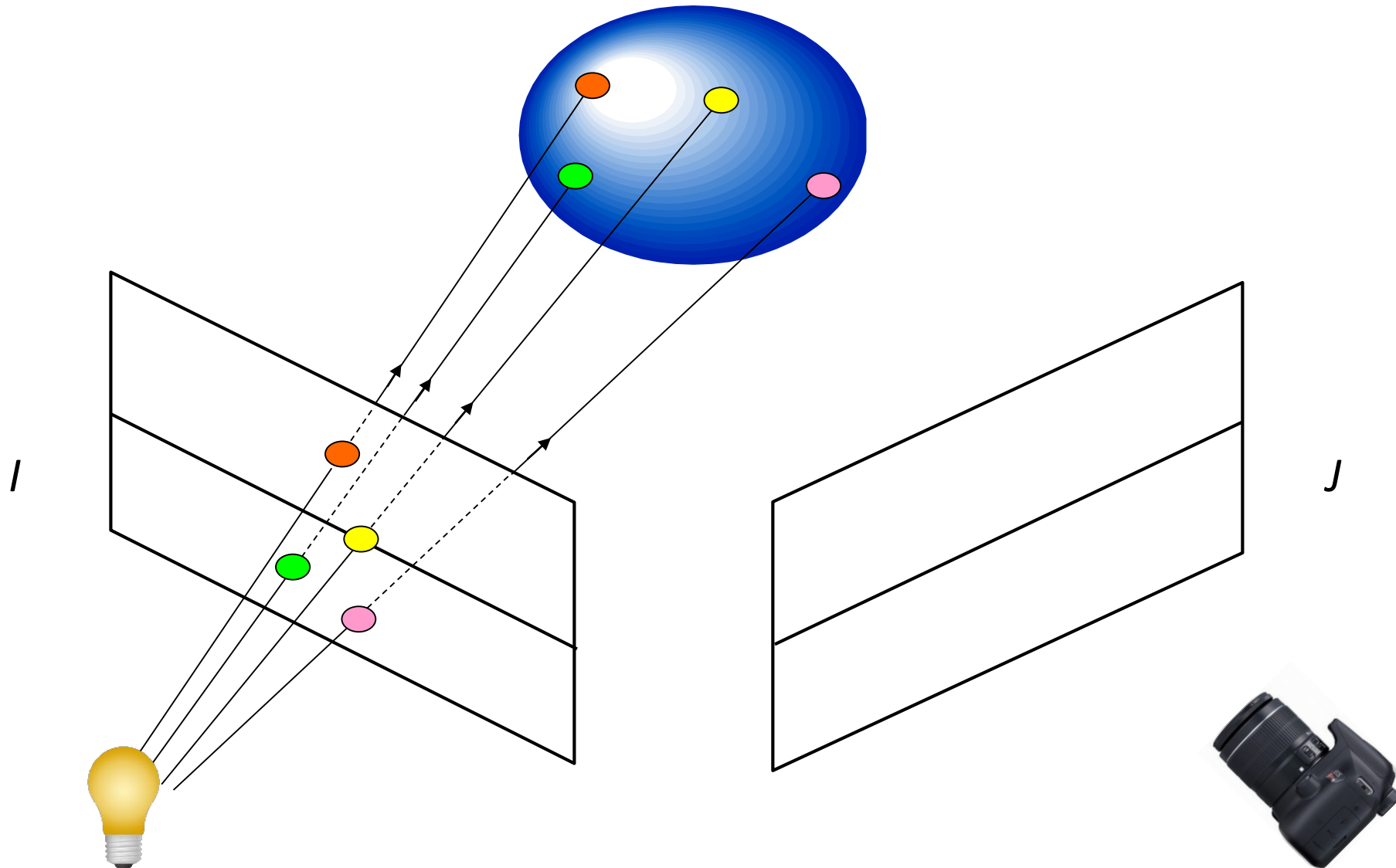
Motivation - 3D Vision / Scanning



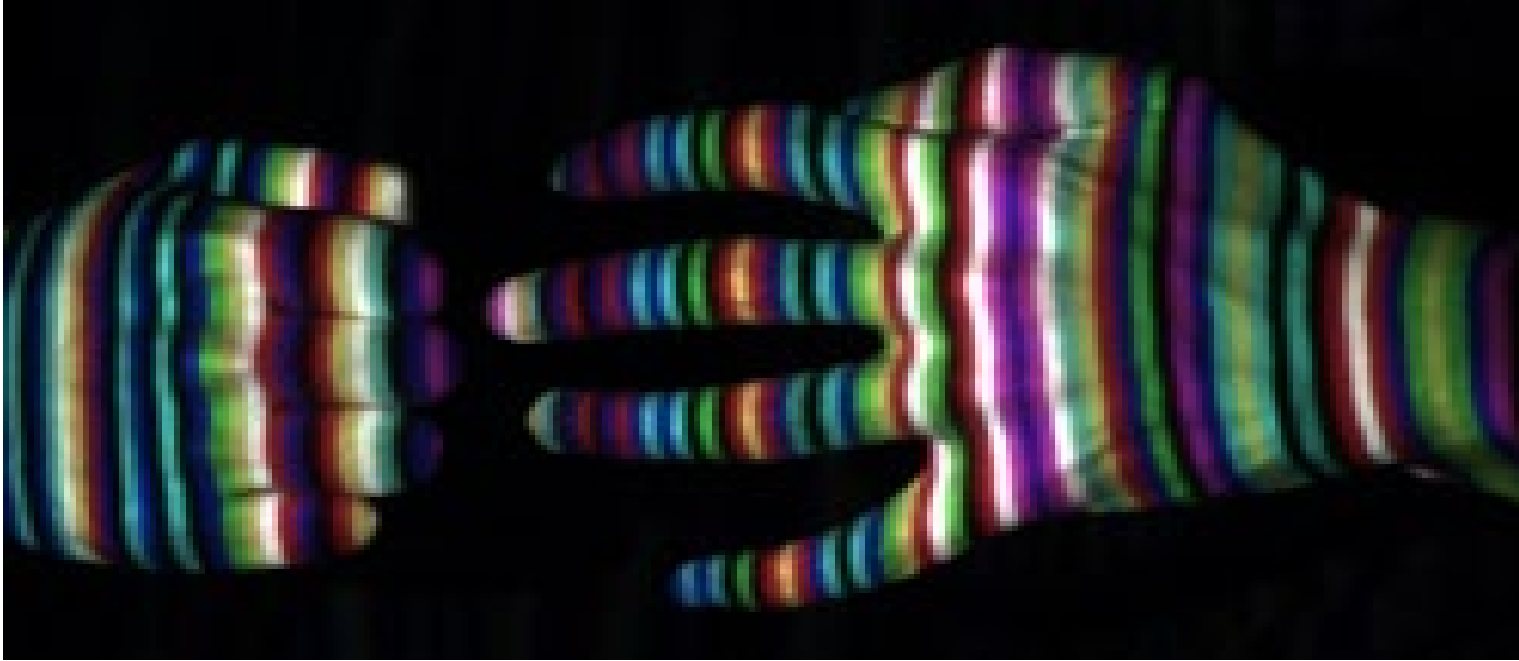
3D Vision – *Multiview Stereo*



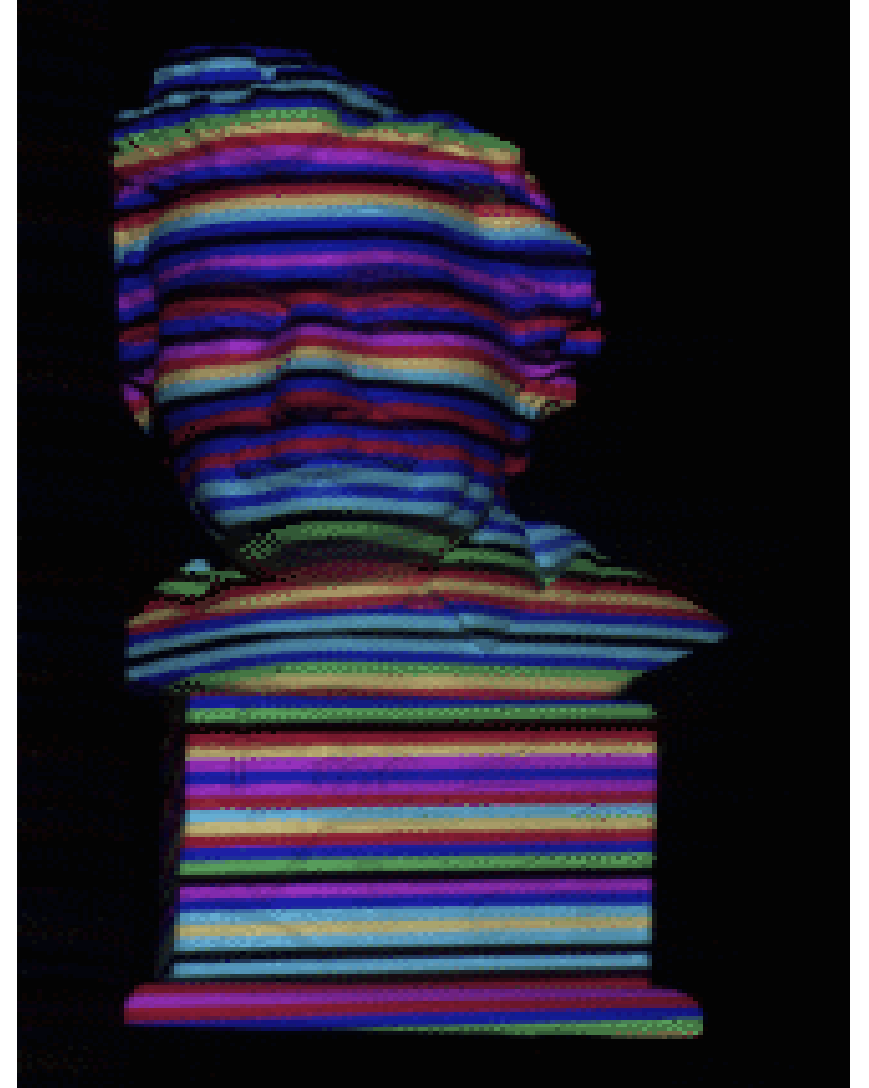
3D Vision – *But what if....*



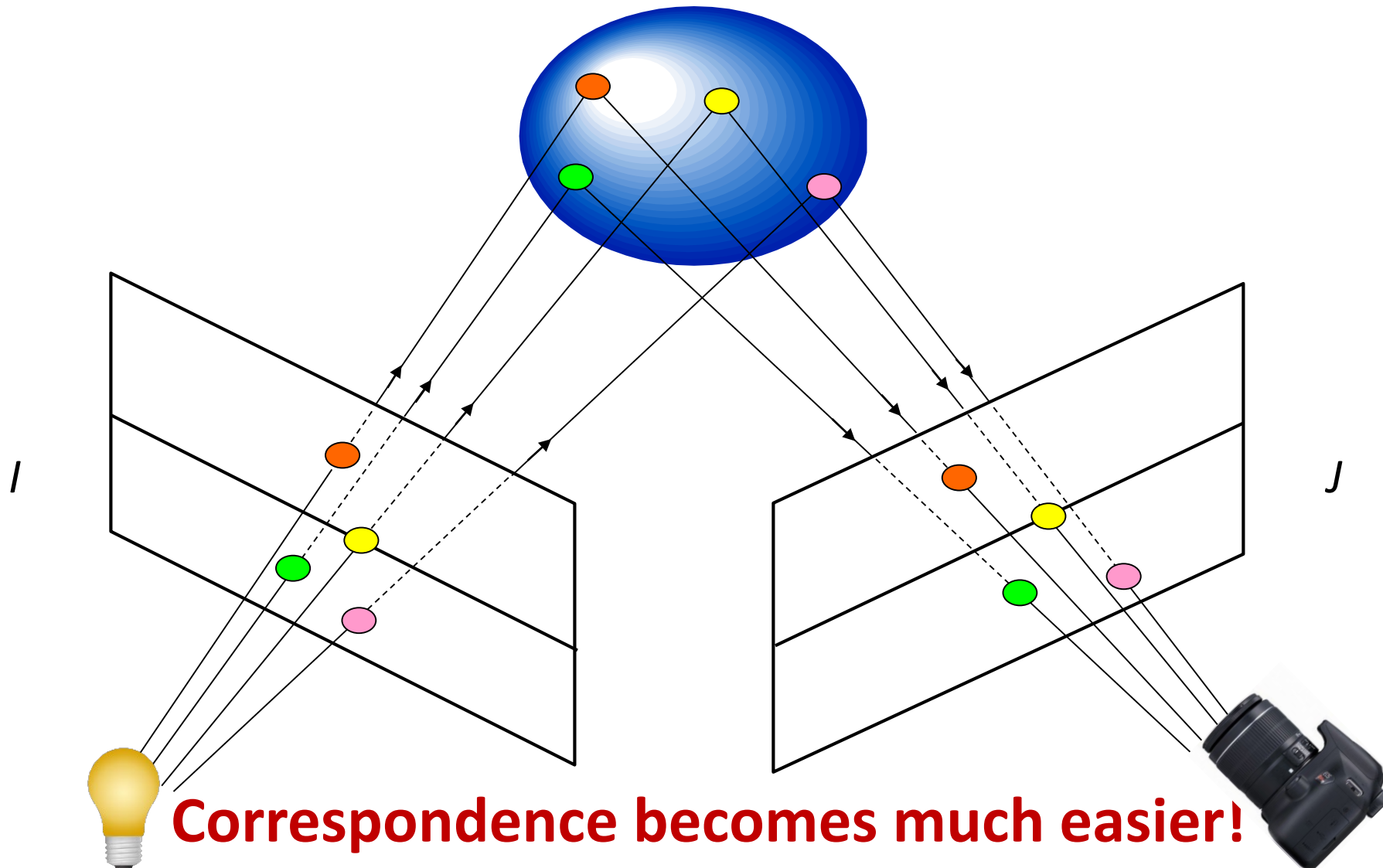
Structured Light



- **What:** Cleverly illuminate the scene to extract scene properties
- **How:** Project an spatio-temporal pattern on a surface or volume.



3D Vision – *Structured Light*



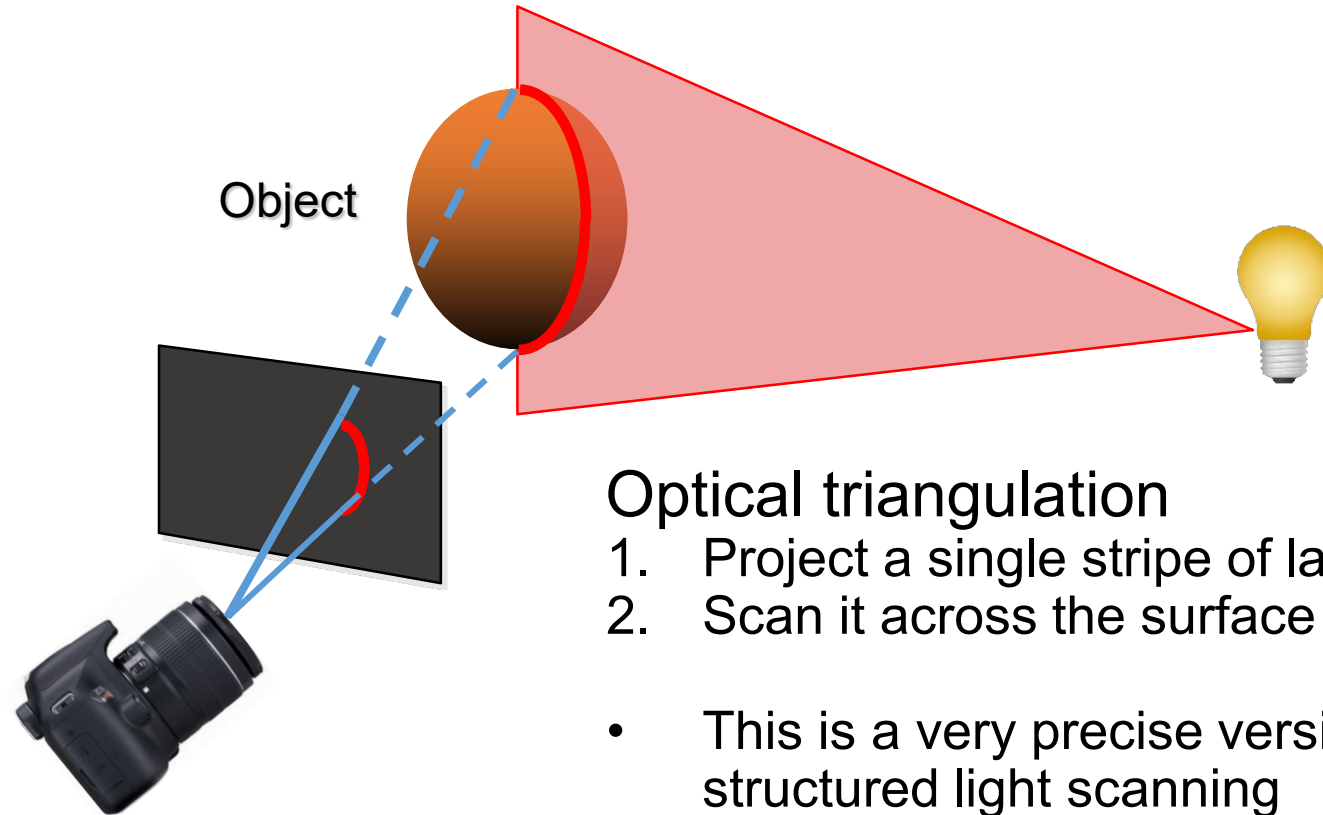
Correspondence becomes much easier!

Structured Light – *What pattern do we project?*

Wild guesses, please?

Structured Light – *What pattern do we project?*

- A single stripe



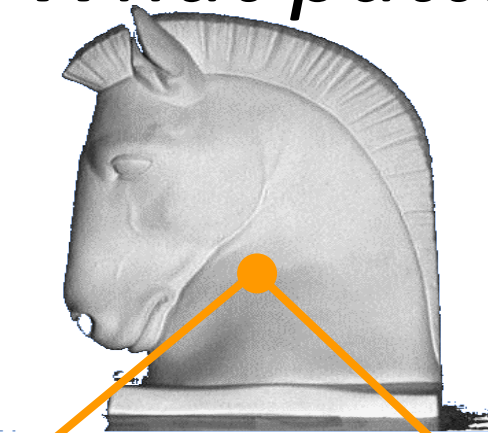
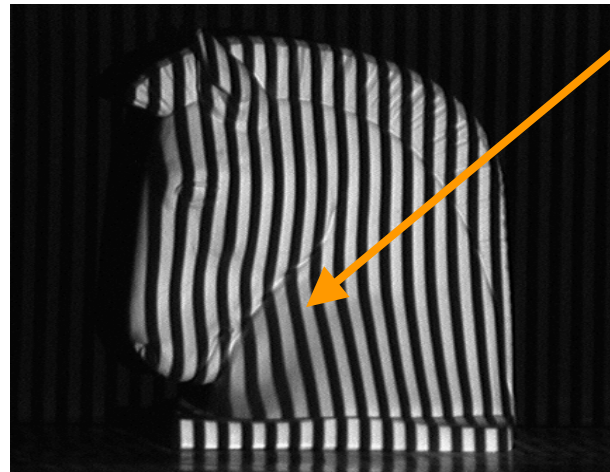
Optical triangulation

1. Project a single stripe of laser light
 2. Scan it across the surface of the object
- This is a very precise version of structured light scanning
 - Good for high resolution 3D, but needs many images and takes time

Can we do better?

Structured Light – *What pattern do we project?*

- Binary coding
Can we do better?

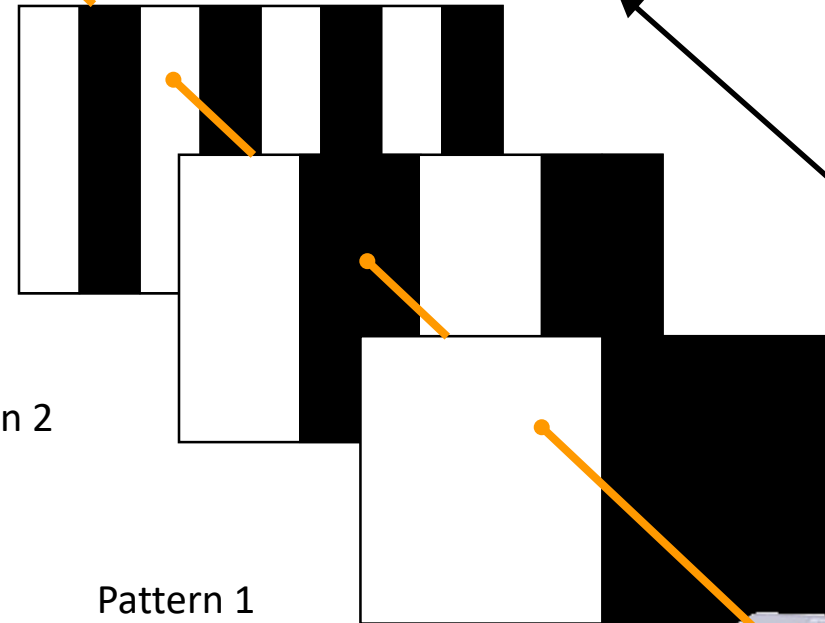


...

Pattern 3

Pattern 2

Pattern 1



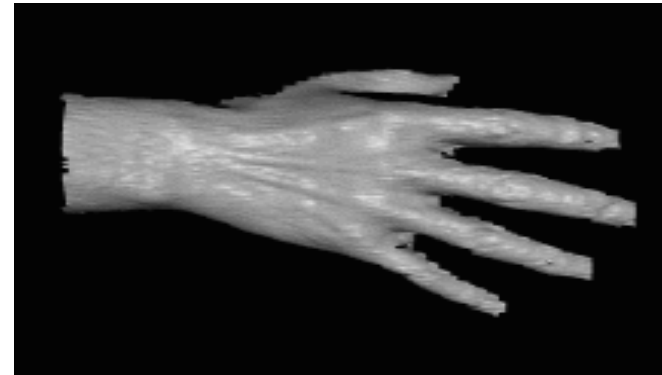
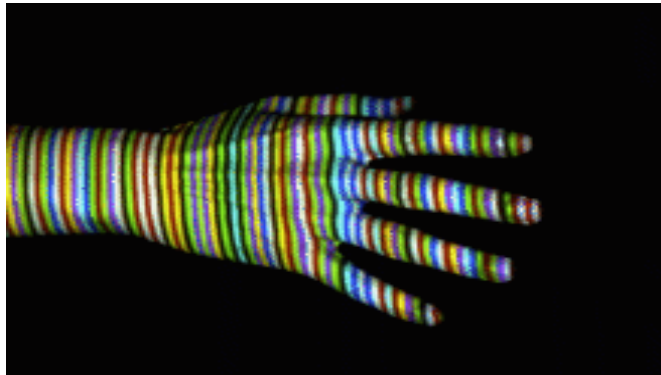
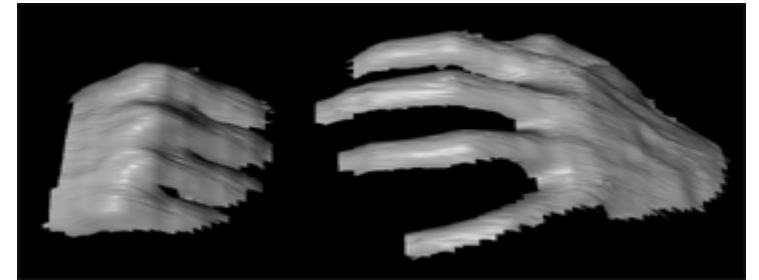
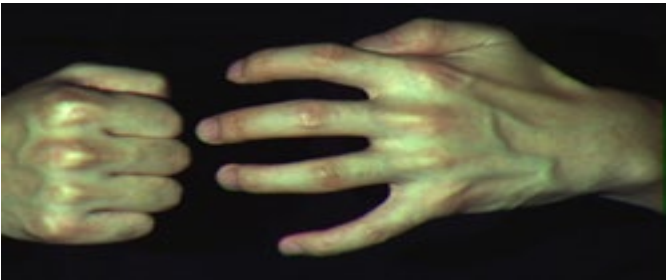
Example: 7 binary patterns proposed by Posdamer & Altschuler

Projected over time



Structured Light – *What pattern do we project?*

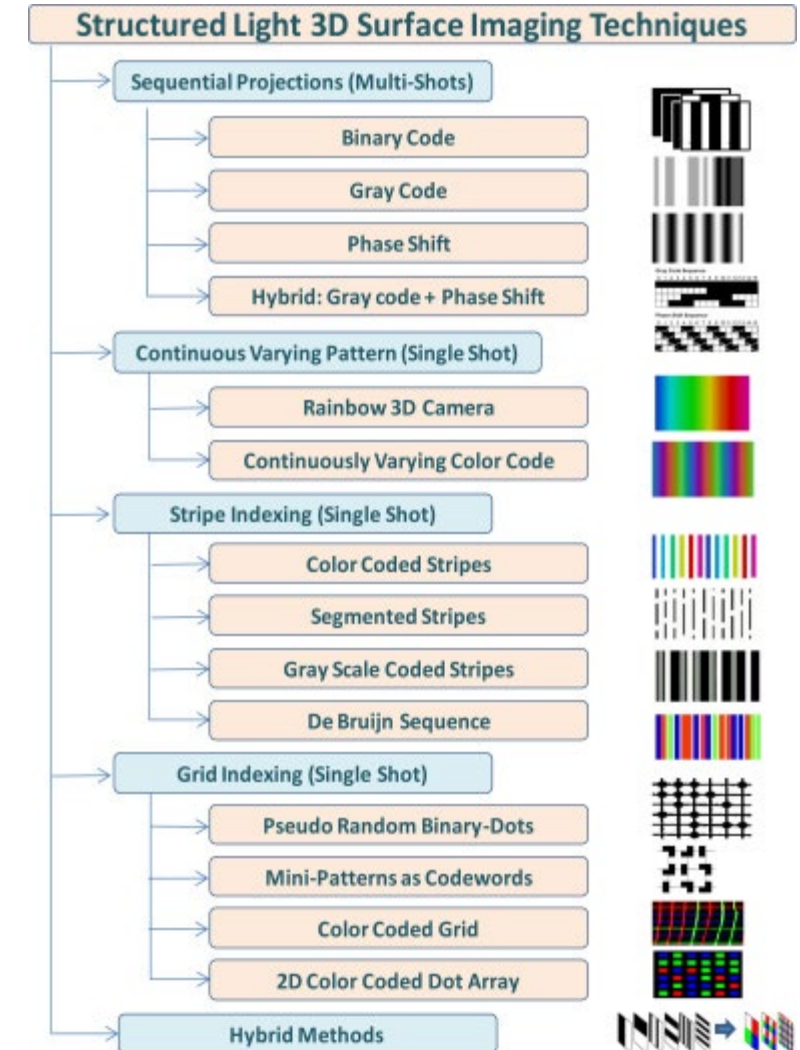
- Complex continuous (non-binary) patterns



- Need very few images (one or two), real-time and dynamic scenes
- But needs a more complex correspondence algorithm

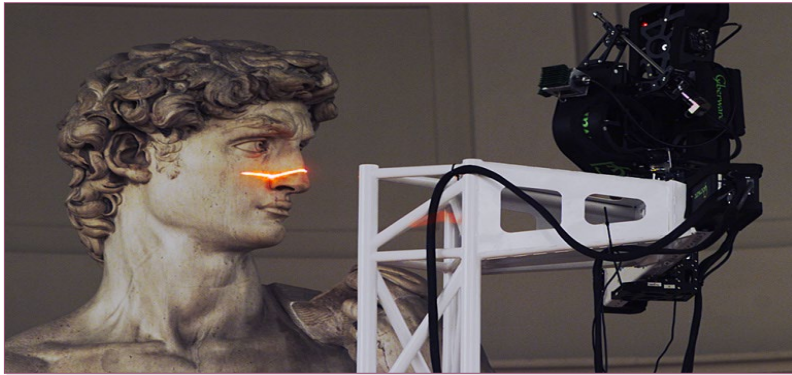
Structured Light – *What pattern do we project?*

- Complex patterns



[Geng 2011]

Structured Light – *What pattern do we project?*

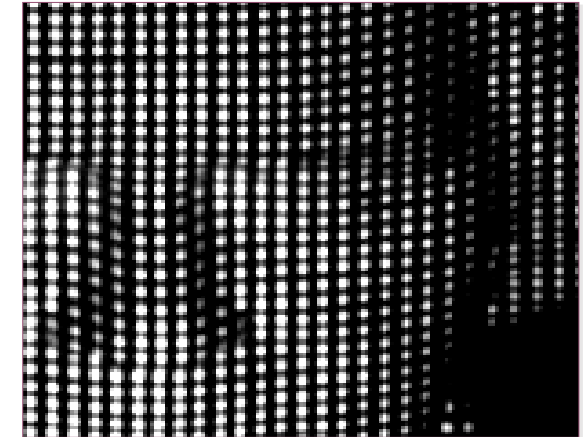


Single-stripe

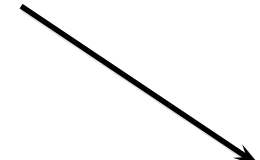


Slow, robust

Multi-stripe
Multi-frame



Single-frame



Fast, fragile



Questions?

Active Illumination + Computation - *Recap*

- **Example I:** Flash-No-Flash Photography

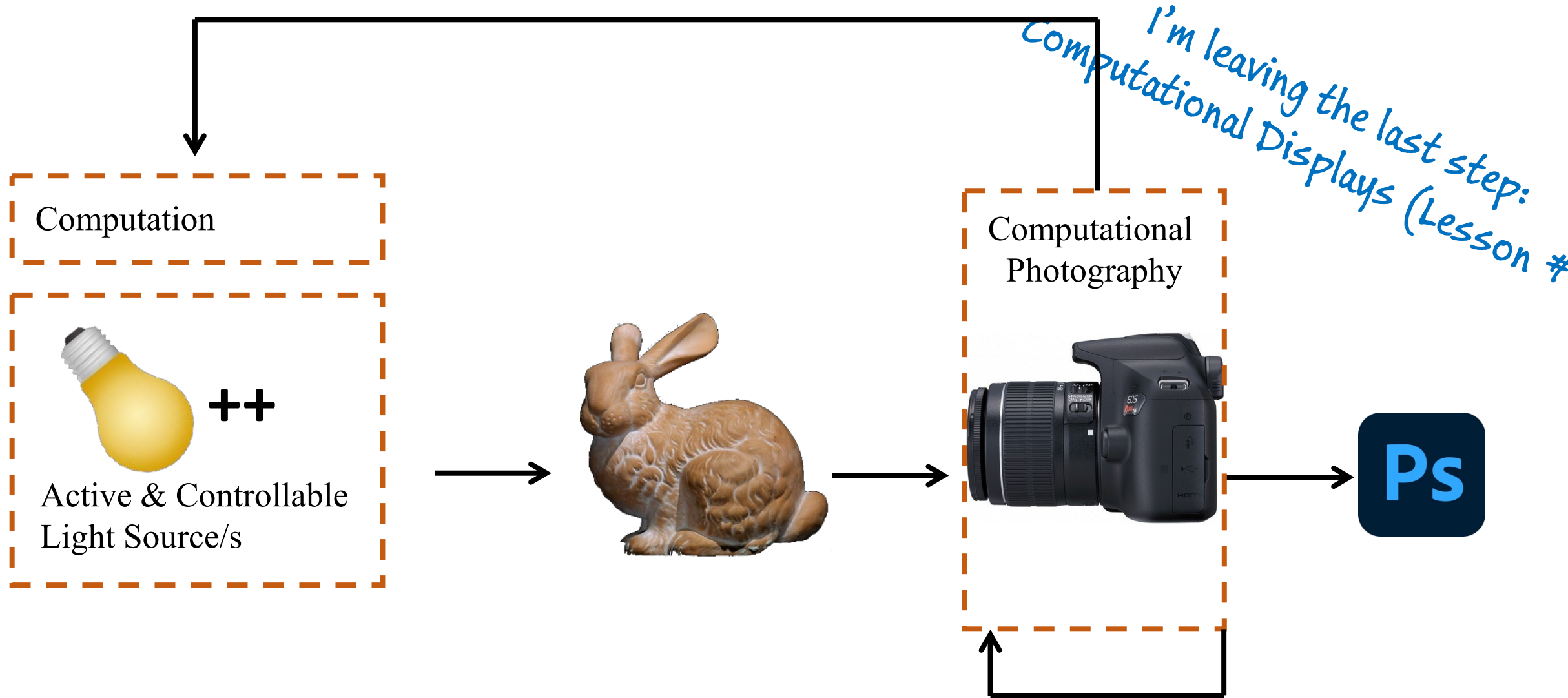
By adding a little bit of computation after capture, and coding illumination, we can significantly improve photography in low-light conditions

- **Example II:** Structured Illumination

By coding the illumination on the directional domain, and potentially taking multiple captures (or not), we can significantly improve 3D range sensing. This was the working principle of MS Kinect 1.

- Can we do more?

The pipeline – *Computational Imaging*

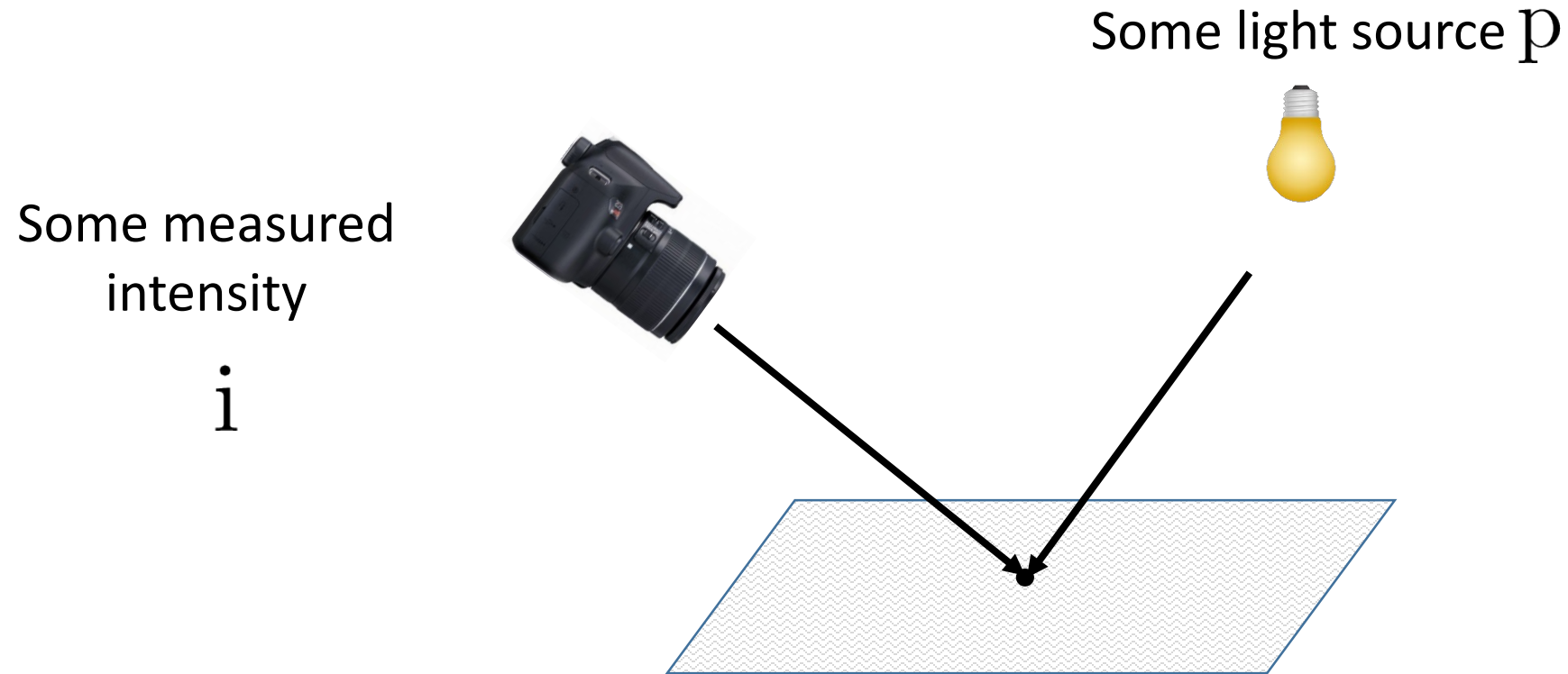








Reminder – *The Light Transport Matrix*

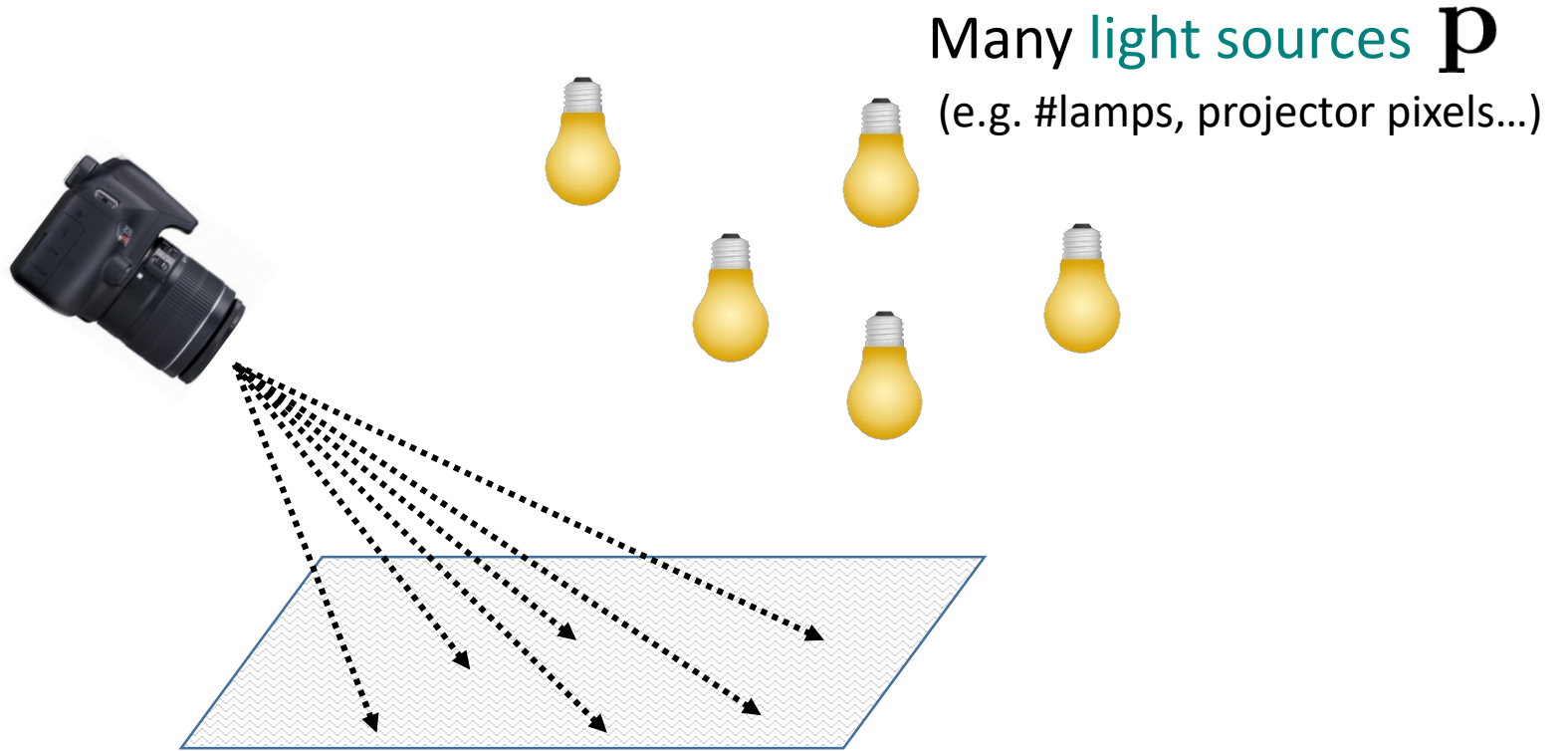


Some coefficient t that **defines light transport linearly**

$$i = tp$$

Reminder – *The Light Transport Matrix*

Many measured
intensities $\mathbf{i}$
(e.g. camera pixels)



Some **coefficients** $\mathbf{T}$ (matrix) that define light transport linearly as

$$\mathbf{i} = \mathbf{T}\mathbf{p} \quad [\text{Debevec et al. 2000}]$$

Reminder – *The Light Transport Matrix*

The LTM $\mathbf{T}$ establishes a linear transport relationship between a set of light sources $\mathbf{p}$ and a set of observations $\mathbf{i}$.

$$\mathbf{i} = \mathbf{T}\mathbf{p}$$

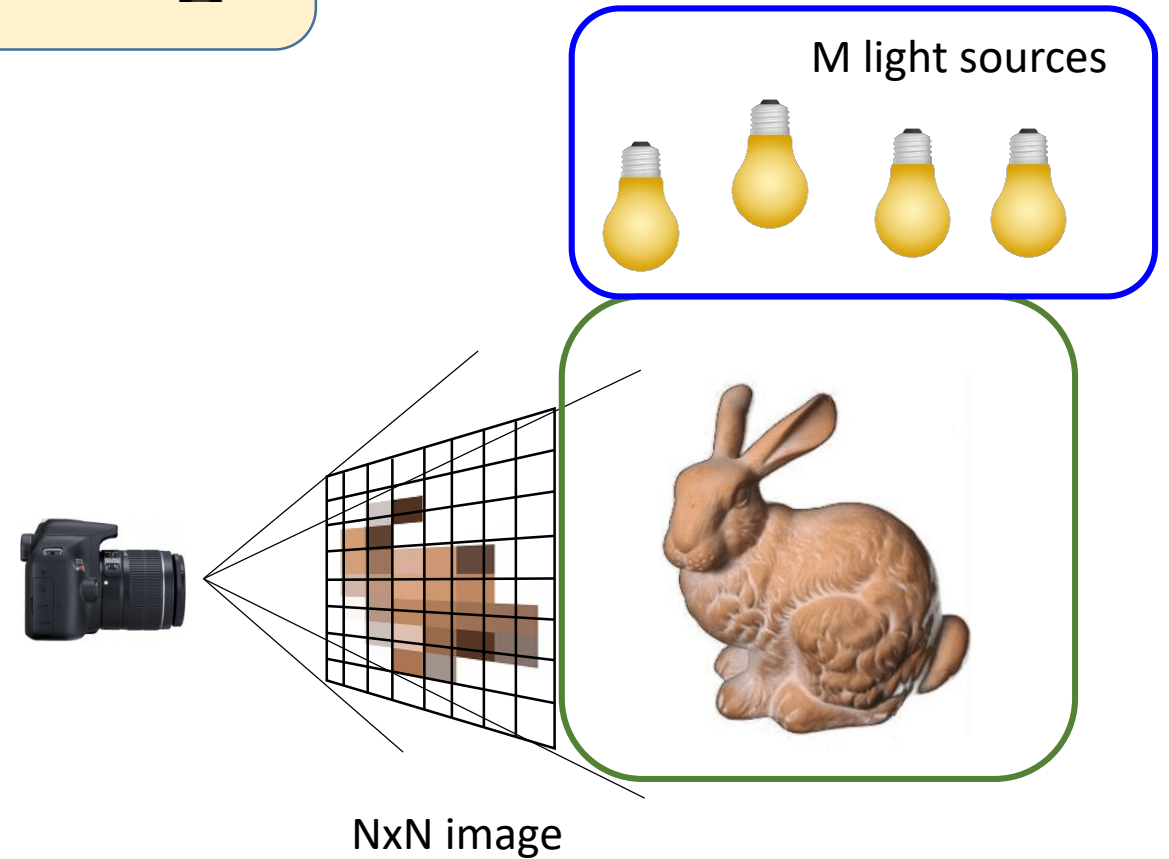
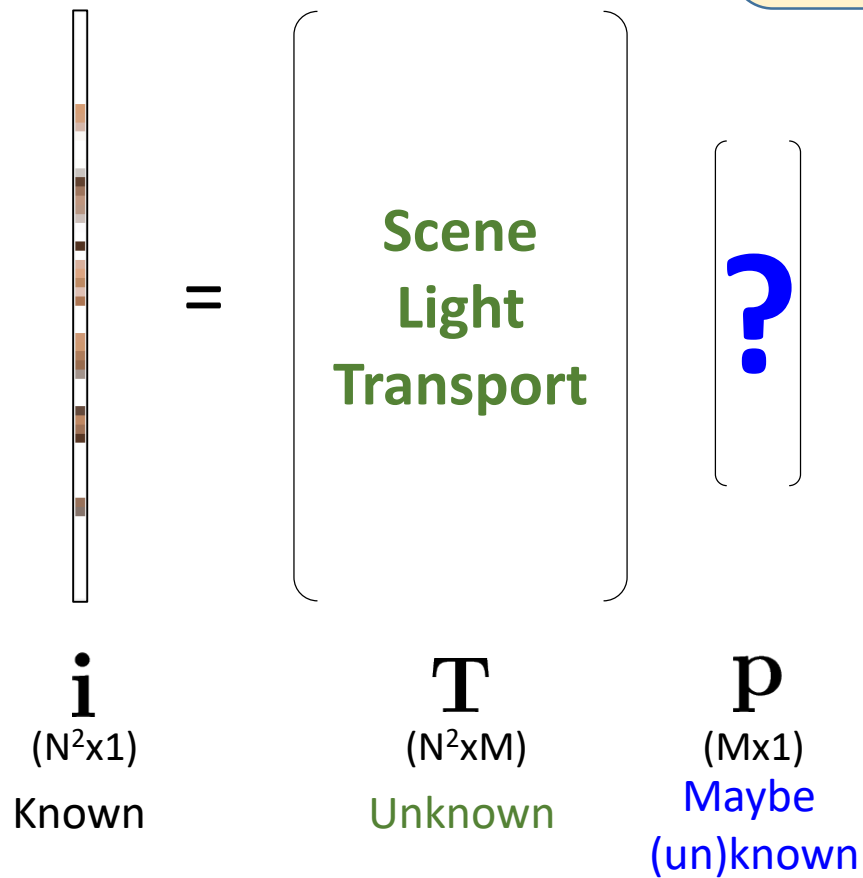
$\mathbf{i}$ - *Observations* ($N \times 1$): Made by a sensor (e.g. image pixels, a LIDAR measurement)

$\mathbf{p}$ - *Light sources* ($M \times 1$): Illuminate the scene (e.g. light bulbs, lasers, projector pixels)

$\mathbf{T}$ - *Light transport matrix* ($N \times M$): Encodes transport through the scene between $\mathbf{p}$ and $\mathbf{i}$

Reminder – *The Light Transport Matrix*

$$\mathbf{i} = \mathbf{T}\mathbf{p}$$



Computational Light Transport

Computational Light Transport

- **What** is it?

The set of techniques aiming to capture/reconstruct the LTM of a scene

- **Why** is it important? (a.k.a. applications)

- **How** do we capture it?

Questions?

Computational Light Transport

- **What** is it?
The set of techniques aiming to capture/reconstruct/exploit the LTM of a scene
- **Why** is it important? (a.k.a. applications)
- **How** do we capture it?

Computational Light Transport - *Applications*

1) Relighting

Number of photos: 40

<https://youtu.be/fVBICVBEGVU?t=161>

[O'Toole & Kutulakos 2010]



Computational Light Transport - *Applications*

1) Relighting

Avatar © 2009 20th Century Fox & Weta Digital Ltd
Source: [Pantaleoni et al. SIG 2010]



Computational Light Transport - *Applications*

1) Relighting

Halo 3 ©Bungie

Source: [Chen and Liu, SIG 2008]

Computational Light Transport - *Applications*

1) Relighting

What else? Wild guesses,
please...

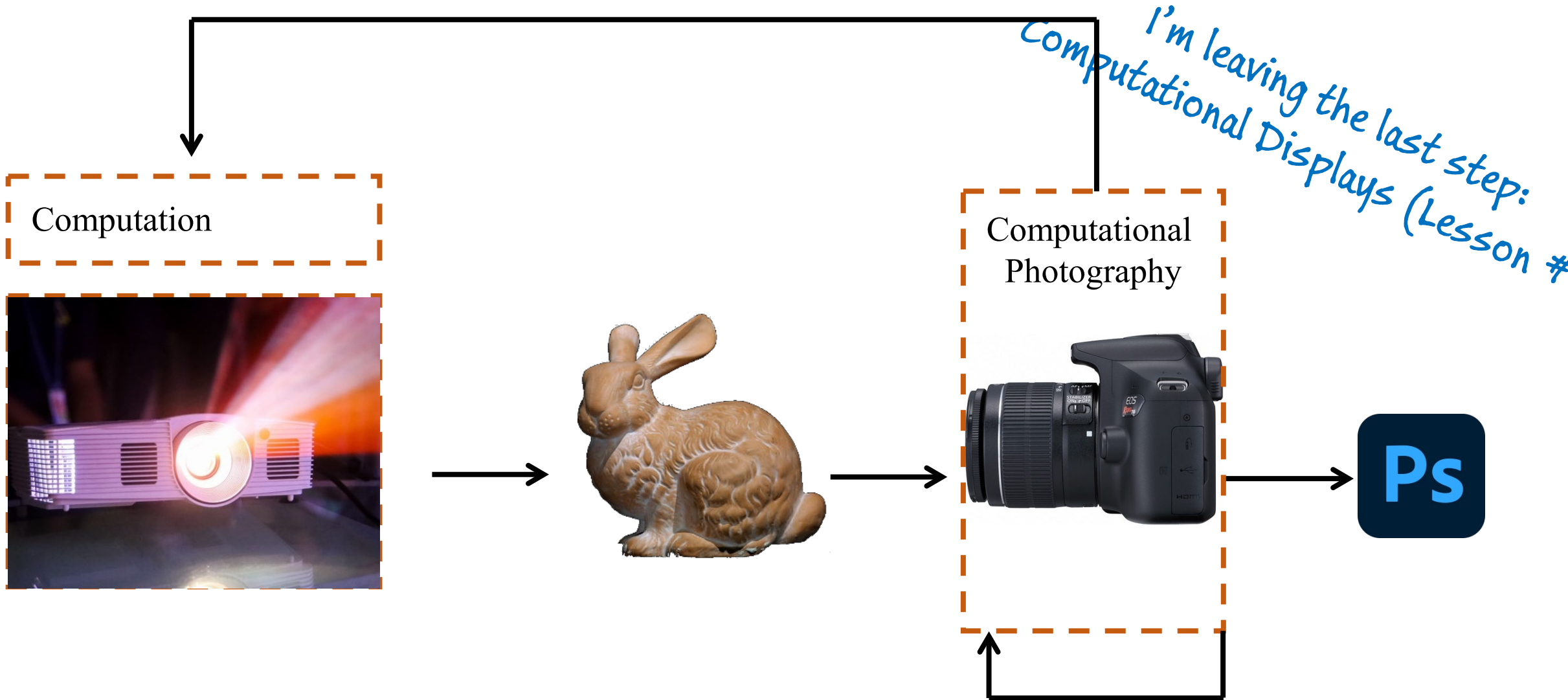
Computational Light Transport - *Applications*

1) Relighting

What else? Wild guesses,
please...

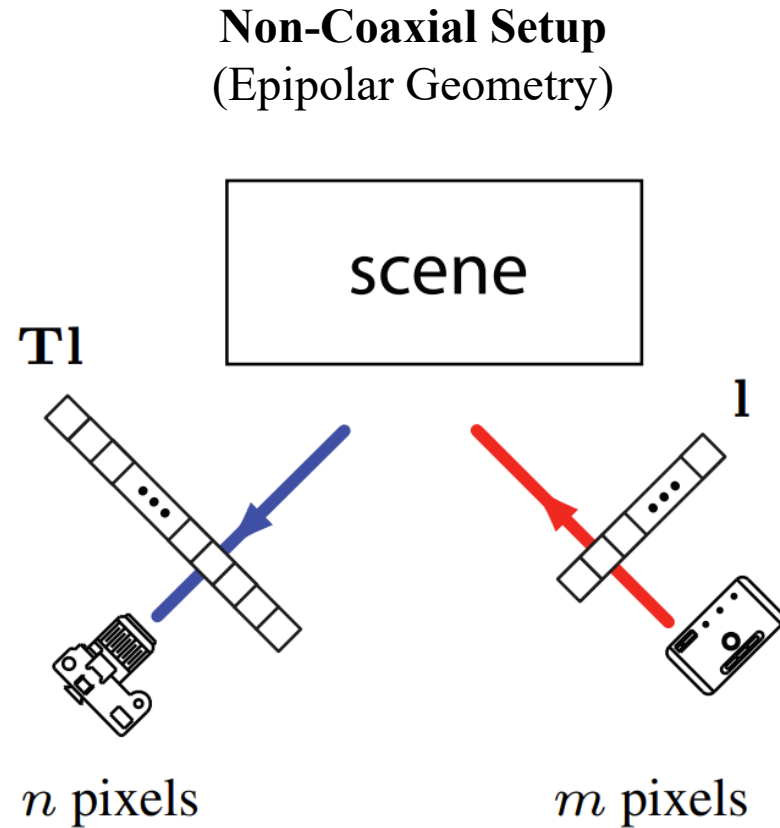
Let's analyze the LTM

The pipeline – *Computational Imaging*

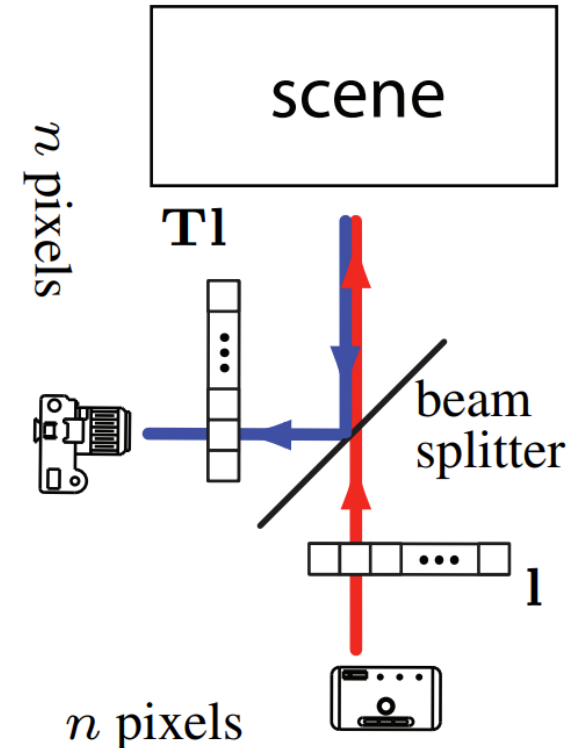


Analyzing the LTM

- Two possible projector/camera pair setups

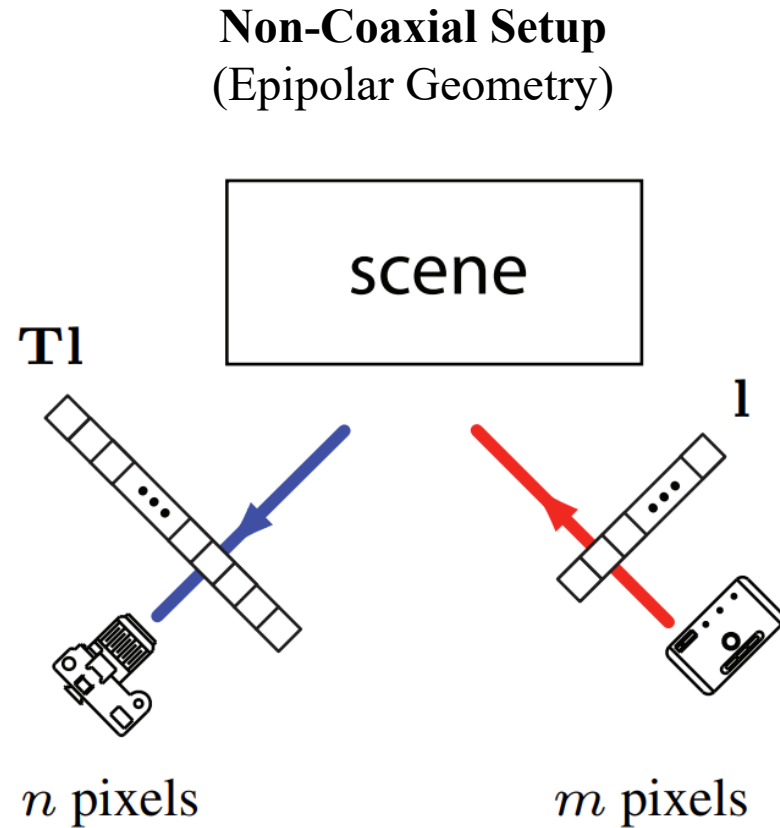


Coaxial Setup
(Camera and Projector share focal plane i.e. same pixels)



Analyzing the LTM

- Two possible projector/camera pair setups



$$\begin{bmatrix} \mathbf{i} \end{bmatrix} = \begin{bmatrix} \text{red line} \\ \text{blue line} \end{bmatrix} \begin{bmatrix} \mathbf{p} \end{bmatrix}$$

$\mathbf{i}$ $(N \times 1)$ $\mathbf{T}$ $(N \times M)$ $\mathbf{p}$ $(M \times 1)$

Non-symmetric LTM

Analyzing the LTM

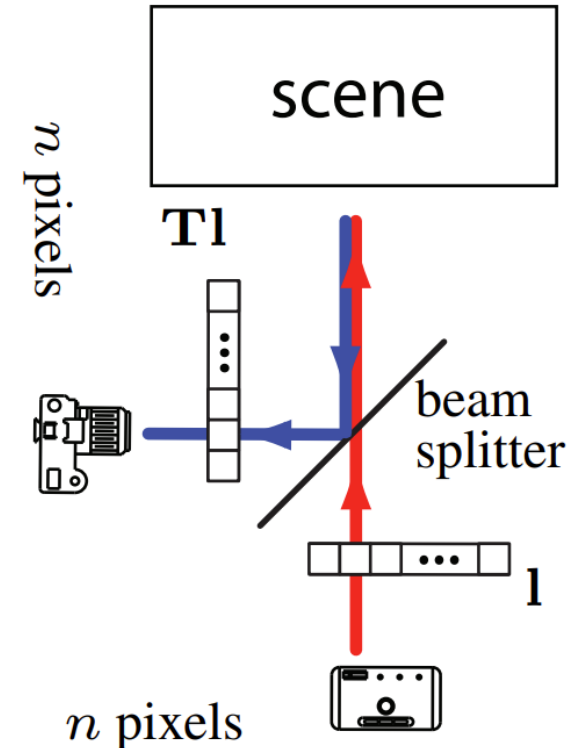
- Two possible projector/camera pair setups

$$\begin{bmatrix} \mathbf{i} \end{bmatrix} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \\ & & & \\ & & & \\ & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \begin{bmatrix} \mathbf{p} \end{bmatrix}$$

$\mathbf{i}$ $\mathbf{T}$ $\mathbf{p}$
(N×1) (N×N) (N×1)

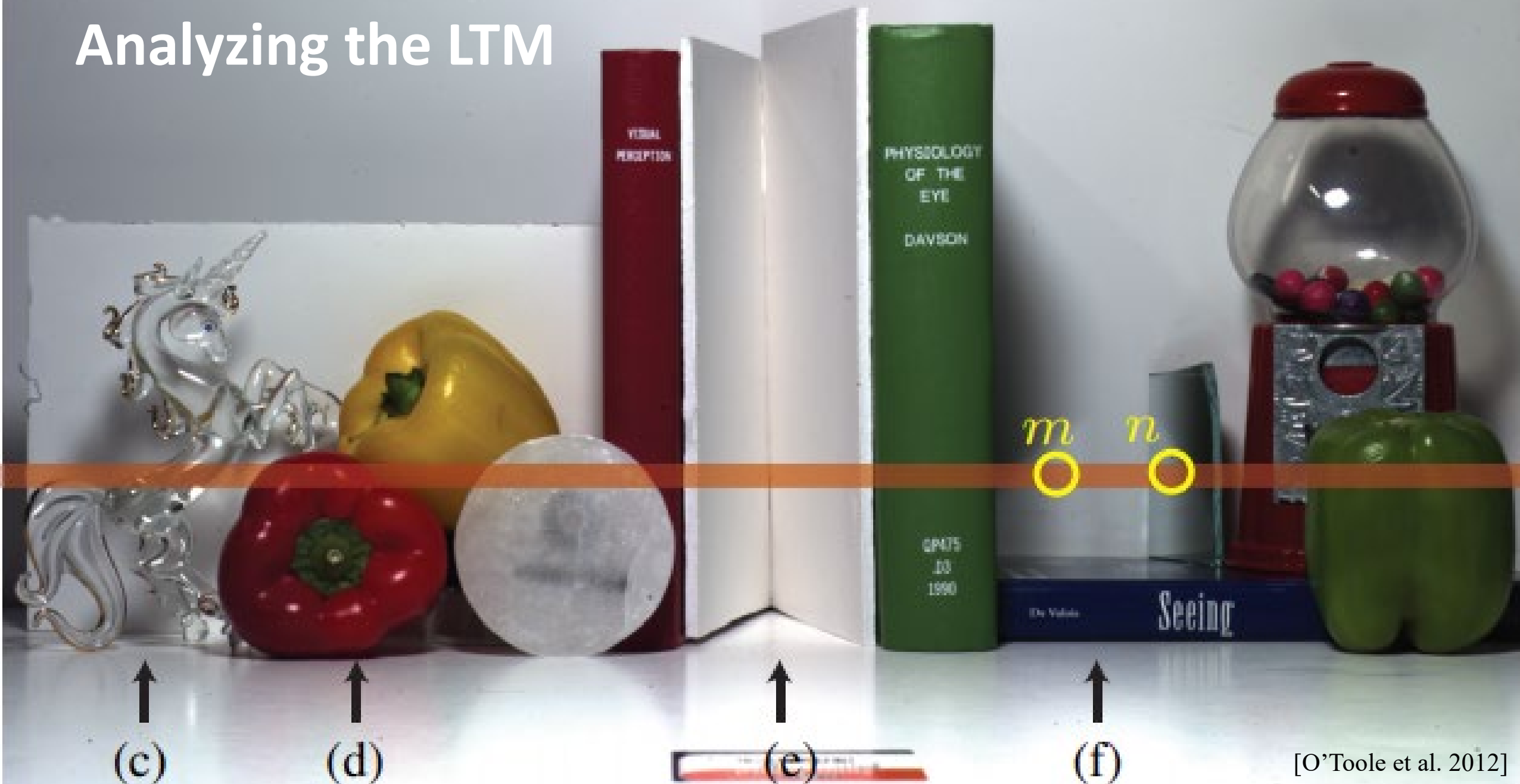
Symmetric LTM

Coaxial Setup
(Camera and Projector share focal plane i.e. same pixels)



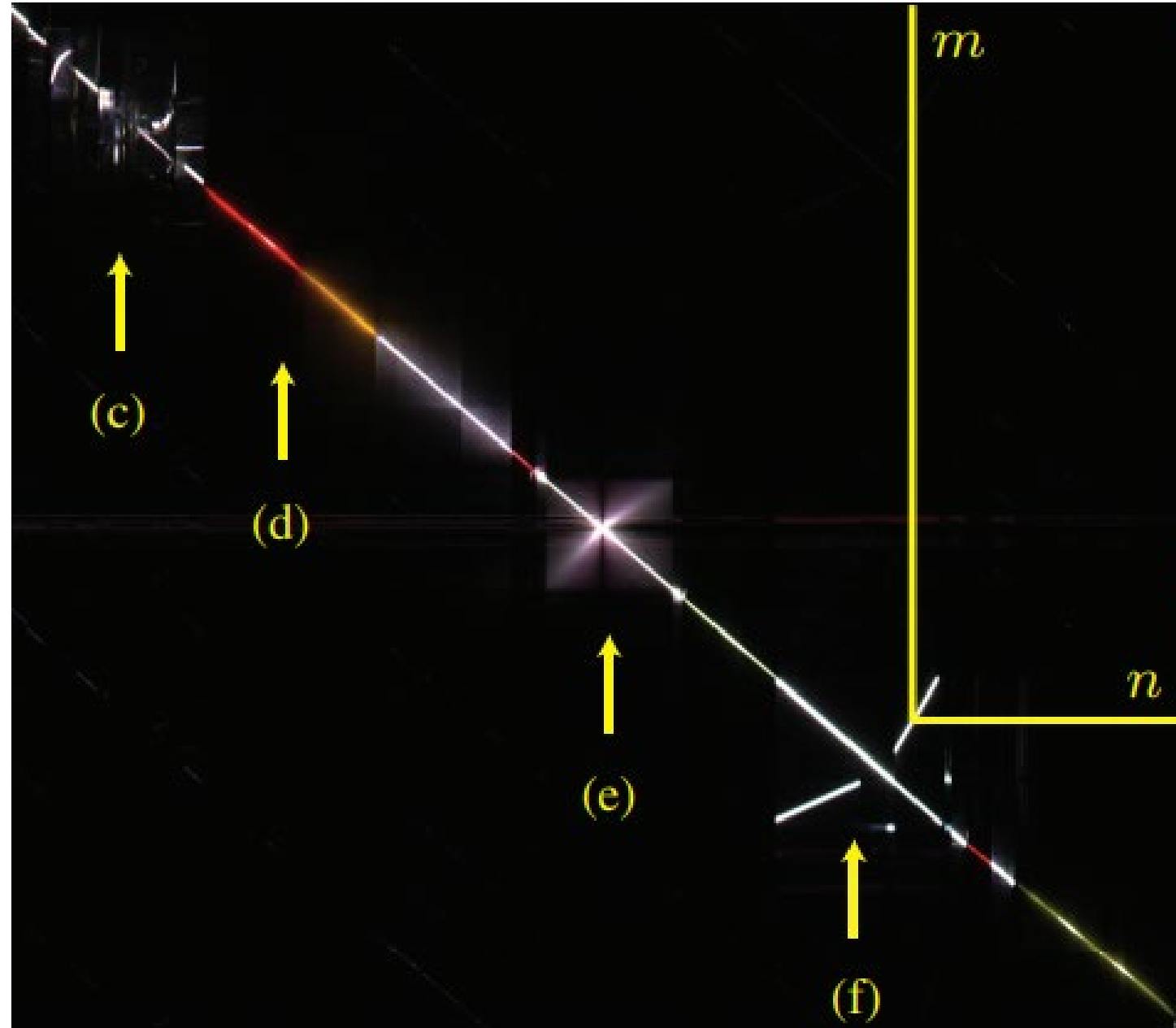
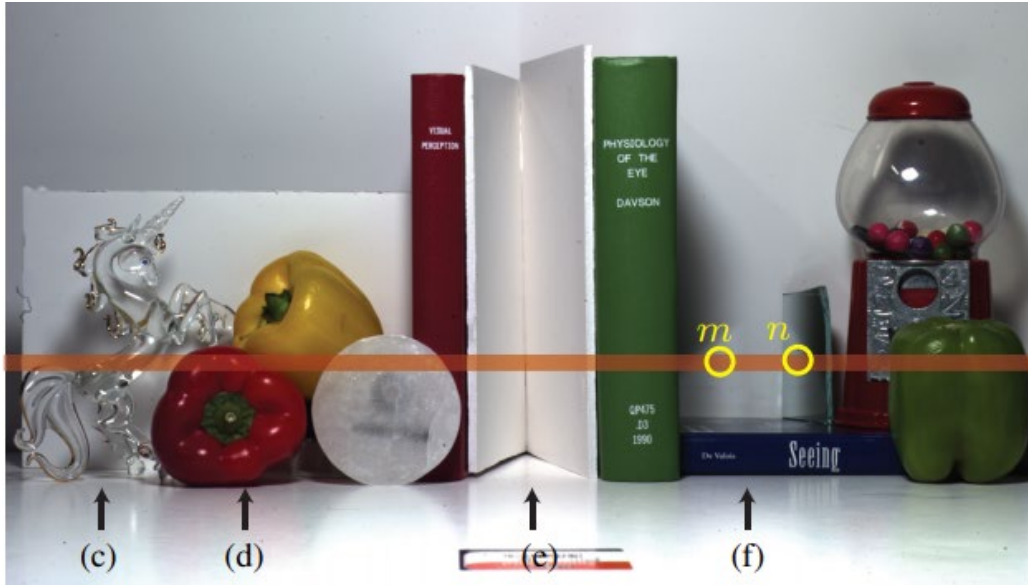
Questions?

Analyzing the LTM



Analyzing the LTM

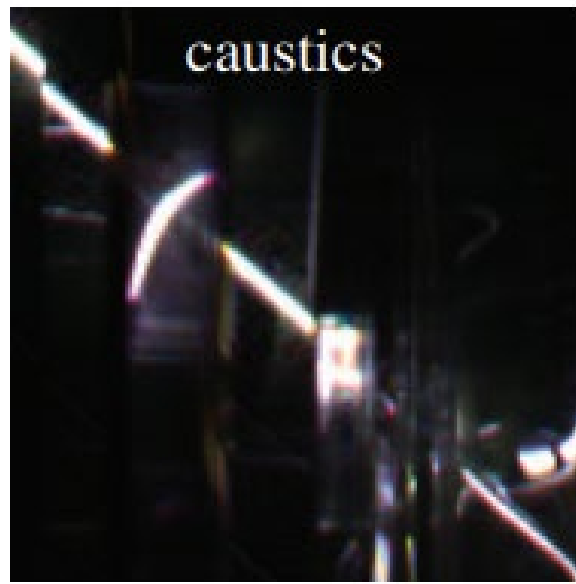
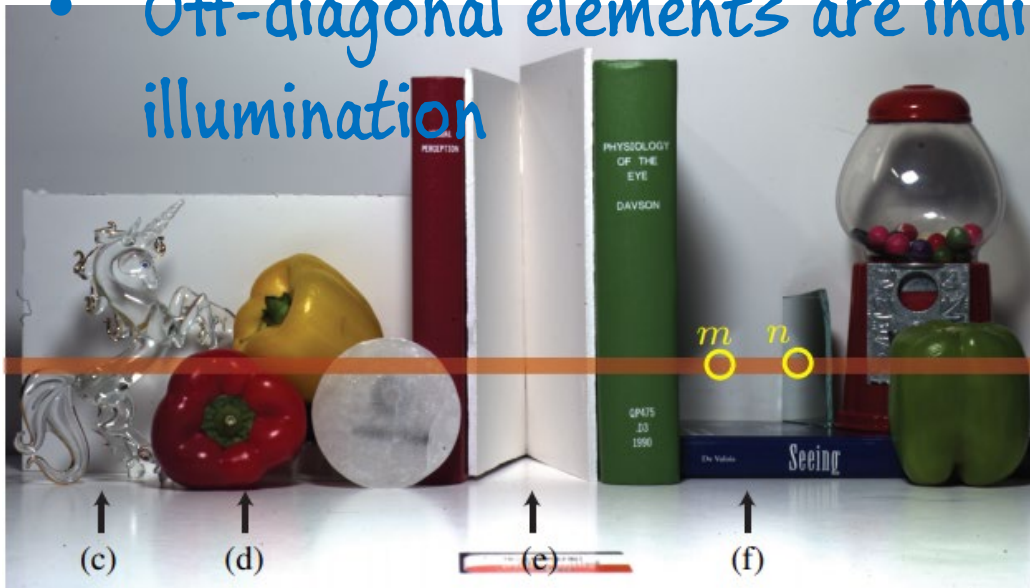
Assuming for now a coaxial setup



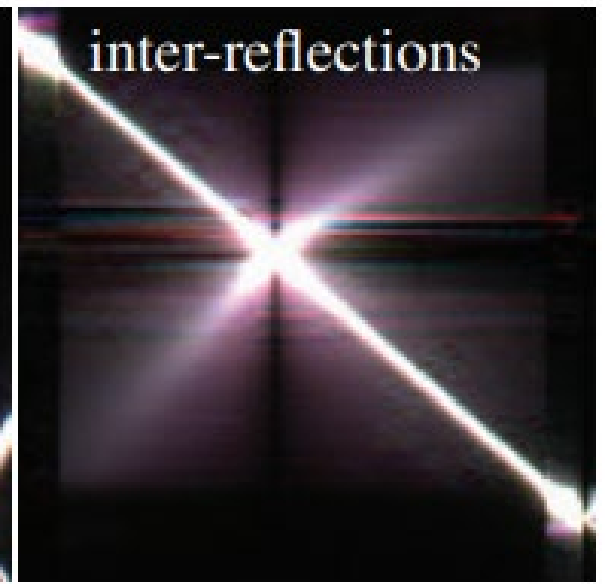
Analyzing the LTM

Assuming for now a coaxial setup

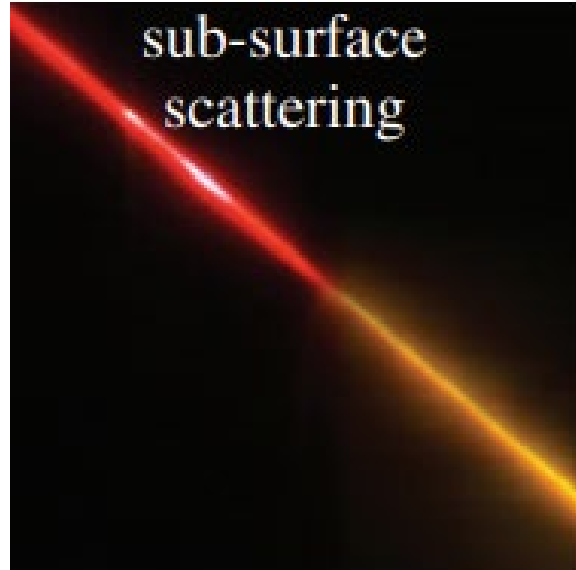
- Diagonal elements of the LTM are direct illumination and backscattering
- Off-diagonal elements are indirect illumination



(c)



(e)



(d)



(f)

Source: [O'Rourke et al. 2012]

Using the LTM - *Illumination Components Separation*

conventional



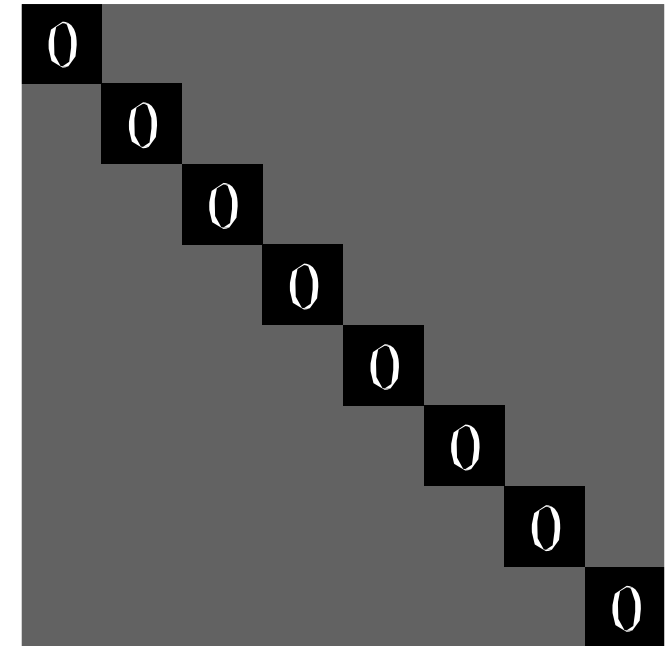
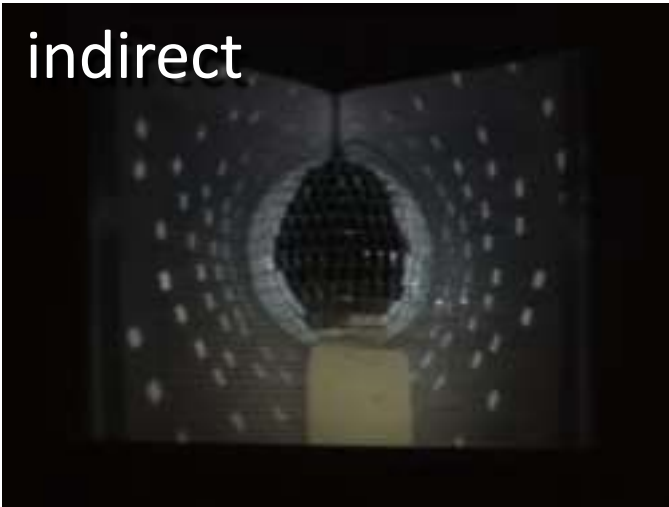
all light paths

Using the LTM - *Illumination Components Separation*

conventional



indirect



indirect light paths

Using the LTM - *Illumination Components Separation*

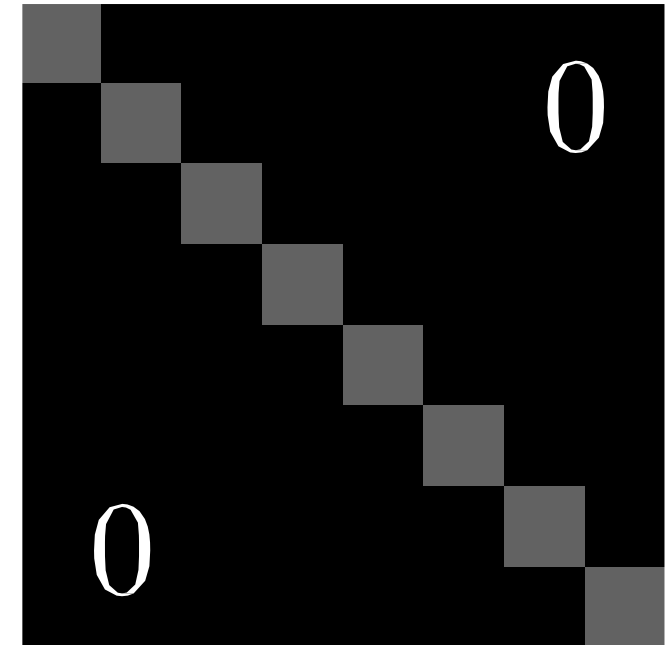
conventional

=

indirect

+

direct + backscatter



direct + back-scatter
light paths

Using the LTM - *Illumination Components Separation*

conventional



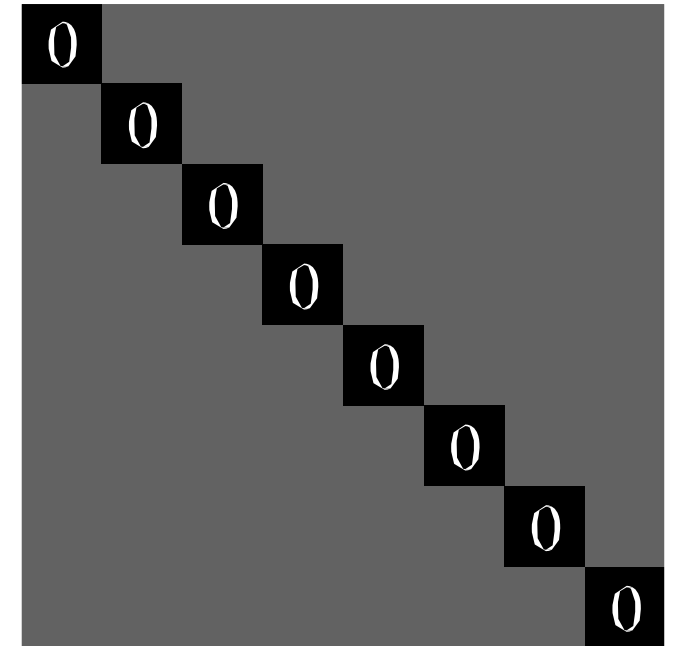
all light paths

Using the LTM - *Illumination Components Separation*

conventional



indirect



indirect light paths

Using the LTM - *Illumination Components Separation*

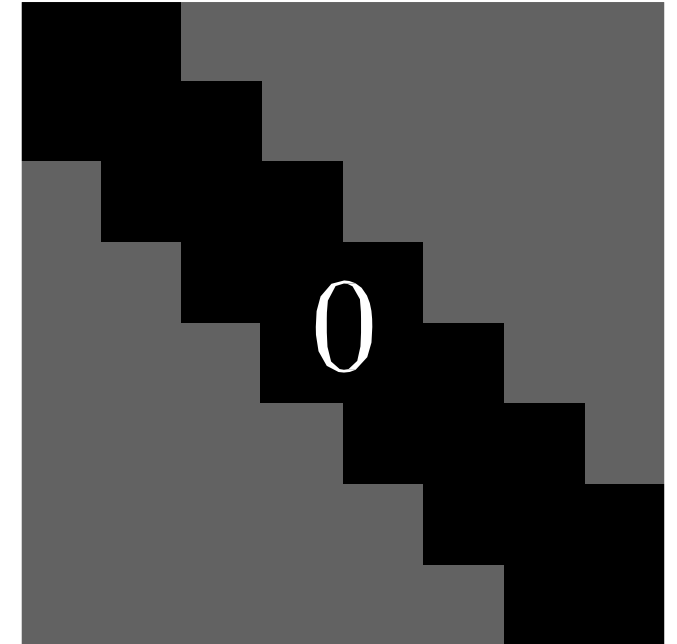
conventional



indirect



mid-range indirect



medium to long range
indirect light paths

Using the LTM - *Illumination Components Separation*

conventional



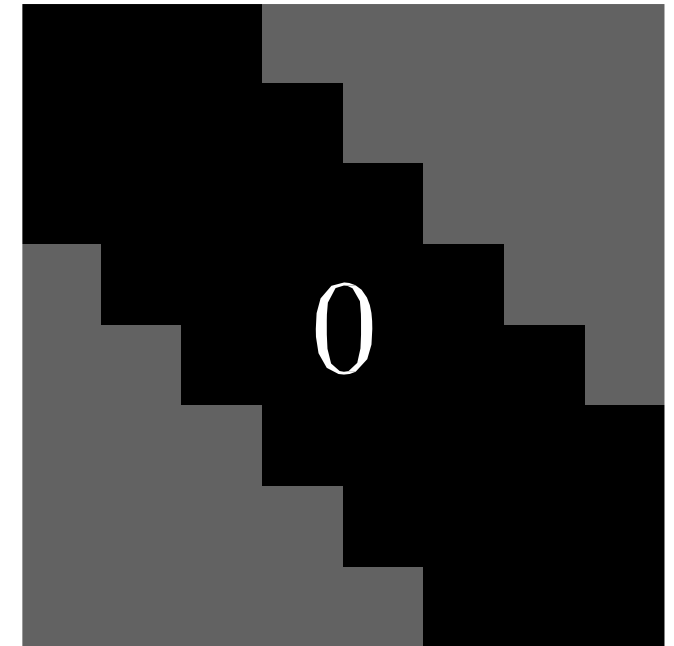
long-range indirect



indirect



mid-range indirect



long range indirect
light paths

Using the LTM - *Descattering*

conventional



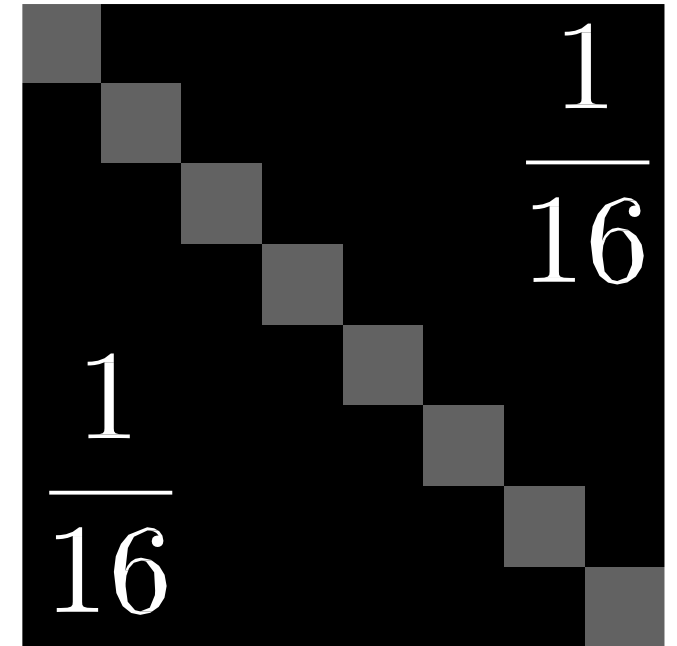
all light paths

Using the LTM - *Descattering*

conventional



direct + $\frac{1}{16}$ indirect



direct + $\frac{1}{16}$ indirect
light paths

Using the LTM - *Descattering*

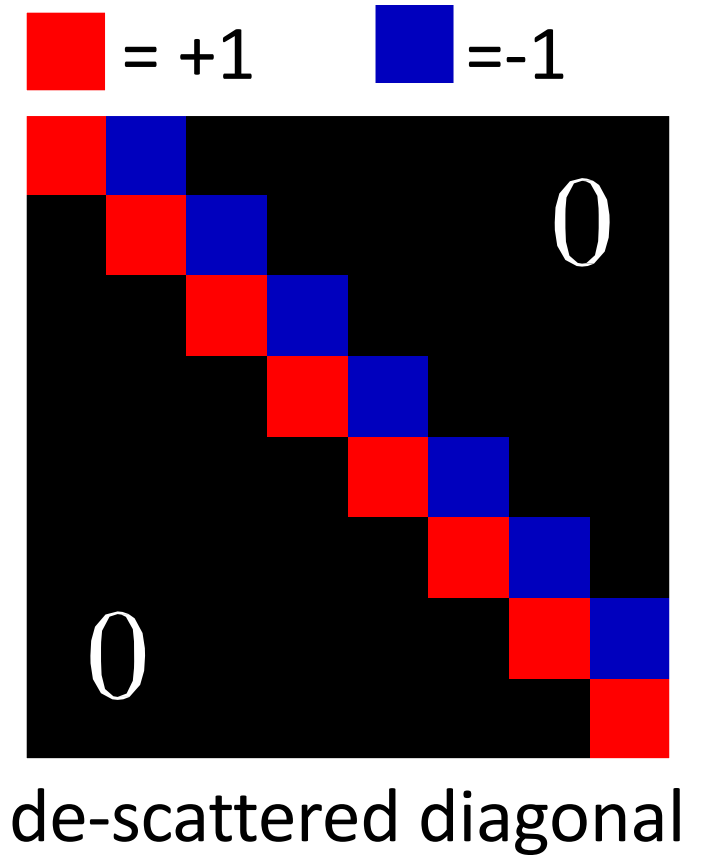
conventional



de-scattered image



direct + 1/16 indirect

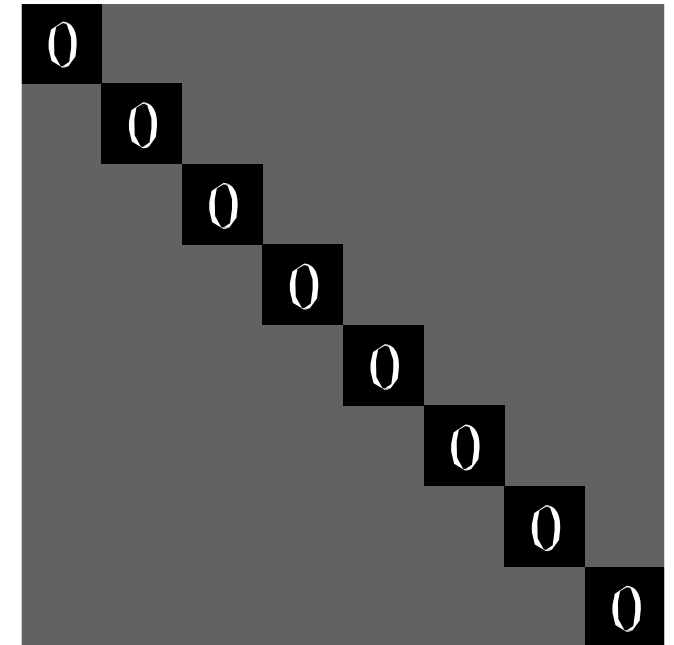


Using the non-symmetric LTM – *Removing 3D regions*



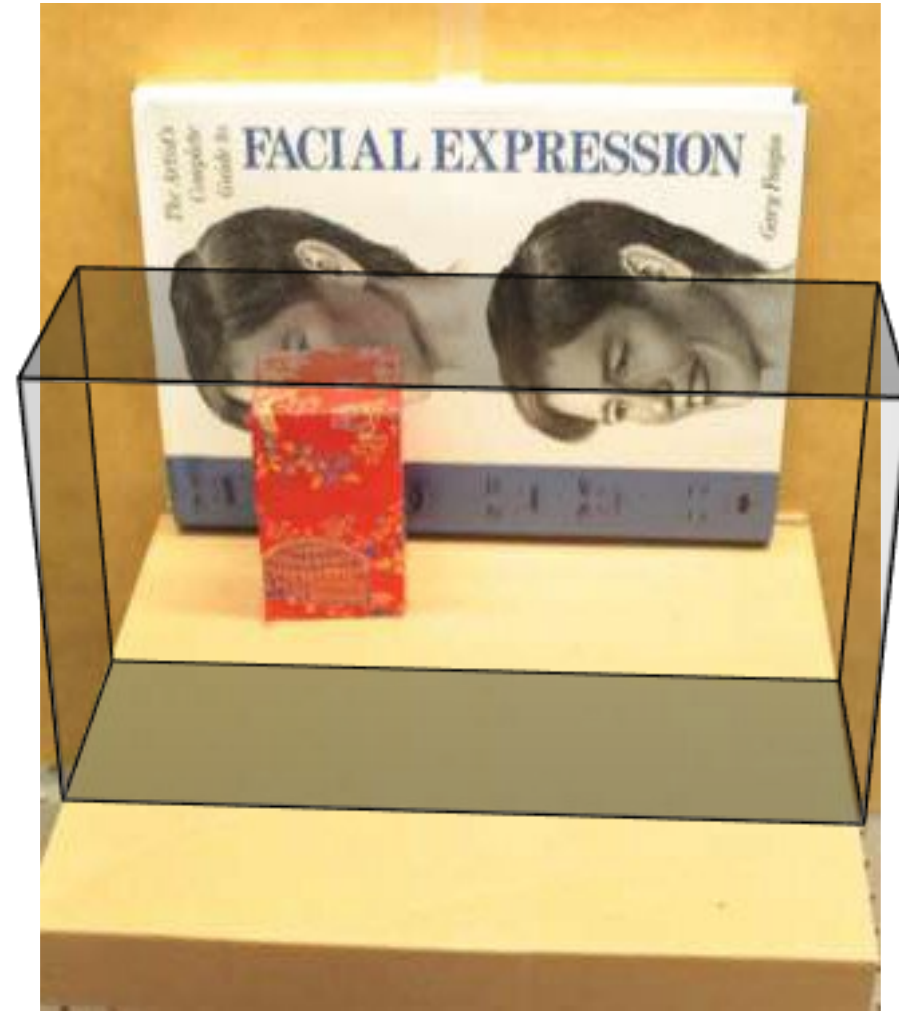
all light paths

Using the non-symmetric LTM – *Removing 3D regions*

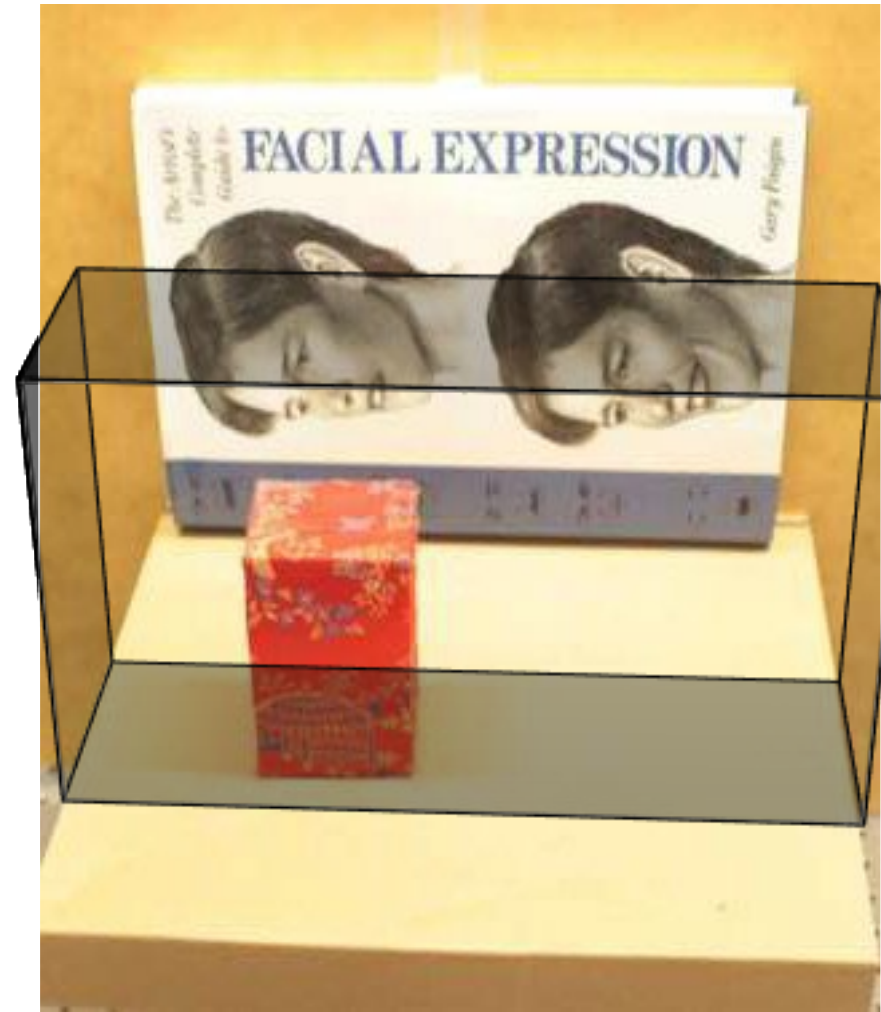


“privacy zone”

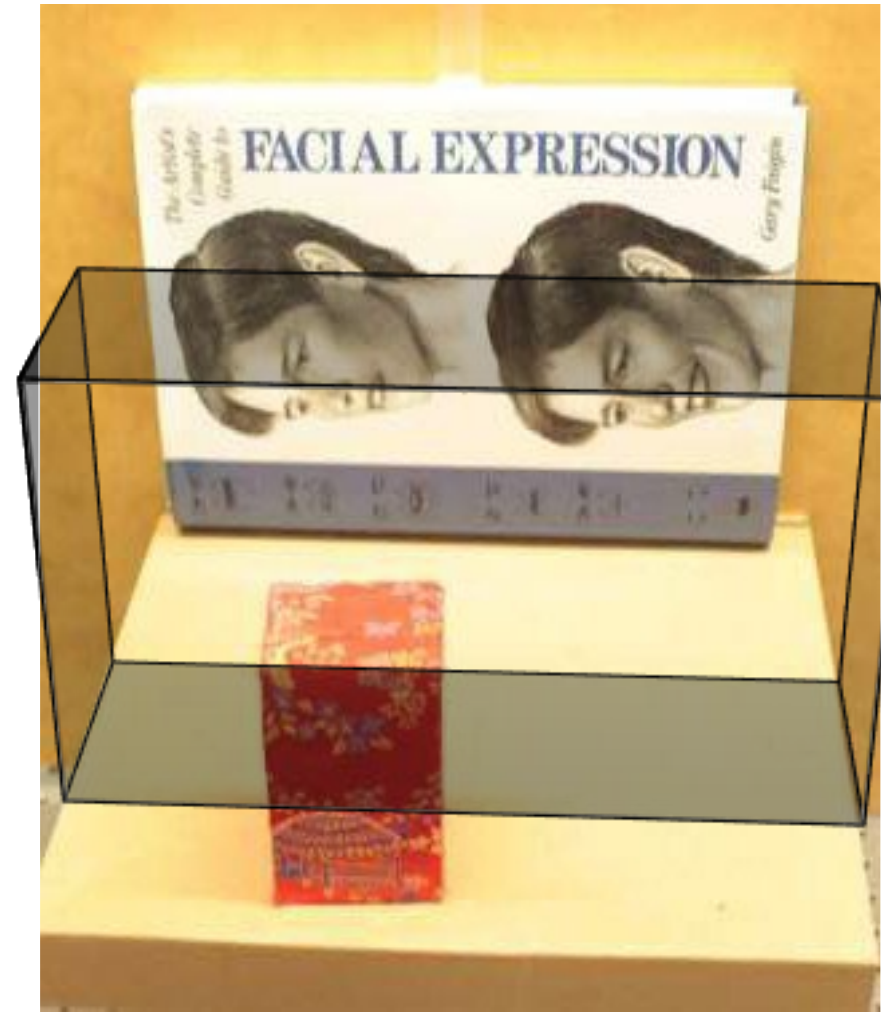
Using the non-symmetric LTM – *Removing 3D regions*



Using the non-symmetric LTM – *Removing 3D regions*



Using the non-symmetric LTM – *Removing 3D regions*



Questions?

Computational Light Transport - *Applications*

- 1) Relighting
- 2) Descattering

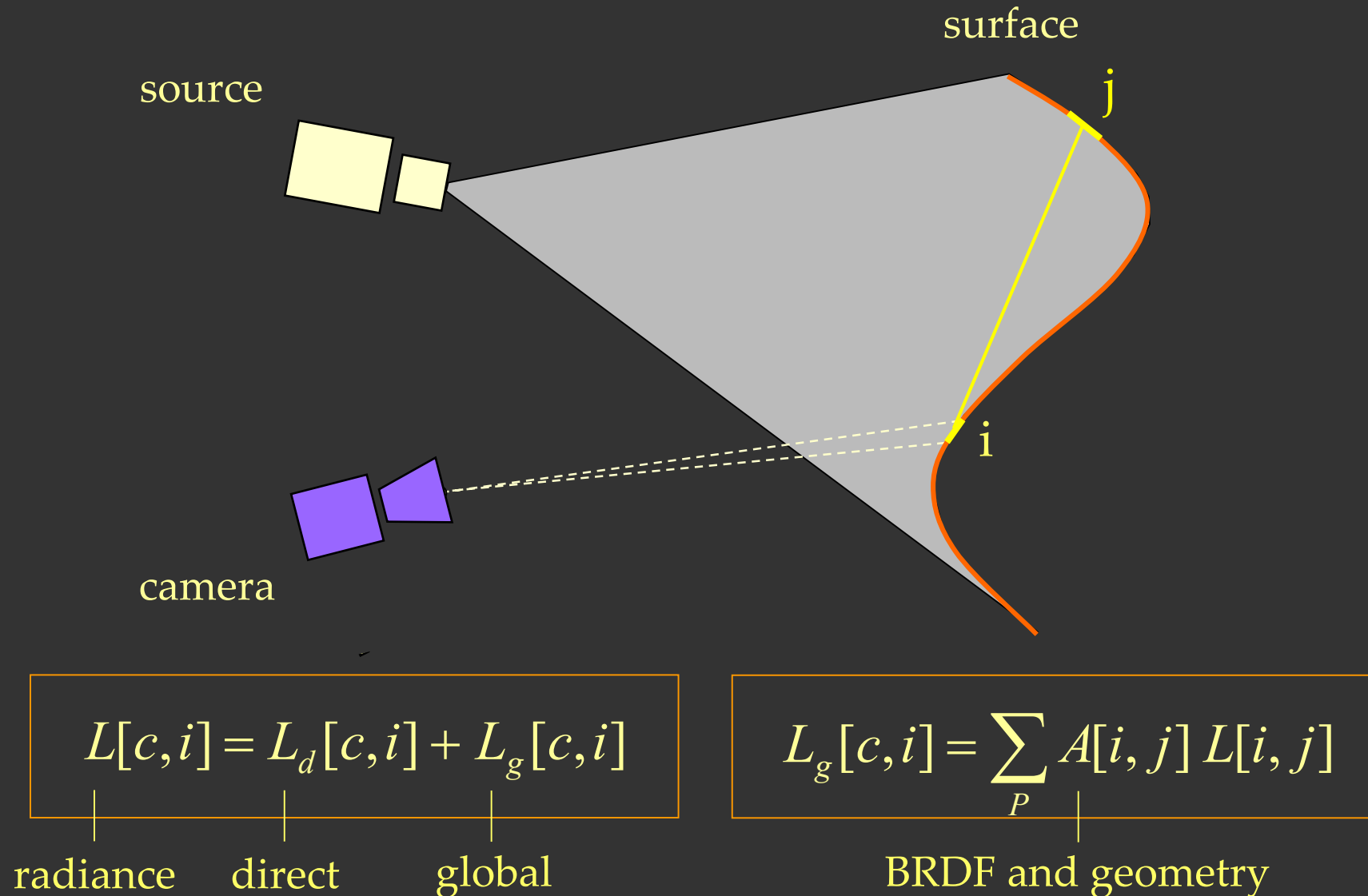
If you can descatter, you can also obtain the

- 3) Making regions “invisible”
info of the medium!

- 4) Direct-Indirect Separation

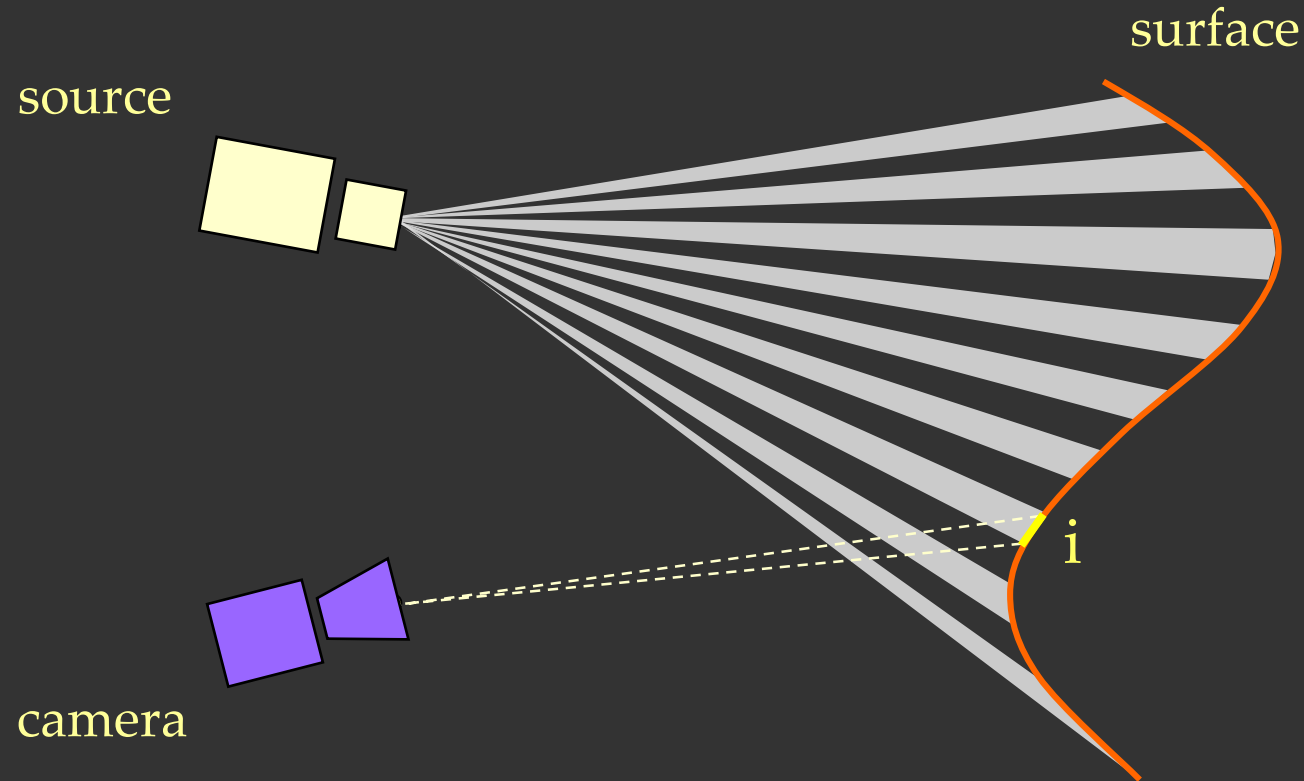
Question: Do we need the full LTM
for direct-indirect separation?

Fast Direct-Indirect Illumination Separation [Nayar et al. 2006]



Fast Direct-Indirect Illumination Separation

[Nayar et al. 2006]

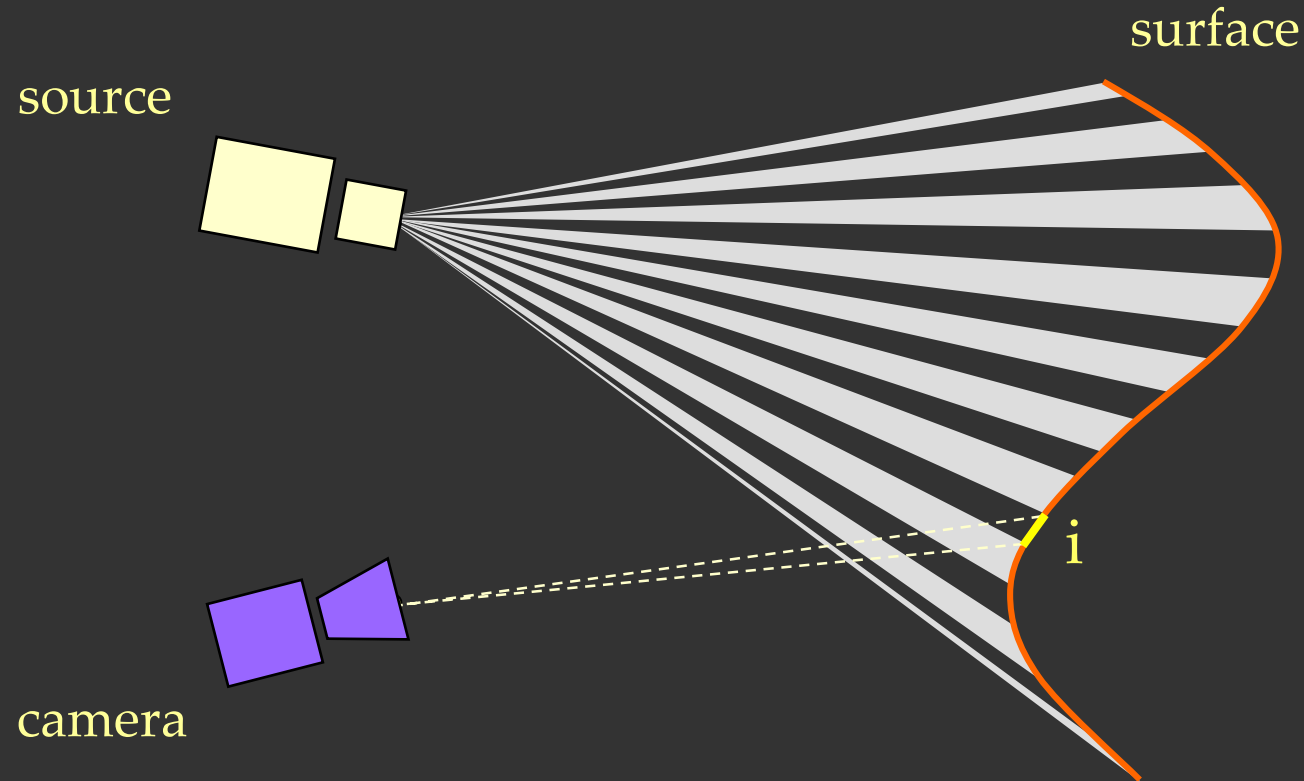


$$L^+[c, i] = L_d[c, i] + \alpha L_g[c, i]$$

fraction of activated source elements

Fast Direct-Indirect Illumination Separation

[Nayar et al. 2006]



[Nayar et al. 2006]

$$L^+[c, i] = L_d[c, i] + \alpha L_g[c, i]$$

$$\bar{L}^-[c, i] = (1 - \alpha) L_g[c, i]$$

fraction of activated source elements

Fast Direct-Indirect Illumination Separation

[Nayar et al. 2006]

$$\alpha = \frac{1}{2}:$$

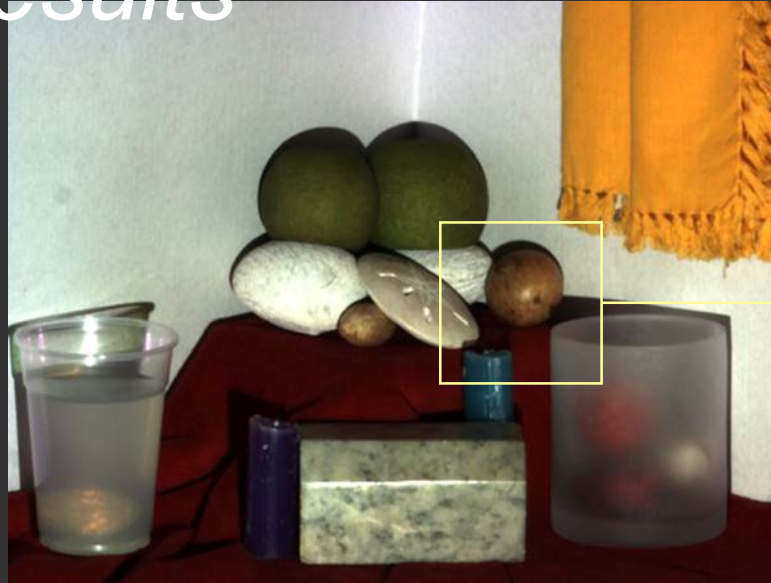
$$L_d = L_{\max} - L_{\min}, \quad L_g = 2L_{\min}$$

direct

global

GI Separation - *Results*

[Nayar et al. 2006]



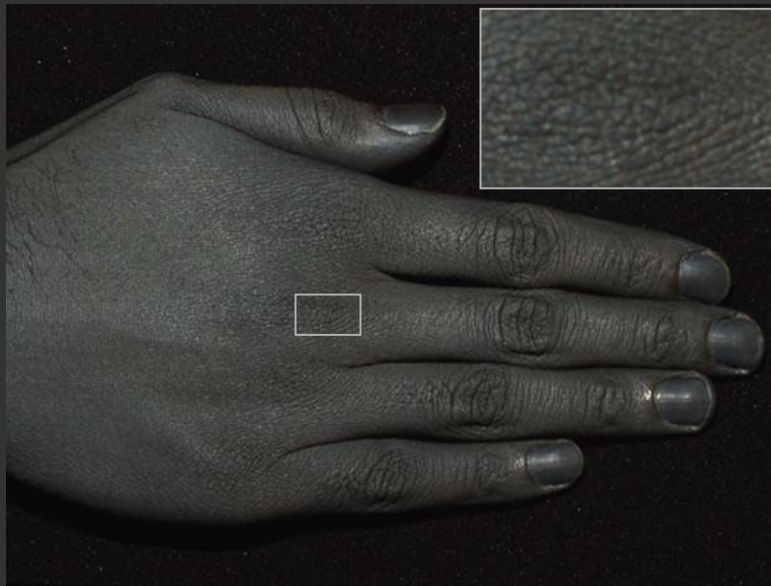
Direct



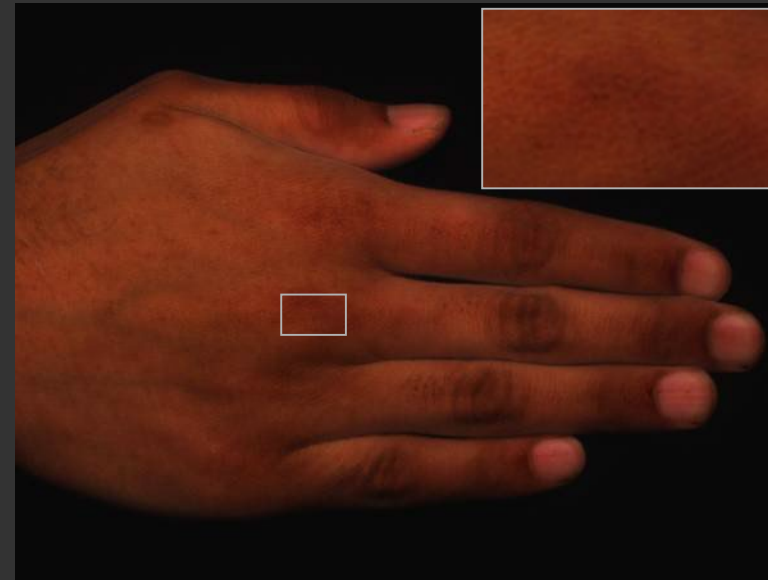
Global

GI Separation - *Results*

[Nayar et al. 2006]



Direct



Global

GI Separation - *Results*

[Nayar et al. 2006]

Without Makeup



Direct



Global



With Makeup



Direct



Global



Questions?

Computational Light Transport - *Applications*

- 1) Relighting
- 2) Descattering

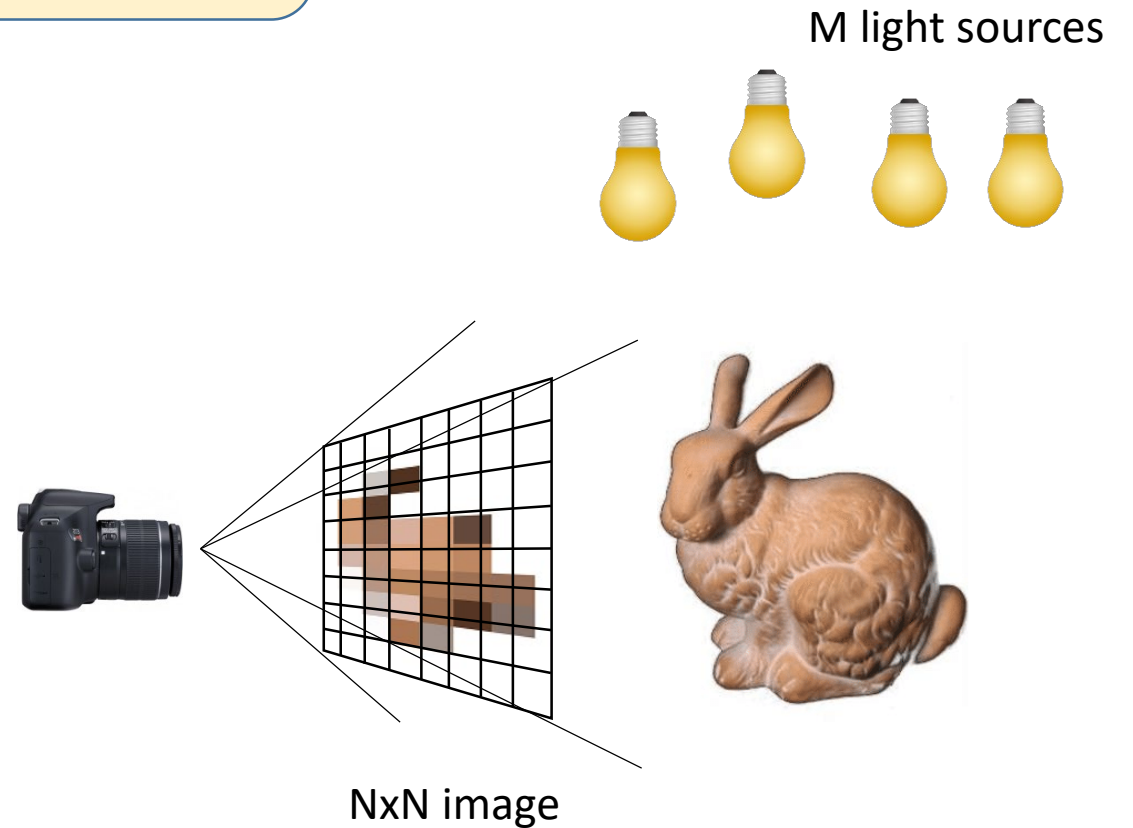
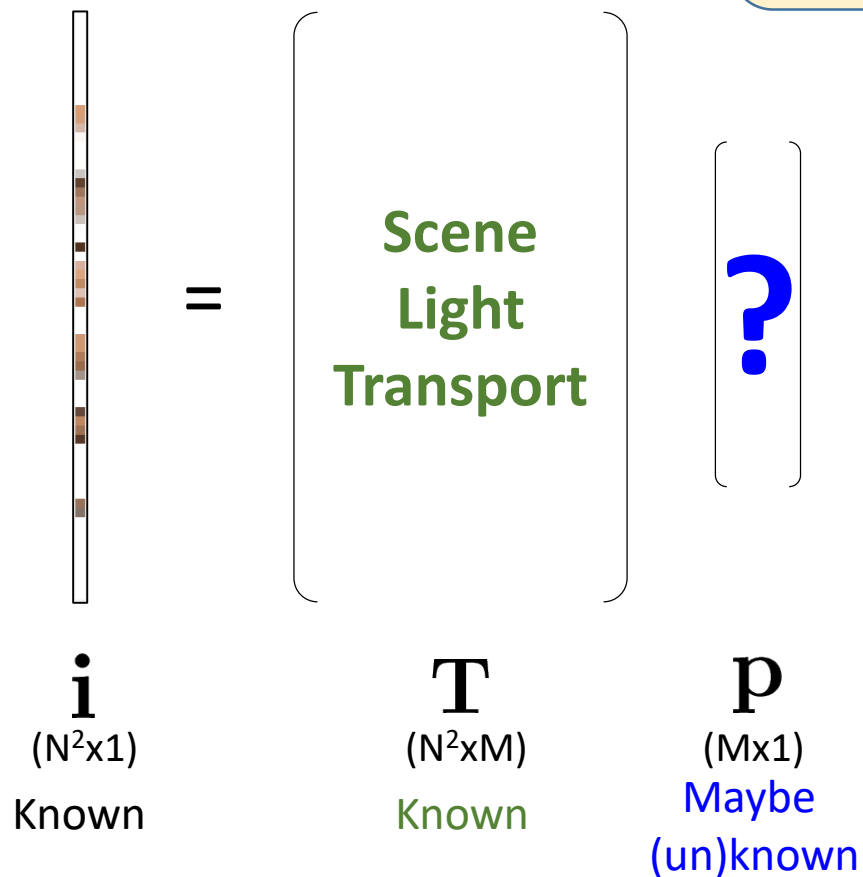
If you can descatter, you can also obtain the

- 3) Making regions “invisible”
- 4) Direct-Indirect Separation

What if we invert the LTM?

The Light Transport Matrix – *Inverting the LTM*

$$\mathbf{i} = \mathbf{T}\mathbf{p}$$



The Light Transport Matrix – *Inverting the LTM*

$$\mathbf{p} = \mathbf{T}^{-1} \mathbf{i}$$

Scene Reciprocal
Light Transport

$\mathbf{T}^T$
($M \times N^2$)
Known

$\mathbf{i}$
($N^2 \times 1$)
Known

$\mathbf{p}$
($M \times 1$)
Unknown

= ?



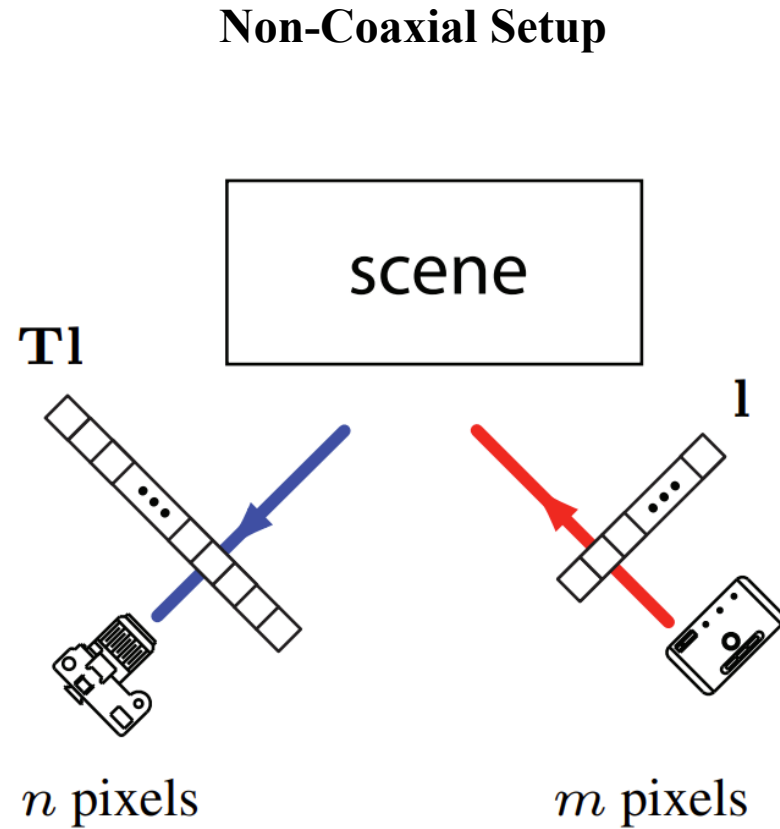
Mx1 image



NxN light sources



Dual Photography [Sen et al. 2005]



Dual Photography [Sen et al. 2005]



Dual Photography [Sen et al. 2005]



Computational Light Transport - *Applications*

- 1) Relighting
- 2) Descattering

If you can descatter, you can also obtain the

- 3) Making regions “invisible”
info of the medium!
- 4) Direct-Indirect Separation
- 5) Dual Photography

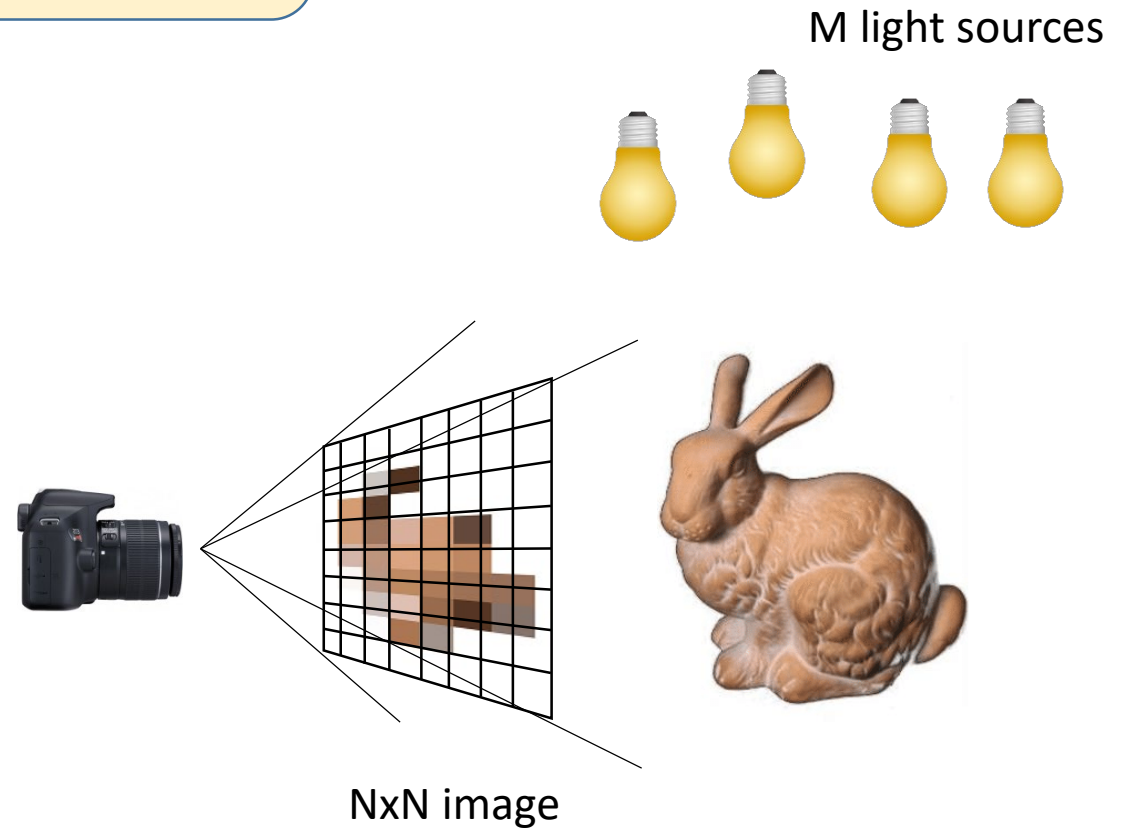
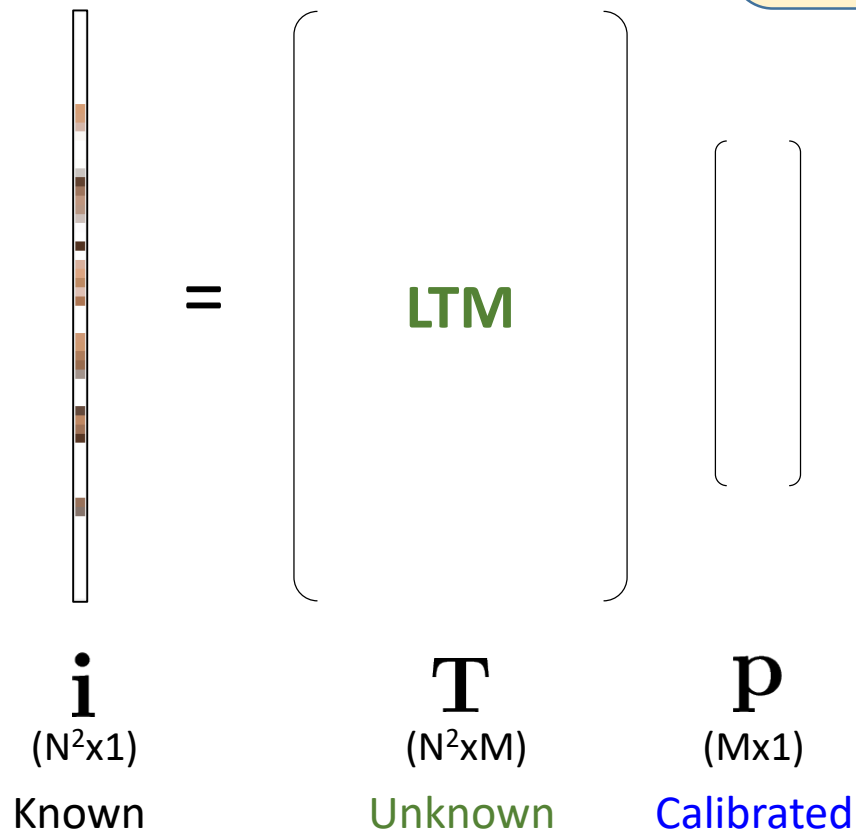
Questions?

Computational Light Transport

- **What** is it?
The set of techniques aiming to capture/reconstruct/exploit the LTM of a scene
- **Why** is it important? (a.k.a. applications)
Relighting, light transport probing (i.e. component separation), descattering...
- **How** do we capture it?

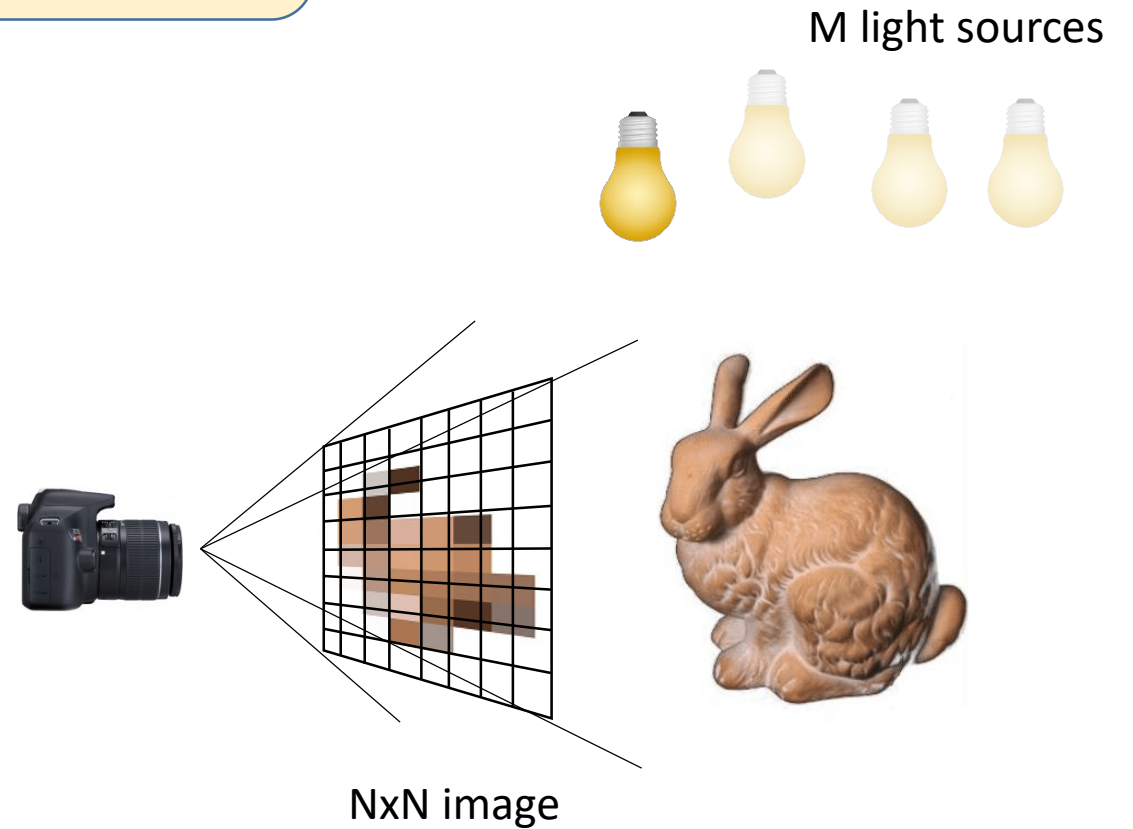
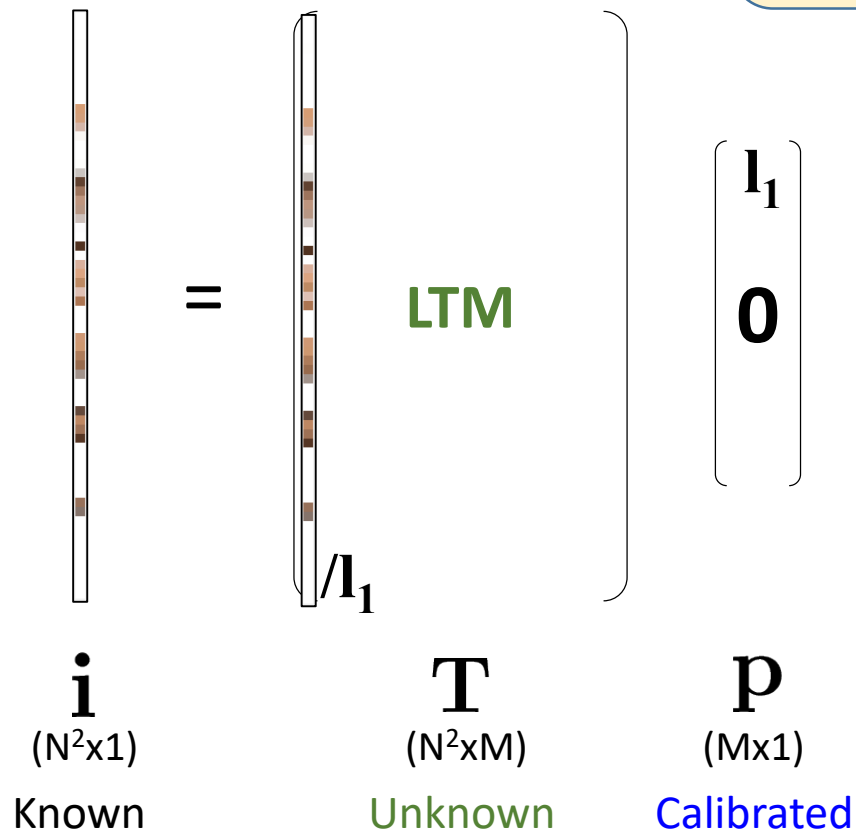
Capturing the Light Transport Matrix

$$\mathbf{i} = \mathbf{T}\mathbf{p}$$



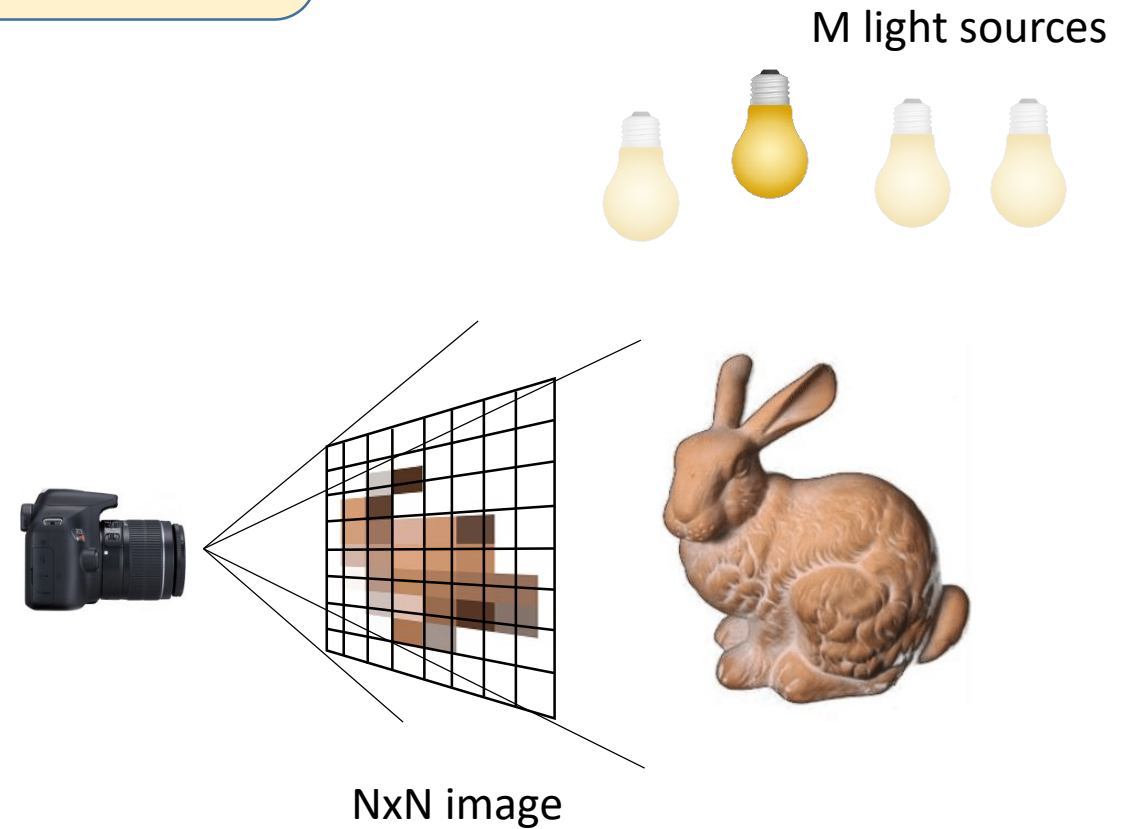
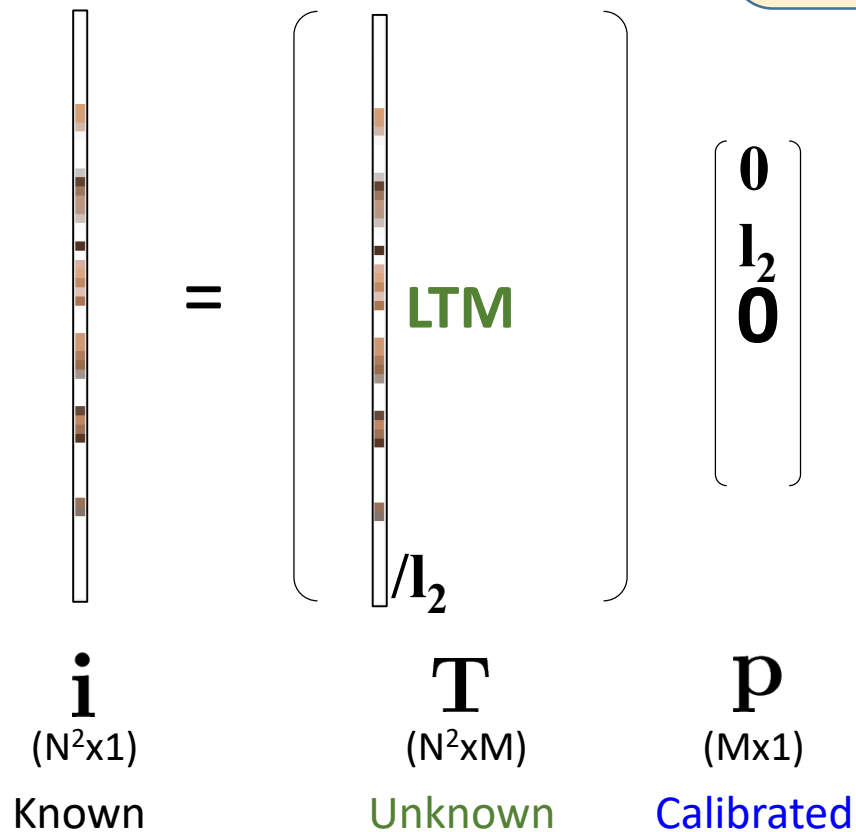
Capturing the Light Transport Matrix

$$\mathbf{i} = \mathbf{T}\mathbf{p}$$



Capturing the Light Transport Matrix

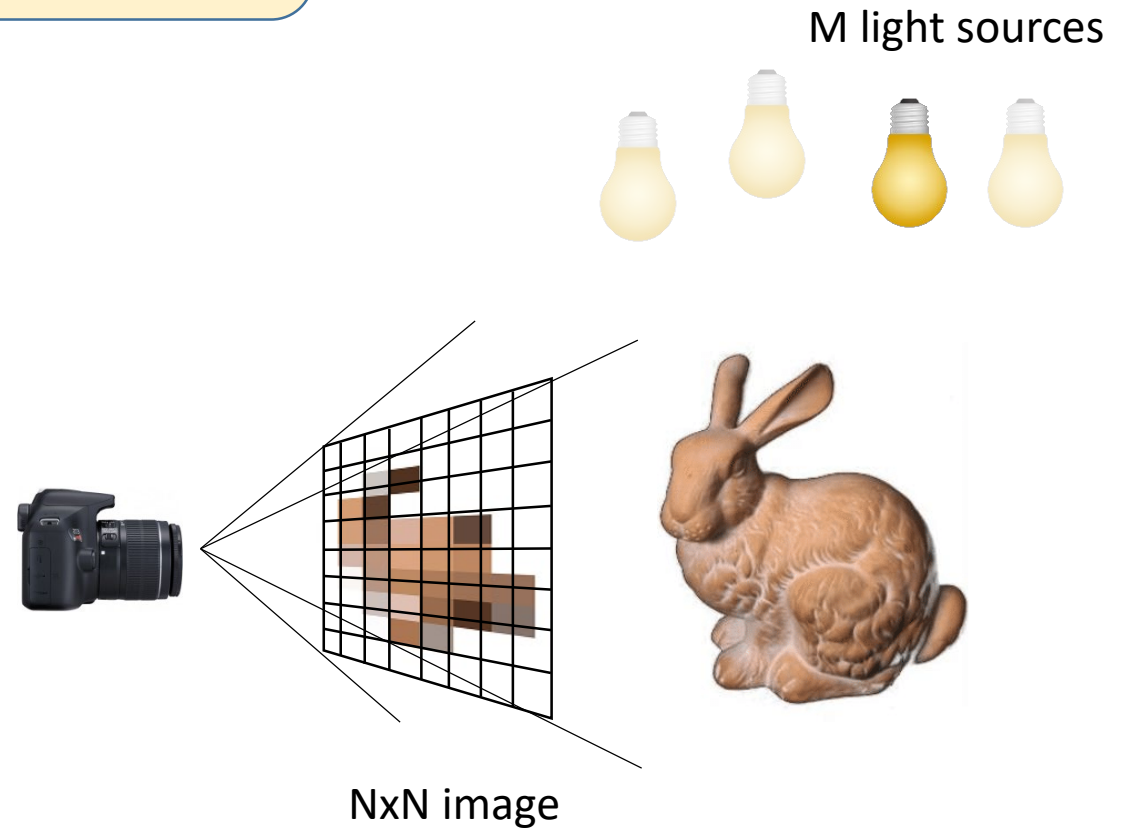
$$\mathbf{i} = \mathbf{T}\mathbf{p}$$



Capturing the Light Transport Matrix

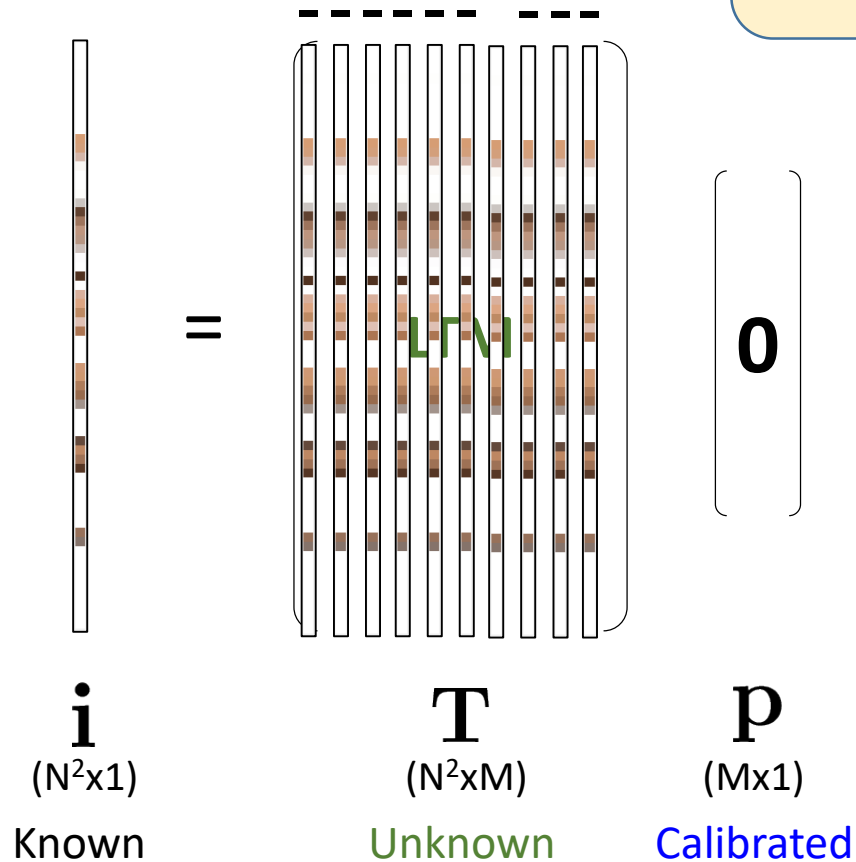
$$\mathbf{i} = \mathbf{T}\mathbf{p}$$

$$\begin{array}{c} \mathbf{i} \\ (N^2 \times 1) \\ \text{Known} \end{array} = \begin{array}{c} \left[\begin{array}{c} \text{LTM} \\ \mathbf{I}_3 \end{array} \right] \begin{array}{c} \mathbf{p} \\ (M \times 1) \\ \text{Calibrated} \end{array} \\ (N^2 \times M) \\ \text{Unknown} \end{array}$$

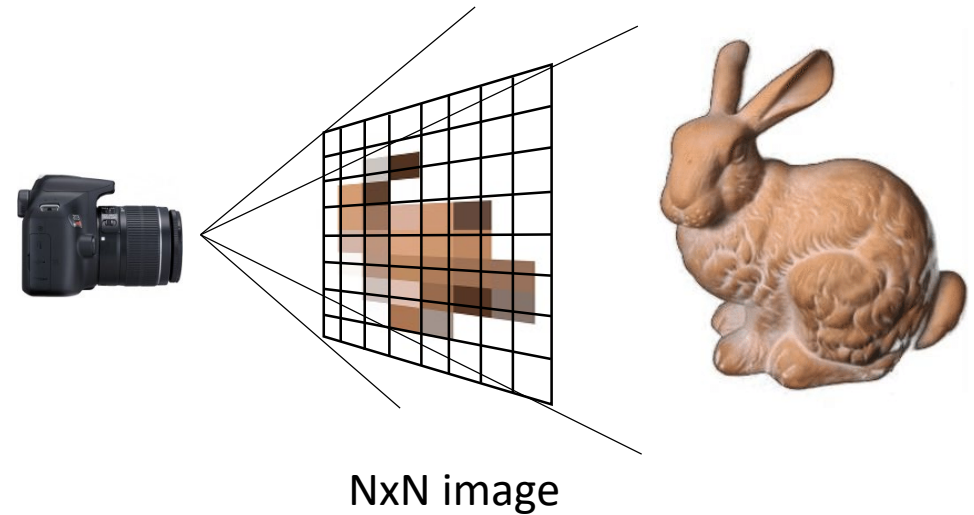


Capturing the Light Transport Matrix

$$\mathbf{i} = \mathbf{T}\mathbf{p}$$



M light sources

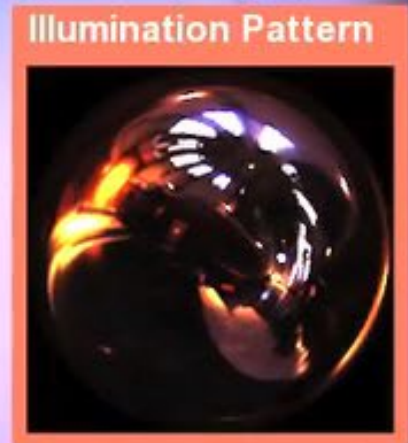


Computational Light Transport – *Capturing the LTM*

- Naïve approach – *Exhaustive scanning the LTM*
- Coded illumination
 - Low-Rank Matrix Approximations (Kernel Nystrom)
 - Compressed Sensing
 - Machine learning

Number of photos: 40

<https://youtu.be/fVBICVBEGVU?t=26>

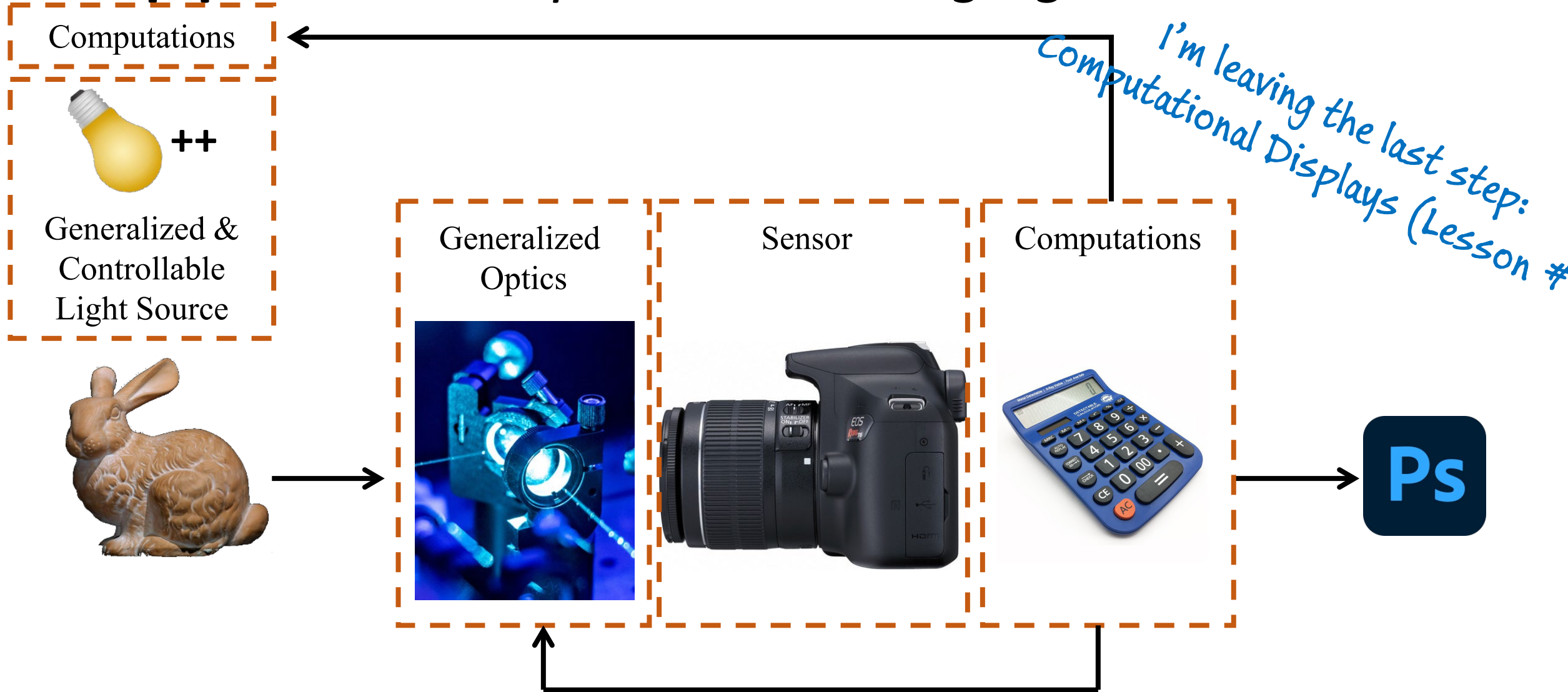


Questions?

Recap

- **Active Illumination** is very useful for a good number of applications.
 - **Example I:** Flash-No-Flash Photography
 - **Example II:** Structured Illumination
- **Computational Illumination** allows to analyze light transport in a scene
 - Many very cool applications, including relighting, light decomposition, dehazing or dual photography.

The pipeline – Computational Imaging



What's next?

- **Transient Imaging:** *A special form of active illumination for densely sampling the temporal domain*
- **Computational Displays**
- **Open Problems in Computational Imaging**
- **Final Project:** *Presentation and mentoring*

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